

Homotopy Theory

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1 Simplicial Sets

1.1 Simplicial complexes

Recall that a **simplex** in $\mathbb{R}^N$ is a subspace,

$$[v_0, \dots, v_n] = \left\{ \sum_i \lambda_i v_i \mid \sum_i \lambda_i = 1, 0 \leq \lambda_i \in \mathbb{R} \right\},$$

where $v_i \in \mathbb{R}^N$ are **geometrically independent** (if $\sum_i \lambda_i v_i = 0$ and $\sum_i \lambda_i = 0$ then $\lambda_i = 0$). The **faces** of a simplex $[v_0, \dots, v_n]$ are the simplices generated by a subset of $\{v_0, \dots, v_n\}$. The **standard n -simplex** is the one generated by the standard basis vectors of $\mathbb{R}^{n+1}$:

$$|\Delta^n| = [e_0, \dots, e_n] = \left\{ (x_0, \dots, x_n) \in \mathbb{R}^{n+1} \mid \sum_i x_i = 1, x_i \geq 0 \right\} \subseteq \mathbb{R}^{n+1}.$$

A **simplicial complex** in $\mathbb{R}^N$ is a collection of simplices K such that the face of any simplex in K is also in K , and the intersection of any two simplices in K is a face of both of them. Given a simplicial complex K , we can define its **geometric realization** $|K|$ to be the union of all the simplices in K . A subset $F \subseteq |K|$ is closed iff $F \cap \sigma$ is closed in the simplex σ for all $\sigma \in K$.

Simplicial complexes are a good starting point, but they are insufficient for our purposes. A straightforward first generalization is as follows.

1.1.1 Definition (Abstract simplicial complexes)

An **abstract simplicial complex** is a collection of sets $K = \{X^n\}_{n < \omega}$, where X^k is a collection of subsets of X^0 of cardinality $k + 1$. Furthermore, it must satisfy the condition that any subset of X^k of size $j + 1$ belongs to X^j .

Elements of X^0 are called the **vertices** of the complex.

Alternatively an abstract simplicial complex is a set K of finite subsets of X^0 which is closed under taking subsets. Then X^n is just the collection of sets in K of cardinality $n + 1$.

In a simplicial complex of $\mathbb{R}^N$ we required also that the intersection of simplices be a simplex, but this is because we were dealing with simplices embedded in an ambient space.

The standard n -simplex $|\Delta^n|$ gives rise to the abstract simplicial complex of non-empty subsets of $\{0, \dots, n\}$.

1.1.2 Definition (Face maps)

We will assume that an abstract simplicial complex K comes equipped with a total ordering on its vertices. Then for an n -simplex $\{v_0, \dots, v_n\} \in K$ we order its vertices and write $[v_0, \dots, v_n]$ to mean $v_0 < \dots < v_n$. We further define the **face maps** of the simplicial complex K to be maps $d_0, \dots, d_n: X^n \rightarrow X^{n-1}$ given by

$$d_i[v_0, \dots, v_n] = [v_0, \dots, \hat{v}_i, \dots, v_n]$$

where the hat means the vertex was removed.

For each n we have maps d_i for $i \leq n$, to differentiate the face maps between simplices of differing dimension we could write d_i^n instead of merely d_i , but this is an overload of information that is not necessary.

There is a fundamental relation between face maps which is immediately clear:

$$d_i d_j = d_{j-1} d_i, \quad \text{for } i < j. \tag{1}$$

Indeed, both sides map $[v_0, \dots, v_n]$ to $[v_0, \dots, \hat{v}_i, \dots, \hat{v}_j, \dots, v_n]$.

1.1.3 Definition (Simplicial maps)

If K, L are simplicial complexes, a **simplicial map** $f: K \rightarrow L$ is a function from vertices of K to vertices of L such that if $\{v_0, \dots, v_n\} \in K$ then $\{f(v_i)\}_i \in L$. Note that a simplicial map need not be injective.

1.2 Simplicial sets

Simplicial sets, like simplicial complexes, can be given a combinatorial definition. Unfortunately the combinatorial definition is lengthier and instead has a more concise category-theoretic definition.

1.2.1 Definition (Simplicial sets)

The **simplex category** Δ is the category whose objects are finite ordinals with order-preserving maps between them. Explicitly, its objects are $[n] = \{0, \dots, n\}$ and a morphism $f: [n] \rightarrow [m]$ is a function which preserves order (not necessarily strictly).

A **simplicial set** is a functor $\Delta^{\text{op}} \rightarrow \mathbf{Set}$. Then the category of simplicial sets is the functor category $\mathbf{sSet} = [\Delta^{\text{op}}, \mathbf{Set}]$. Morphisms are natural transformations.

Explicitly, a simplicial set X is a collection of sets $X_n = X([n])$ along with functions $f: X_n \rightarrow X_m$ for every order-preserving $f: [m] \rightarrow [n]$. Elements of X_0 are called the **vertices** of the simplicial set.

To understand simplicial sets, it is worthwhile to understand the simplex category in more depth. There are some obvious maps in the simplex category Δ :

- (1) The **coface map** $\delta^i: [n-1] \rightarrow [n]$ (for $i \leq n$) is the unique injection whose image leaves out only i . Explicitly, $\delta^i(j) = j$ for $j < i$ and $\delta^i(j) = j+1$ for $j \geq i$.
- (2) The **codegeneracy map** $\sigma^i: [n+1] \rightarrow [n]$ (for $i \leq n$) is the unique surjection which picks out i twice. Explicitly, $\sigma^i(j) = j$ for $j \leq i$ and $\sigma^i(j) = j-1$ for $j > i$.

The following identities are then easily verified:

$$\begin{aligned} \delta^j \circ \delta^i &= \delta^i \circ \delta^{j-1} \text{ for } i < j, \\ \sigma^j \circ \sigma^i &= \sigma^i \circ \sigma^{j+1} \text{ for } i \leq j, \\ \sigma^j \circ \delta^i &= \begin{cases} \delta^i \circ \sigma^{j-1} & i < j \\ \text{id} & i = j \text{ or } j+1 \\ \delta^{i-1} \circ \sigma^j & j+1 < i \end{cases}. \end{aligned}$$

Then if X is a simplicial set, we write $X(\delta^i) = d_i: X_n \rightarrow X_{n-1}$ for the image of the coface maps, and $X(\sigma^j) = s_j: X_n \rightarrow X_{n+1}$ for the image of the codegeneracy maps under X . These are called the **face** and **degeneracy** maps. Note that due to the contravariance of X , the face maps “shrink” and the degeneracy maps “grow”, while the reverse is true for the coface and codegeneracy maps.

The identities in the simplex category then dualize to

$$\begin{aligned} d_i d_j &= d_{j-1} d_i \text{ for } i < j, \\ s_i s_j &= s_{j+1} s_i \text{ for } i \leq j, \\ d_i s_j &= \begin{cases} s_{j-1} d_i & i < j \\ \text{id} & i = j \text{ or } j+1 \\ s_j d_{i-1} & j+1 < i \end{cases}. \end{aligned} \tag{1}$$

Note that the first identity is the same as the one in (1.1.1).

It is not hard to see that every morphism in the simplex category can be written as a composition of face and degeneracy maps. Thus we can equivalently define a simplicial set to be any collection of sets with face and degeneracy maps satisfying the identities in (1). And a morphism of simplicial sets $X \rightarrow Y$ is a sequence of functions $f_n: X_n \rightarrow Y_n$ which commute with the face and degeneracy maps: $d \circ f = f \circ d$ and $s \circ f = s \circ f$.

If K is a simplicial complex, then it defines a simplicial set X where

$$\begin{aligned} X_n &= \{[v_0, \dots, v_n] \mid \text{The } v_i \text{ form a simplex in } K \text{ with } v_0 \leq \dots \leq v_n\}, \\ d_i[v_0, \dots, v_n] &= [v_0, \dots, \hat{v}_i, \dots, v_n], \\ s_i[v_0, \dots, v_n] &= [v_0, \dots, s_i, s_i, \dots, v_n]. \end{aligned}$$

Note that the elements of X_n can be *degenerate* simplices. For example if $[0, 1, 2]$ is a simplex of K , then $[0, 0, 1, 2, 2, 2]$ is a simplex of X . Essentially the simplicial set associated with a simplicial complex is obtained by adjoining all of these degenerate simplices.

We can generalize what we mean by a degenerate simplex.

1.2.2 Definition

Let X be a simplicial set. A **degenerate simplex** of X is a simplex $x \in X$ which is in the image of some degeneracy map.

The prototypical simplicial set is

$$\Delta^n = \text{hom}_\Delta(\bullet, [n]).$$

We can better characterize Δ^n as follows. Notice that an increasing map $f: [k] \rightarrow [n]$ can be represented as a (degenerate) k -simplex $[f(0), f(1), \dots, f(k)]$. Thus we see that this definition of Δ^n coincides with the one obtained from taking the simplicial complex defined by $|\Delta^n|$ and adjoining degenerate simplices.

Notice by the Yoneda lemma, for a simplicial set X ,

$$\text{hom}_{\text{Set}}(\Delta^n, X) = \text{Nat}(\text{hom}_\Delta(\bullet, [n]), X) \cong X([n]) = X_n$$

where the isomorphism is obtained by mapping $f: \Delta^n \rightarrow X$ to $f(\text{id}_n)$ where $\text{id}_n: [n] \rightarrow [n]$ is the identity.

1.2.3 Definition

For a simplicial set X and an n -simplex $x \in X_n$, we define $\iota_x: \Delta^n \rightarrow X$ to be the unique simplicial map where $\iota_x(\text{id}_n) = x$. This is called the **representing map** of x .

1.3 Geometric realization

Consider the functor $|\Delta^\bullet|: \Delta \rightarrow \mathbf{Top}$ which maps $[n]$ to $|\Delta^n|$, and an order preserving map $\varphi: [n] \rightarrow [m]$ to the continuous map $|\Delta^n| \rightarrow |\Delta^m|$ given by

$$(x_0, \dots, x_n) \in |\Delta^n| \mapsto (y_0, \dots, y_m) \in |\Delta^m|, \quad y_j = \sum_{\varphi(i)=j} x_i.$$

For example, the face maps δ^i define maps $|\Delta^{n-1}| \rightarrow |\Delta^n|$ by

$$\delta^i(x_0, \dots, x_{n-1}) = (x_0, \dots, 0, \dots, x_{n-1}),$$

and the degeneracy maps $\sigma^i: |\Delta^n| \rightarrow |\Delta^{n-1}|$ give

$$\sigma^i(x_0, \dots, x_n) = (x_0, \dots, x_i + x_{i+1}, \dots, x_n).$$

This then defines a functor

$$\text{hom}_{\mathbf{Top}}(|\Delta^\bullet|, \bullet): \Delta^{\text{op}} \times \mathbf{Top} \rightarrow \mathbf{Set}.$$

In particular for every topological space X , we have a simplicial set

$$\mathcal{S}(X) = \text{hom}_{\mathbf{Top}}(|\Delta^\bullet|, X),$$

this is called the **singular set** of X . The face and degeneracy maps of the singular set are then

$$d_i(\rho) = \rho \circ \delta^i, \quad s_i(\rho) = \rho \circ \sigma^i.$$

Note:

The singular set of X plays a central role in defining the singular homology of X . Recall that X 's chain complex is defined at each coordinate to be the free Abelian group generated by $\mathcal{S}(X)_n$: $C_n(X) = Z[\mathcal{S}(X)_n]$. The differentials are then defined in terms of the face maps: $\partial: C_n(X) \rightarrow C_{n-1}(X)$ is defined on a singular n -simplex $\sigma \in \mathcal{S}(X)_n$ by

$$\partial\sigma = \sum_{i=0}^n (-1)^i d_i\sigma,$$

and ∂ is extended to all of $C_n(X)$ by linearity.

Geometrically, we view a simplicial set X as a collection of (possibly degenerate) n -simplices X_n for all $n < \omega$. If we attach the geometric n -simplex $|\Delta^n|$ to each n -simplex X_n we obtain $X_n \times |\Delta^n|$. We give each X_n the discrete topology, so this becomes a topological space. The geometric realization of X as a whole should be the union of all these simplices, but this is not enough. For one, degenerate simplices should not be counted; and we want to glue the faces of simplices together.

So we define the geometric realization as follows.

1.3.1 Definition

For a simplicial set X , we define its **geometric realization** to be the space

$$|X| = \coprod_{n < \omega} X_n \times |\Delta^n| / \sim$$

where each X_n has the discrete topology and $\sim$ is the equivalence generated by the relations

$$(x, \delta^i p) \sim (d_i x, p), \quad (x, \sigma^i p) \sim (s_i x, p).$$

Here $\delta^i: |\Delta^{n-1}| \rightarrow |\Delta^n|$ and $\sigma^i: |\Delta^n| \rightarrow |\Delta^{n-1}|$ are the face and degeneracy maps respectively of the standard simplices.

For $x \in X_n$, we denote by $(x, |\Delta^n|)$ the image of $\{x\} \times |\Delta^n|$ in $|X|$.

1.3.2 Lemma

If X is a simplicial set, then every degenerate simplex of X is the degeneracy of a unique non-degenerate simplex. That is, if $x \in X$ is degenerate then there is a unique non-degenerate $y \in X$ such that $x = s_{i_1} \cdots s_{i_k} y$ for some i_j s.

Proof

Clearly if x is degenerate there must exist *some* y which satisfies this. Now suppose $x = Sy = S'y'$ for y, y' non-degenerate and S, S' a composition of degeneracy maps. Suppose $S = s_{i_1} \cdots s_{i_k}$, then define $D = d_{i_k} \cdots d_{i_1}$. This is a left inverse for S , and thus $y = DS'y'$.

Degeneracy and face maps commute according to the identities laid out in (1.2.1). Thus $y = DS'y' = \tilde{S}\tilde{D}y'$ for some composition of degeneracy maps and face maps. Since y is by assumption non-degenerate, $\tilde{S}$ must be empty so that $y = \tilde{D}y'$.

By symmetry, $y' = \bar{D}y$ for some composition of face maps as well. Since face maps lower the dimension, this can only happen if $\bar{D}, \tilde{D}$ are empty. Thus $y = y'$. ■

Similarly we have the following result.

1.3.3 Lemma

Every point $v \in |\Delta^n|$ can be expressed uniquely as $v = \delta^{i_1} \cdots \delta^{i_k} u$ where u is an interior point (all of its coordinates are positive).

Call a point $(x, v) \in |X|$ **non-degenerate** if x is non-degenerate and v is interior.

1.3.4 Lemma

For a simplicial set X , every point in its geometric realization $(x, v) \in |X|$ is equivalent to a unique non-degenerate point of $|X|$.

Proof

Define $\lambda: |X| \rightarrow |X|$ as follows. For $x \in X_n$ take $i_1, \dots, i_k$ and y non-degenerate such that $x = s_{i_1} \cdots s_{i_k} y$. Then define

$$\lambda(x, v) = (y, \sigma^{i_k} \cdots \sigma^{i_1} v).$$

Similarly define $\mu: |X| \rightarrow |X|$, which for (x, v) maps

$$\mu(x, v) = (d_{i_k} \cdots d_{i_1} x, u)$$

where $v = \delta^{i_1} \cdots \delta^{i_k} u$.

Clearly λ, μ both map points to equivalent points. Then $\lambda \circ \mu$ maps a point to an equivalent non-degenerate point.

Uniqueness is because it can be verified that if $p \sim p'$ then $\lambda\mu(p) = \lambda\mu(p')$. ■

An important fact is the following.

1.3.5 Theorem

The geometric realization of a simplicial set X is a CW complex.

Proof

Take the n -cells of $|X|$ to be the images of the non-degenerate n -simplices of X . That is, simplices $(x, |\Delta^n|)$ for $x \in X_n$ non-degenerate. By the previous lemma, these cover $|X|$, and it can be verified that these make $|X|$ into a CW complex. ■

1.3.6 Theorem

There is an adjunction $|\bullet|: \mathbf{sSet} \rightleftarrows \mathbf{Top}: \mathcal{S}$ between the geometric realization and the singular set functors.

Proof

For a simplicial set X and a topological space Y , we define maps

$$\Phi: \mathbf{hom}_{\mathbf{Top}}(|X|, Y) \rightleftarrows \mathbf{hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y)): \Psi$$

as follows. For a continuous $f: |X| \rightarrow Y$, restricting f to a non-degenerate simplex $(x, |\Delta^n|)$ restricts to a continuous map $\Phi(f)_n: |\Delta^n| \rightarrow Y$.

Conversely for a map $g: X \rightarrow \mathcal{S}(Y)$, for each n -simplex $x \in X$ we have a continuous function $\rho_x: |\Delta^n| \rightarrow Y$. So define $\Psi(g): |X| \rightarrow Y$ to act on the simplex $(x, |\Delta^n|) \in |X|$ by applying ρ_x to $|\Delta^n|$. ■

1.4 Kan complexes

Recall that $\mathbf{sSet} = [\Delta^{\text{op}}, \mathbf{Set}]$. Since $\mathbf{Set}$ is complete and cocomplete, so too is $\mathbf{sSet}$. That is to say, $\mathbf{sSet}$ has all (small) limits and colimits. The co/limit of a functor is taken coordinate-wise, so such limits and colimits are easy to describe.

Most important for now is the product of simplicial sets. If K, L are simplicial sets, then their product $K \times L$ is the simplicial set where

$$(K \times L)_n = K_n \times L_n \text{ for } n < \omega, \quad (K \times L)\varphi = (K\varphi) \times (L\varphi) \text{ for } \varphi: [n] \rightarrow [m].$$

Explicitly, the face and degeneracy maps are

$$d_i(x, y) = (d_i(x), d_i(y)), \quad s_i(x, y) = (s_i(x), s_i(y)).$$

If X is a simplicial set, a **simplicial subset** of X is a simplicial set Y such that $Y_n \subseteq X_n$ for all n and the face and degeneracy maps of X restrict to those of Y . This is equivalent to Y being a subobject of X in the functor category $\mathbf{sSet} = [\Delta^{\text{op}}, \mathbf{Set}]$ (see the following note).

Note:

In a functor category $[\mathcal{C}, \mathcal{D}]$ there are two notions of a “subfunctor”. Let $F, G: \mathcal{C} \rightarrow \mathcal{D}$ be functors, then we may say that F is a subfunctor of G iff it is a subobject of G in the functor category. This means that there is a monic natural transformation $F \Rightarrow G$.

Another, generally easier to validate, definition is that F is a subfunctor of G iff $F(c)$ is a subobject of $G(c)$ in a natural way. That is to say, there is a natural transformation $\alpha: F \Rightarrow G$ whose components $\alpha_c: F(c) \rightarrow G(c)$ are monic.

In general clearly the latter implies the former: a pointwise monic natural transformation is clearly monic. Under relatively weak conditions, that $\mathcal{D}$ has pullbacks, the former also implies the latter.

Recall that a morphism $f: X \rightarrow Y$ is monic iff its pullback with itself is X (with projections given by the identity). That is, iff the following square is a pullback

$$\begin{array}{ccc} X & \xrightarrow{\quad \text{id} \quad} & X \\ \text{id} \downarrow & & \downarrow f \\ X & \xrightarrow{\quad f \quad} & Y \end{array}$$

Thus if $\alpha: F \Rightarrow G$ is monic, then its pullback with itself is F . Limits in the functor category are computed pointwise, so this means that α_c ’s pullback with itself is Fc . This means that α_c is monic for all c , as desired.

Notice that if $\{X^i\}_{i \in I}$ is a collection of simplicial subsets of X , then so is their intersection and union. This is because the X^i are invariant under any $\varphi: X_m \rightarrow X_n$, and thus so too is their intersection and union.

1.4.1 Definition

If $S \subseteq X$ is an arbitrary subset (meaning a subset of $\bigcup_n X_n$), then the simplicial subset it **generates** is the smallest simplicial subset of X containing S (meaning it contains $S \cap X_n$ as n -simplices). Such a minimal simplicial subset exists as we can take the intersection of all simplicial subsets containing S .

For a morphism in the simplex category $\varphi: [n] \rightarrow [m]$, let us write $X_\varphi: X_m \rightarrow X_n$ for the induced map. Then if $S \subseteq X$ is an arbitrary set, then the simplicial subset it generates is just

$$(S) = \{X_\varphi(x) \mid x \in S\}.$$

This is a not-so-formal way of saying that $(S)_n$ is the collection of all $X_\varphi(x)$ where $x \in S \cap X_k$ and $\varphi: [n] \rightarrow [k]$. Recall the standard n -simplex $\Delta^n = \text{hom}_\Delta(\bullet, [n])$. Its k -simplices Δ_k^n are then order-preserving maps $[k] \rightarrow [n]$. In particular, Δ_{n-1}^n consists only of $\delta^0, \dots, \delta^n$. Recall that the k -simplex in Δ^n corresponding to $\varphi: [k] \rightarrow [n]$ is given by $[\varphi(0), \dots, \varphi(k)]$. In particular δ^i corresponds to $[0, \dots, \hat{k}, \dots, n]$.

1.4.2 Definition

The **boundary** of Δ^n , $\partial\Delta^n$, is the simplicial subset of Δ^n generated by $\delta^0, \dots, \delta^n$.

The **simplicial n -horn** Λ_k^n is the simplicial subset of Δ^n generated by $\delta^0, \dots, \hat{\delta}^k, \dots, \delta^n$ (all coface maps except for δ^k).

The i th face of Δ^n is the simplicial subset $\partial_i\Delta^n \subseteq \Delta^n$ generated by δ^i . Since the union of simplicial subsets is a simplicial subset, we see that

$$\partial\Delta^n = \bigcup_{i \leq n} \partial_i\Delta^n, \quad \Lambda_k^n = \bigcup_{i \neq k} \partial_i\Delta^n.$$

In more generality, for a set of indices $I \subseteq [n]$ we can define Δ^I to be the simplicial subset of Δ^n consisting of functions $f: [k] \rightarrow [n]$ whose image lies in I . This is indeed a simplicial subset because precomposing f with $\varphi: [m] \rightarrow [k]$ still has its image lie in I .

Then $\partial_i\Delta^n = \Delta^{[n]-i}$. Indeed, δ^i by definition “misses” i . And if $\varphi: [k] \rightarrow [n]$ misses i , then we can find $\psi: [k] \rightarrow [n-1]$ such that $\varphi = \delta^i \circ \psi$ so that φ is generated by δ^i . Then

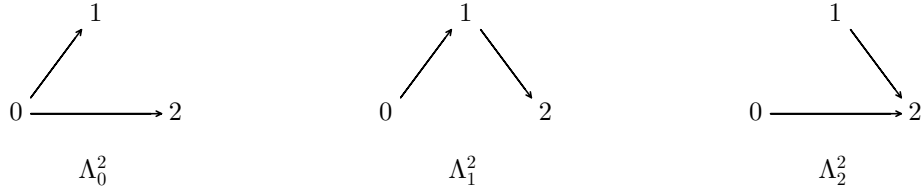
$$\partial\Delta^n = \bigcup_{i=0}^n \Delta^{[n]-i}$$

so that its k th simplices consists of maps $\varphi: [k] \rightarrow [n]$ which are not surjective (as it must miss some i).

Similarly

$$\Lambda_k^n = \bigcup_{i \neq k} \Delta^{[n]-i}, \quad (\Lambda_k^n)_j = \left\{ f: [j] \rightarrow [n] \mid f \text{ does not have } 0, \dots, \hat{k}, \dots, n \text{ in its image} \right\}.$$

Since δ^k is the unique map without k in its image, geometrically we view Λ_k^n as the union of the faces of Δ^n except for the one opposite the k th vertex. For $n = 2$ for example, the three horns are



Note that there exists a coequalizer diagram

$$\coprod_{0 \leq i < j \leq n} \Delta^{n-2} \rightrightarrows \coprod_{i=0}^n \Delta^{n-1} \rightarrow \partial\Delta^n$$

where the first morphisms have coordinates $\Delta^{n-2} \rightarrow \coprod_i \Delta^{n-1}$ correspond to δ^{j-1} and δ^i respectively. That is,

$$\begin{array}{ccccc} \Delta^{n-2} & \xrightarrow{\delta^{j-1}} & \Delta^{n-1} & & \\ \downarrow \text{inc}_{i < j} & & \downarrow \text{inc}_i & \searrow \delta^i & \\ \coprod_{i < j} \Delta^{n-2} & \rightrightarrows & \coprod_i \Delta^{n-1} & \longrightarrow & \partial\Delta^n \\ \uparrow \text{inc}_{i < j} & & \uparrow \text{inc}_j & \nearrow \delta^j & \\ \Delta^{n-2} & \xrightarrow{\delta^i} & \Delta^{n-1} & & \end{array}$$

Similarly there is a coequalizer

$$\begin{array}{ccccc}
 \Delta^{n-2} & \xrightarrow{\delta^{j-1}} & \Delta^{n-1} & & \\
 \text{inc}_{i < j} \downarrow & & \downarrow \text{inc}_i & \searrow \delta^i & \\
 \coprod_{i < j} \Delta^{n-2} & \xRightarrow{\quad} & \coprod_{i \neq k} \Delta^{n-1} & \xrightarrow{\quad} & \Lambda_k^n \\
 \text{inc}_{i < j} \uparrow & & \uparrow \text{inc}_j & \nearrow \delta^j & \\
 \Delta^{n-2} & \xrightarrow{\delta^i} & \Delta^{n-1} & &
 \end{array}$$

Using the universal property of coequalizers, the following lemma is immediate.

1.4.3 Lemma

Let X be an arbitrary simplicial set.

- (1) A simplicial map $\partial\Delta^n \rightarrow X$ is the same as a tuple $(y_0, \dots, y_n)$ of $(n-1)$ -simplices of X such that $d_i y_j = d_{j-1} y_i$ for $i < j$.
- (2) A simplicial map $\Lambda_k^n \rightarrow X$ is the same as a tuple $(y_0, \dots, \hat{y}_k, \dots, y_n)$ of $(n-1)$ -simplices of X such that $d_i y_j = d_{j-1} y_i$ for $i < j$, neither equal to k .

Proof

A map out of a coequalizer $A \rightrightarrows B$ is just a map $B \rightarrow X$ whose composition with either arrow in the coequalizer diagram is equal. Couple this with the fact that $\text{hom}_{\text{Set}}(\Delta^n, X) \cong X_n$ and you have the desired result. $\blacksquare$

1.4.4 Definition

Say that a simplicial set X has dimension $\leq n$ if all k -simplices are degenerate for $k > n$. The **n -skeleton** of X is the largest simplicial subset of X with dimension $\leq n$ (which exists as we can take unions). We denote it by $\text{sk}_n(X)$.

Alternatively, the n -skeleton of X is the simplicial subset generated by all non-degenerate simplices of dimension $\leq n$.

For example, $\text{sk}_{n-1}(\Delta^n) = \partial\Delta^n$. This is because $\partial\Delta_k^n$ consists only of degenerate simplices for $k \geq n$, and for $k < n$ we have $\partial\Delta_k^n = \Delta_k^n$.

In general we have a filtration of a simplicial set through its skeletons,

$$\emptyset = \text{sk}_{-1}(X) \subseteq \text{sk}_0(X) \subseteq \text{sk}_1(X) \subseteq \dots, \quad X = \bigcup_n \text{sk}_n(X).$$

1.4.5 Definition

A simplicial map $p: X \rightarrow Y$ is a **fibration** if for every commutative diagram of solid arrows, there is a dashed arrow making it commute

$$\begin{array}{ccc}
 \Lambda_k^n & \xrightarrow{\quad} & X \\
 \downarrow & \nearrow \text{dashed} & \downarrow p \\
 \Delta^n & \xrightarrow{\quad} & Y
 \end{array}$$

The arrow $\Lambda_k^n \hookrightarrow \Delta^n$ is the inclusion map.

Fibrations are also called **Kan fibrations**.

Equivalently, for every tuple of $(n-1)$ -simplices $(x_0, \dots, \hat{x}_k, \dots, x_n)$ such that $d_i x_j = d_{j-1} x_i$ for $i < j$ not equal to k , if there is an n -simplex $y \in Y_n$ such that $d_i y = p(x_i)$, then there is an n -simplex $x \in X_n$ such that $d_i x = x_i$ and $p(x) = y$.

There is a similar concept of fibration for topological spaces. A continuous map $f: T \rightarrow U$ is a **Serre fibration** if a dashed arrow exists for any commutative diagram of continuous maps

$$\begin{array}{ccc} |\Lambda_k^n| & \xrightarrow{\quad} & T \\ \downarrow & \nearrow \text{dashed} & \downarrow f \\ |\Delta^n| & \xrightarrow{\quad} & U \end{array}$$

Because $\mathcal{S}(|\Delta^n|) = \Delta^n$ and $\mathcal{S}(|\Lambda_k^n|) = \Lambda_k^n$, and geometric realization is left-adjoint to $\mathcal{S}$, such diagrams can be identified with diagrams

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{\quad} & \mathcal{S}(T) \\ \downarrow & \nearrow \text{dashed} & \downarrow \mathcal{S}(f) \\ \Delta^n & \xrightarrow{\quad} & \mathcal{S}(U) \end{array}$$

So that $f: T \rightarrow U$ is a Serre fibration iff $\mathcal{S}(f): \mathcal{S}(T) \rightarrow \mathcal{S}(U)$ is a (Kan) fibration.

Notice that for any topological space X and one-point space $*$, $X \rightarrow *$ is a Serre fibration. This is because $|\Lambda_k^n| \hookrightarrow |\Delta^n|$ is a strong deformation retract. Thus $\mathcal{S}(X) \rightarrow \mathcal{S}(*)$ is a fibration of simplicial sets. Note that $\mathcal{S}(*) = \Delta^0 = *$ is the terminal simplicial set (with a single simplex of each dimension).

This leads to the following definition.

1.4.6 Definition

A simplicial set X is **fibrant**, or a **Kan complex**, if the map $X \rightarrow *$ is fibrant.

Simplicial sets which arise from geometric simplicial complexes are not Kan complexes, thus Δ^n is not. Neither is $\partial\Delta^n$.

The following lemma characterizes equivalent conditions for a simplicial set being fibrant, and is immediate.

1.4.7 Lemma

Let X be a simplicial set, then the following are equivalent.

- (1) X is fibrant.
- (2) Every map $f: \Lambda_k^n \rightarrow X$ extends to a map $g: \Delta^n \rightarrow X$ for all n, k .
In a commutative diagram:

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{\quad} & X \\ \downarrow & \nearrow \text{dashed} & \\ \Delta^n & & \end{array}$$

- (3) For every tuple of $(n-1)$ -simplices $(x_0, \dots, \hat{x}_k, \dots, x_n)$ of X such that $d_i x_j = d_{j-1} x_i$ for $i < k$ not k , there is an n -simplex $x \in X_n$ such that $d_i x = x_i$.

Fibrant simplicial sets are extremely important, as we will see soon.

An interesting result is that the underlying simplicial set of every simplicial group is fibrant. A simplicial group is a simplicial object in the category of groups, i.e. a functor $\Delta^{\text{op}} \rightarrow \text{Grp}$. By underlying simplicial set, we mean the composition of the functor with the forgetful functor $\text{Grp} \rightarrow \text{Set}$.

1.4.8 Lemma (Moore)

If $H: \Delta^{\text{op}} \rightarrow \text{Grp}$ is a simplicial group, its underlying set is fibrant.

1.5 Anodyne extensions

The homotopy theory of simplicial sets revolves around fibrations. Originally this was developed in terms of the extension condition, which results in tedious combinatorial manipulations. Fortunately with a more modern viewpoint we can suppress most of these combinatorial arguments.

1.5.1 Definition

A class M of monomorphisms of $\mathbf{sSet}$ is called **saturated** if the following conditions are satisfied.

- (1) M contains all isomorphisms.
- (2) M is closed under pushouts in the sense that if

$$\begin{array}{ccc} A & \xrightarrow{\quad} & C \\ i \downarrow & & \downarrow i_* \\ B & \xrightarrow{\quad} & B \cup_A C \end{array}$$

is a pushout square and $i \in M$, then $i_* \in M$ as well.

- (3) Each retract of an element of M is in M . That is, given the following commutative diagram

$$\begin{array}{ccccc} A' & \xrightarrow{\quad} & A & \xrightarrow{\quad} & A' \\ i' \downarrow & & \downarrow i & & \downarrow i' \\ B' & \xrightarrow{\quad} & B & \xrightarrow{\quad} & B' \end{array}$$

where the horizontal arrows compose to the identity. If $i \in M$ then $i' \in M$.

- (4) M is closed under countable many compositions and arbitrary direct sums. This means that if there is a countable chain

$$A_1 \xrightarrow{i_1} A_2 \xrightarrow{i_2} \dots$$

with all $i_j \in M$, then the canonical map $A_1 \rightarrow \text{colim } A_i$ is in M .

Similarly if $i_j: A_j \rightarrow B_j$ for $j \in I$ are all M , then so is

$$\coprod_j i_j: \coprod_{j \in I} A_j \rightarrow \coprod_{j \in I} B_j.$$

Note that the maps in (2), (3), (4) may not obviously be monic. But these are all easily checked to be so. We do this for (2), denote $f: A \rightarrow C$ so that $B \cup_A C$ is the disjoint union $B \amalg C$ where we identify $f(a)$ and $i(a)$. The map i_* then maps $c \in C$ to its equivalence class $[c]$. Notice that because i is monic, we cannot identify $f(a), f(a')$ as that would mean $i(a) = i(a')$ and thus $a = a'$. Thus no two distinct elements of C are identified and thus i_* is monic, as desired.

1.5.2 Definition

A map $p: X \rightarrow Y$ is said to have the **right lifting property (RLP)** with respect to a class of morphisms M if in any solid arrow diagram

$$\begin{array}{ccc} A & \xrightarrow{\quad} & X \\ i \downarrow & \nearrow \text{dashed} & \downarrow p \\ B & \xrightarrow{\quad} & Y \end{array}$$

where $i \in M$, a dashed arrow making the diagram commute exists.

Dually we define the **left lifting property**.

1.5.3 Lemma

The class of monomorphisms M_p having the left lifting property (LLP) with respect to a fixed map $p: X \rightarrow Y$ is saturated.

Proof

We prove condition (2). Suppose we have a commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{\quad} & C & \xrightarrow{\quad} & X \\ i \downarrow & & \downarrow j & & \downarrow p \\ B & \xrightarrow{\quad} & B \cup_A C & \xrightarrow{\quad} & Y \end{array}$$

where the left square is a pushout and $i \in M_p$; we wish to show $j \in M_p$. Because $i \in M_p$, there is a $\theta: B \rightarrow X$ so that

$$\begin{array}{ccc} A & \xrightarrow{\quad} & X \\ i \downarrow & \nearrow \theta & \downarrow p \\ B & \xrightarrow{\quad} & Y \end{array}$$

commutes. But then this defines maps $B \rightarrow X, C \rightarrow X$ which commute and thus a map $\theta_*: B \cup_A C \rightarrow X$ which commutes with the necessary morphisms. Thus $j \in M_p$ as desired. ■

Clearly the intersection of saturated classes is saturated. Thus for any class of monomorphisms B , we have a saturated class M_B of all saturated classes containing B . This is called the saturated class **generated** by B . Alternatively M_B is called the **saturation** of B .

Now consider the following classes of monomorphisms.

B₁ The set of all inclusions $\Lambda_k^n \subseteq \Delta^n$ for $0 \leq k \leq n, n > 0$.

B₂ The set of all inclusions

$$(\Delta^1 \times \partial\Delta^n) \cup (\{e\} \times \Delta^n) \subseteq (\Delta^1 \times \Delta^n), \quad e = 0, 1$$

where $\{e\}$ corresponds to the simplicial set consisting of the vertex e and all of its degeneracies.

B₃ The set of all inclusions

$$(\Delta^1 \times Y) \cup (\{e\} \times X) \subseteq (\Delta^1 \times X), \quad e = 0, 1$$

where $Y \subseteq X$ is an inclusion.

1.5.4 Proposition

The classes $\mathbf{B}_1$, $\mathbf{B}_2$, and $\mathbf{B}_3$ have the same saturation.

Proof

Long and tedious. ■

1.5.5 Definition

The saturation of $\mathbf{B}_1$ (equivalently $\mathbf{B}_2, \mathbf{B}_3$) is called the class of **anodyne extensions**.

1.5.6 Corollary

A fibration is a map $p: X \rightarrow Y$ which has the right lifting property with respect to all anodyne extensions.

Proof

Because the class of monics for which p has the right lifting property is saturated, and contains $\Lambda_k^n \subseteq \Delta^n$ by definition, we obtain the desired result. ■

1.5.7 Corollary

If $i: K \hookrightarrow L$ is an anodyne extension, and $Y \hookrightarrow X$ is an arbitrary inclusion, then

$$(K \times X) \cup (L \times Y) \rightarrow (L \times X)$$

is an anodyne extension.

Proof

Consider the class of monics $K' \rightarrow L'$ such that

$$(K' \times X) \cup (L' \times Y) \rightarrow (L' \times X)$$

is anodyne. This is a saturated set.

Now consider the inclusion

$$(\Delta^1 \times Y') \cup (\{e\} \times X') \subseteq (\Delta^1 \times X'), \quad Y' \subseteq X'$$

We want to show that this is in the class, as it would imply all anodyne morphisms are. It is in the class iff the following inclusion is anodyne:

$$((\Delta^1 \times Y') \cup (\{e\} \times X')) \times X \cup (\Delta^1 \times X' \times Y) \rightarrow (\Delta^1 \times X' \times X)$$

which is just isomorphic to

$$(\Delta^1 \times ((Y' \times X) \cup (X' \times Y))) \cup (\{e\} \times X' \times X) \rightarrow \Delta^1 \times X' \times X$$

which is anodyne as desired.

Thus all anodyne extensions are in the class, in particular i . ■

Note:

In the above corollary, the simplicial set $(K \times X) \cup (L \times Y)$ is the union when viewing both terms as subsets of $L \times X$. Categorically, this is the simplicial set $(K \times X) \cup_{K \times Y} (L \times Y)$.

1.6 Function complexes

In this section we will discuss some constructions in $\mathbf{sSet}$.

We first define quotient simplicial objects. First we define the quotient of a set by a subset: if $A \subseteq B$ is a subset, then $B/A = (B - A) \sqcup \{*\}$; we identify A with a single point and all other elements of B with themselves. We will also denote the additional point $*$ by $[A]$.

Now if $Y \subseteq X$ is a simplicial subset, then we can form the **quotient** simplicial set X/Y as follows. Then for $n < \omega$ define

$$(X/Y)_n = X_n/Y_n,$$

and let $q_n: X_n \rightarrow X_n/Y_n$ be the quotient map. Then for an order-preserving $\varphi: [n] \rightarrow [m]$ (which we identify with $X_m \rightarrow X_n$), there is a unique $\bar{\varphi}: (X/Y)_m \rightarrow (X/Y)_n$ such that

$$\bar{\varphi} \circ q_m = q_n \circ \varphi$$

which maps $x \in X_m - Y_m$ to $q_n \circ \varphi(x)$ and $y \in Y_m$ to $[Y_n]$. Uniqueness implies that $\varphi \circ \bar{\psi} = \bar{\psi} \circ \bar{\varphi}$, so that this is a functor. Thus X/Y is a simplicial set itself, and $q: X \rightarrow X/Y$ is a morphism.

We can generalize this: let $R \subseteq X \times X$ be a simplicial subset such that $R_n \subseteq X_n \times X_n$ is an equivalence relation for all $n < \omega$. Then we can define the quotient X/R where the set of n -simplices is $(X/R)_n = X_n/R_n$ (quotient by an equivalence relation). As before if we let $q_n: X_n \rightarrow X_n/R_n$ be the quotient map, then for $\varphi: [n] \rightarrow [m]$ there is a unique $\bar{\varphi}: (X/R)_m \rightarrow (X/R)_n$ such that

$$\bar{\varphi} \circ q_m = q_n \circ \varphi.$$

This is well-defined precisely because $R \subseteq X \times X$ so if $q_m(x) = q_m(y)$, meaning $(x, y) \in R_m$, then $(\varphi(x), \varphi(y)) \in R_n$ and thus $\bar{\varphi} \circ q_m(x) = q_n \circ \varphi(x)$ is well-defined. Thus X/R is a simplicial set and $q: X \rightarrow X/R$ is a morphism.

The quotient X/Y is a special case of quotienting by an equivalence. For each $n < \omega$, Y_n defines an equivalence on $X_n \times X_n$ where $x_0 \sim x_1$ iff they are equal or both in Y_n . It is readily checked that this equivalence is simplicial, and then X/Y and X/R are equal.

A **pointed simplicial set** is a simplicial set X with a distinguished vertex $* \in X_0$. We denote by $*$ also all of $* \in X_0$'s degeneracies. A morphism of pointed simplicial sets is a morphism of the underlying simplicial sets which preserves the base point. Note that it is sufficient to check that $* \in X_0$ is mapped to $* \in Y_0$, as then naturality implies $S*$ is mapped to $S*$. We denote the category of pointed simplicial sets by $\mathbf{sSet}_*$.

For two pointed simplicial sets X, Y , we define their **wedge sum** $X \vee Y$ to be the simplicial subset of $X \times Y$ consisting of all $(x, *)$ and $(*, y)$. This is clearly a simplicial subset, and we define the **smash product** of X and Y to be the quotient $X \wedge Y = X \times Y / X \vee Y$. It is easy to see that the wedge sum $X \vee Y$ is the pushout of the diagram $X \leftarrow * \rightarrow Y$, so the quotient of $X \amalg Y$ by identifying $* \in X$ with $* \in Y$.

If $f: X \rightarrow Y$ is a simplicial map and $K \subseteq Y$ is a simplicial subset, then $f^{-1}K \subseteq X$ defined to be all the simplices of X which map to K via f , is a simplicial subset. Indeed, if $\varphi: [m] \rightarrow [n]$ is a morphism, and $x \in X_n$ maps to $f(x) \in K_n$ then $f(\varphi_*x) = \varphi_*f(x) \in K$ and so $\varphi_*x \in f^{-1}K$ as desired.

We can view this fiber as the following pullback

$$\begin{array}{ccc} f^{-1}K & \xrightarrow{\quad} & K \\ \downarrow & & \downarrow \\ X & \xrightarrow{\quad f \quad} & Y \end{array}$$

In particular for a vertex $* \in Y$, it generates the simplicial subset consisting of all its degeneracies, and the fiber of the vertex is $f^{-1}*$ (the preimage of this simplicial subset).

Similarly we easily see that the image of a simplicial map is a simplicial subset of the codomain.

Now we focus on function complexes. First, consider the following functor $\Delta^\bullet: \Delta \rightarrow \mathbf{sSet}$ given by $[n] \mapsto \Delta^n$. This is just the Yoneda embedding; it maps $\varphi: [n] \rightarrow [m]$ to postcomposition with φ , which we identify with

$\varphi: \Delta^n \rightarrow \Delta^m$. Thus we can view $\Delta^\bullet$ as a simplicial object in $\mathbf{sSet}$ (a simplicial simplicial set). Since taking products is also functorial, for any simplicial set $X \in \mathbf{sSet}$, this defines a functor

$$X \times \Delta^\bullet, \quad [n] \mapsto X \times \Delta^n, \quad \varphi \mapsto 1 \times \varphi_*.$$

For $Y \in \mathbf{sSet}$, we also have a functor $\mathrm{hom}_{\mathbf{sSet}}(\bullet, Y): \mathbf{sSet}^{\mathrm{op}} \rightarrow \mathbf{Set}$. Composing these functors we get a simplicial set $\mathrm{hom}_{\mathbf{sSet}}(X \times \Delta^\bullet, Y): \Delta^{\mathrm{op}} \rightarrow \mathbf{Set}$.

We spell this construction out explicitly.

1.6.1 Definition (function complex)

Let X and Y be simplicial sets. Their **function complex** $\mathbf{Hom}(X, Y)$ to be the simplicial set whose simplices are

$$\mathbf{Hom}(X, Y)_n = \mathrm{hom}_{\mathbf{sSet}}(X \times \Delta^n, Y)$$

and for $\varphi: [m] \rightarrow [n]$, it induces the map

$$\varphi^*: \mathbf{Hom}(X, Y)_n \rightarrow \mathbf{Hom}(X, Y)_m$$

defined by mapping $f: X \times \Delta^n \rightarrow Y$ to the composite

$$X \times \Delta^m \xrightarrow{1 \times \varphi} X \times \Delta^n \xrightarrow{f} Y.$$

The construction of the function complex is functorial in both X and Y . That is the function complex is a functor

$$\mathbf{Hom}: \mathbf{sSet}^{\mathrm{op}} \times \mathbf{sSet} \rightarrow \mathbf{sSet}.$$

A morphism $f: X \rightarrow X'$ defines $f^*: \mathbf{Hom}(X', Y) \rightarrow \mathbf{Hom}(X, Y)$ given by precomposition with $f \times \mathrm{id}$. That is, it maps $g: X' \times \Delta^n \rightarrow Y$ to

$$f^*(g) = X \times \Delta^n \xrightarrow{f \times \mathrm{id}} X' \times \Delta^n \xrightarrow{g} Y = g \circ (f \times \mathrm{id}).$$

Similarly $f: Y \rightarrow Y'$ defines $f_*: \mathbf{Hom}(X, Y) \rightarrow \mathbf{Hom}(X, Y')$ by postcomposition with f . So $g: X \times \Delta^n \rightarrow Y$ is mapped to $f \circ g$.

There is a natural **evaluation map**

$$\mathrm{ev}: X \times \mathbf{Hom}(X, Y) \rightarrow Y, \quad (x, f) \mapsto f(x, \mathrm{id}_n)$$

where $x \in X_n$ and $f: X \times \Delta^n \rightarrow Y$. For a simplicial structure map $\varphi: [m] \rightarrow [n]$ we have that

$$\mathrm{ev}(\varphi(x, f)) = \mathrm{ev}(\varphi x, \varphi f) = \varphi f(\varphi x, \mathrm{id}_n) = f(\varphi x, \varphi(\mathrm{id})) = \varphi f(x, \mathrm{id}) = \varphi \mathrm{ev}(x, f).$$

Thus ev is a simplicial map.

1.6.2 Proposition (exponential law)

The function

$$\mathrm{ev}_*: \mathrm{hom}_{\mathbf{sSet}}(K, \mathbf{Hom}(X, Y)) \rightarrow \mathrm{hom}_{\mathbf{sSet}}(X \times K, Y)$$

defined by mapping $g: K \rightarrow \mathbf{Hom}(X, Y)$ to the composite

$$X \times K \xrightarrow{1 \times g} X \times \mathbf{Hom}(X, Y) \xrightarrow{\mathrm{ev}} Y$$

is a bijection natural in K, X, Y .

Proof

The inverse of ev_* is the map

$$\text{hom}_{\mathbf{sSet}}(X \times K, Y) \rightarrow \text{hom}_{\mathbf{sSet}}(K, \mathbf{Hom}(X, Y))$$

obtained by mapping $g: X \times K \rightarrow Y$ to $g_*: K \rightarrow \mathbf{Hom}(X, Y)$ where for $x \in K_n$,

$$g_*(x) = X \times \Delta^n \xrightarrow{1 \times \iota_x} X \times K \xrightarrow{g} Y.$$

Indeed, $\text{ev}_*(g_*)$ maps $(x, k) \in X_n \times K_n$ to

$$(x, k) \mapsto \text{ev}(x, g_*(k)) = g_*(k)(x, \text{id}_n) = g(x, \iota_k \text{id}_n) = g(x, k)$$

so that $\text{ev}_*(g_*) = g$. And for $g: K \rightarrow \mathbf{Hom}(X, Y)$, $\text{ev}_*(g)_*$ maps $k \in K_n$ to $\text{ev}_*(g)_*(k) \in \text{hom}_{\mathbf{sSet}}(X \times \Delta^n, Y)$ defined by

$$(x, \varphi) \mapsto \text{ev}_*(g)(x, \iota_k \varphi) = \text{ev}(x, g(\iota_k \varphi)) = g(\iota_k \varphi)(x, \text{id}_n)$$

where $\varphi \in \Delta^n$. But $\iota_k(\varphi) = \varphi(k)$ and so this is equal to

$$= g(\varphi(k))(x, \text{id}_n) = (\varphi g(k))(x, \text{id}_n)$$

and recall that in the definition of $\mathbf{Hom}(X, Y)$, φ acts on it by

$$= g(k)(x, \varphi(\text{id}_n)) = g(k)(x, \varphi)$$

so that $\text{ev}_*(g)_* = g$ as desired. ■

Thus $\mathbf{sSet}$ is **cartesian closed**: its product has a left adjoint. The function complex $\mathbf{Hom}$ is also called the **interior hom**.

Note:

In general for a category $\mathcal{C}$, its presheaf category $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is cartesian closed. The internal hom of presheaves is given much like here. For presheaves $F, G: \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ we define the presheaf

$$\mathbf{Hom}(F, G): \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}, \quad \mathbf{Hom}(F, G)(c) = \text{Nat}(F \times y(c), G)$$

where $y: \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is the Yoneda embedding.

1.6.3 Proposition

If $i: K \hookrightarrow L$ is an inclusion of simplicial sets and $p: X \rightarrow Y$ is a fibration, then the map

$$\mathbf{Hom}(L, X) \xrightarrow{(i^*, p_*)} \mathbf{Hom}(K, X) \times_{\mathbf{Hom}(K, Y)} \mathbf{Hom}(L, Y)$$

which is induced by the diagram

$$\begin{array}{ccc} \mathbf{Hom}(L, X) & \xrightarrow{p_*} & \mathbf{Hom}(L, Y) \\ \downarrow i^* & & \downarrow i^* \\ \mathbf{Hom}(K, X) & \xrightarrow{p_*} & \mathbf{Hom}(K, Y) \end{array}$$

is a fibration.

Proof

Diagrams of the form

$$\begin{array}{ccccc}
 \Lambda_k^n & \xrightarrow{\quad} & \mathbf{Hom}(L, X) & & \\
 \downarrow & & \downarrow (i^*, p_*) & & \\
 \Delta^n & \xrightarrow{\quad} & \mathbf{Hom}(K, X) \times_{\mathbf{Hom}(K, Y)} \mathbf{Hom}(L, Y) & \xrightarrow{\quad} & \mathbf{Hom}(K, X) \\
 & & \downarrow & & \downarrow \\
 & & \mathbf{Hom}(L, Y) & \xrightarrow{\quad} & \mathbf{Hom}(K, Y)
 \end{array}$$

correspond to, according to the exponential law, diagrams of the form

$$\begin{array}{ccccc}
 \Lambda_k^n \times K & \xrightarrow{\quad} & \Lambda_k^n \times L & & \\
 \downarrow & & \downarrow & & \\
 \Delta^n \times K & \xrightarrow{\quad} & (\Lambda_k^n \times L) \cup_{\Lambda_k^n \times K} (\Delta^n \times K) & \xrightarrow{\quad} & X \\
 & & \downarrow j & & \downarrow p \\
 & & \Delta^n \times L & \xrightarrow{\quad} & Y
 \end{array}$$

By corollary 1.5.7 j is anodyne. Since p is fibrant, it has the right lifting property relative to j and thus the desired lift exists in the lower diagram; this corresponds to a lift in the original diagram. ■

1.6.4 Corollary

- (1) If $p: X \rightarrow Y$ is a fibration, so is $p_*: \mathbf{Hom}(K, X) \rightarrow \mathbf{Hom}(K, Y)$.
- (2) If $i: K \hookrightarrow L$ is an inclusion and X is fibrant, then $i^*: \mathbf{Hom}(L, X) \rightarrow \mathbf{Hom}(K, X)$ is a fibration.

Proof

- (1) Consider the inclusion $i: * \hookrightarrow K$ and the isomorphism

$$\mathbf{Hom}(K, Y) \cong * \times_* \mathbf{Hom}(K, Y) \cong \mathbf{Hom}(*, X) \times_{\mathbf{Hom}(*, Y)} \mathbf{Hom}(K, Y).$$

Under this isomorphism, p_* corresponds to (i^*, p_*) which is fibrant by the above proposition.

- (2) Consider the map $p: X \rightarrow *$ which is fibrant by definition. Then as in the previous point,

$$\mathbf{Hom}(K, X) \cong \mathbf{Hom}(K, X) \times_{\mathbf{Hom}(K, *)} \mathbf{Hom}(L, *)$$

under which i corresponds to (i^*, p_*) which is fibrant by the above proposition. ■

1.6.5 Lemma

Let X be fibrant, then $(\delta^k)^*: \mathbf{Hom}(\Delta^n, X) \rightarrow \mathbf{Hom}(\Delta^{n-1}, X)$ has the right-lifting property relative to any inclusion $\partial\Delta^m \subseteq \Delta^m$.

Proof

A commutative square

$$\begin{array}{ccc} \partial\Delta^m & \longrightarrow & \mathbf{Hom}(\Delta^n, X) \\ \downarrow & & \downarrow (\delta^k)^* \\ \Delta^m & \longrightarrow & \mathbf{Hom}(\Delta^{n-1}, X) \end{array}$$

corresponds to, under exponentiation, maps $\partial\Delta^m \times \Delta^n \rightarrow X$ and $\Delta^m \times \Delta^{n-1} \rightarrow X$ which agree on $\partial\Delta^m \times \Delta^{n-1}$. Thus they define a map $(\partial\Delta^m \times \Delta^n) \cup (\Delta^m \times \Delta^{n-1}) \rightarrow X$. Because X is fibrant and $(\partial\Delta^m \times \Delta^n) \cup (\Delta^m \times \Delta^{n-1}) \subseteq \Delta^m \times \Delta^n$ is anodyne, this lifts to $\Delta^m \times \Delta^n \rightarrow X$. Via exponentiation, this gives the desired extension $\Delta^m \rightarrow \mathbf{Hom}(\Delta^n, X)$. ■

1.7 Simplicial homotopy

1.7.1 Definition

Let $f, g: K \rightarrow X$ be simplicial maps. A **simplicial homotopy** from f to g is a map $h: K \times \Delta^1 \rightarrow X$ such that the following diagram commutes

$$\begin{array}{ccc} K \times \Delta^0 = K & & \\ \downarrow 1 \times \delta^1 & \searrow f & \\ K \times \Delta^1 & \xrightarrow{h} & X \\ \uparrow 1 \times \delta^0 & \nearrow g & \\ K \times \Delta^0 = K & & \end{array}$$

If there exists a simplicial homotopy from f to g we write $f \simeq g$.

Recall that a k -simplex in Δ^n is an increasing $f: [k] \rightarrow [n]$ which we can represent as $[f(0), \dots, f(k)]$. In particular its vertices can naturally be written as $0, \dots, n$ according to the image of $f: [0] \rightarrow [n]$.

Note that $\delta^1: \Delta^0 \rightarrow \Delta^1$ corresponds to the vertex of Δ^1 given by $\delta^1: [0] \rightarrow [1]$, which by definition skips 1; its image is 0. Similarly δ^0 corresponds to the vertex 1.

So a homotopy can equivalently be given as a simplicial map $h: K \times \Delta^1 \rightarrow X$ such that

$$h(x, 0) = f(x), \quad h(x, 1) = g(x)$$

for all $x \in K$ (where 0, 1 denote the degeneracies of the vertices 0, 1).

1.7.2 Definition

Suppose $\iota: L \hookrightarrow K$ is an inclusion of simplicial sets, and let $f, g: K \rightarrow X$ be simplicial maps. Further suppose that $f|_L = g|_L$. A homotopy from f to g **relative to** L is a homotopy $h: f \simeq g$ such that the following diagram commutes.

$$\begin{array}{ccc} K \times \Delta^1 & \xrightarrow{h} & X \\ \iota \times \text{id} \uparrow & & \uparrow f|_L = g|_L \\ L \times \Delta^1 & \xrightarrow{\text{proj}_L} & L \end{array}$$

This means that $h(x, t) = f(x) = g(x)$ for all $x \in L, t \in \Delta^1$. A homotopy of the form

$$L \times \Delta^1 \xrightarrow{\text{proj}_L} L \xrightarrow{\alpha} X$$

is called a constant homotopy: it is a homotopy from α to itself.

Note:

Homotopy is not in general an equivalence. Consider the maps $\iota_0, \iota_1: \Delta^0 \rightarrow \Delta^n$ classifying the vertices 0, 1 respectively. A homotopy between them is a map $h: \Delta^1 \rightarrow \Delta^n$ such that $h(0) = 0$ and $h(1) = 1$. This corresponds to the 1-simplex $[0, 1] \in \Delta^n$.

But there is not a homotopy from ι_1 to ι_0 as this would correspond to a simplex $[1, 0]$ which does not exist.

For an n -simplex $\sigma \in X$, denote its **boundary** by

$$\partial\sigma = (d_0\sigma, \dots, d_n\sigma).$$

1.7.3 Lemma

Suppose that X is a fibrant simplicial set. Then the simplicial homotopy of vertices $x: \Delta^0 \rightarrow X$ is an equivalence.

Proof

There is a homotopy $x \leadsto y$ iff there is a 1-simplex $v \in X_1$ such that $d_1v = x$ and $d_0v = y$. That is, iff there is a 1-simplex v with boundary $\partial v = (y, x)$. Note that $\partial(s_0x) = (x, x)$ gives reflexivity.

Now suppose $\partial v_2 = (y, x)$ and $\partial v_0 = (z, y)$ so that $d_0v_2 = d_1v_0 = y$. Thus v_2, v_0 determine a map $(v_0, \cdot, v_2): \Lambda_1^2 \rightarrow X$. Because X is fibrant, we can lift it.

$$\begin{array}{ccc} \Lambda_1^2 & \xrightarrow{(v_0, \cdot, v_2)} & X \\ \downarrow & \nearrow \theta & \\ \Delta^2 & & \end{array}$$

Then

$$\partial(d_1\theta) = (d_0d_1\theta, d_1d_1\theta) = (d_0d_0\theta, d_1d_2\theta),$$

and

$$d_0\theta = \theta \circ \delta^0 = (v_0, \cdot, v_2) \circ \delta^0 = v_0$$

and similar for $d_2\theta$. Thus

$$\partial(d_1\theta) = (d_0v_0, d_1v_2) = (z, x)$$

so that $d_1\theta$ is a homotopy $x \leadsto z$; so homotopy is transitive.

For symmetry, suppose $\partial v_2 = (y, x)$ and let $v_1 = s_0x$. Then $d_1v_1 = d_1v_2 = x$, so these define a map $(\cdot, v_1, v_2): \Lambda_0^2 \rightarrow X$. We extend this to $\theta': \Delta^1 \rightarrow X$ so that

$$\partial(d_0\theta') = (d_0d_0\theta', d_1d_0\theta') = (d_0d_1\theta', d_0d_2\theta') = (x, y).$$

Thus the relation is symmetric. ■

1.7.4 Lemma

The pullback of a fibration is a fibration. Explicitly, suppose we have the following pullback square

$$\begin{array}{ccc}
 F & \xrightarrow{\quad} & X \\
 q \downarrow & & \downarrow p \\
 W & \xrightarrow{\quad} & Y
 \end{array}$$

If p is a fibration, then so is q .

This implies that all the fibers of p are fibrant simplicial sets.

Proof

Consider the diagram

$$\begin{array}{ccccc}
 \Lambda_k^n & \xrightarrow{\quad} & F & \xrightarrow{\quad} & X \\
 \downarrow & & \downarrow & \nearrow \theta & \downarrow p \\
 \Delta^n & \xrightarrow{\quad} & W & \xrightarrow{\quad} & Y
 \end{array}$$

Because p is fibrant, there is a lifting $\theta: \Delta^n \rightarrow X$. So we have maps $\Delta^n \rightarrow X, W$ which commute with the pullback diagram, and thus define a unique map $\tilde{\theta}: \Delta^n \rightarrow F$. This $\tilde{\theta}$ commutes with $F \xrightarrow{q} W$ and $\Delta^n \rightarrow W$ by definition.

It then follows that it commutes with $\Lambda_k^n \hookrightarrow \Delta^n$ and $\Lambda_k^n \rightarrow F$ by the universal property of the pullback. That is, it is easy to see that the composite $\Lambda_k^n \hookrightarrow \Delta^n \xrightarrow{\tilde{\theta}} F$ composes with $F \rightarrow X$ and $F \rightarrow W$ the same way that $\Lambda_k^n \rightarrow F$ does. Because F is the pullback, this means that the composite is equal to $\Lambda_k^n \rightarrow F$ as desired.

The last part of the claim follows because the fibers of p are the pullbacks along $* \rightarrow Y$, and then q is the terminal map $F \rightarrow *$. It being a fibration means that F is fibrant. ■

1.7.5 Corollary

If X is fibrant and $L \subseteq K$ is an inclusion, then

- (1) homotopy of maps $K \rightarrow X$ is an equivalence, and
- (2) homotopy of maps $K \rightarrow X$ (rel L) is an equivalence.

Proof

Clearly (2) implies (1) with $L = \emptyset$. Now we know that $i^*: \mathbf{Hom}(L, X) \rightarrow \mathbf{Hom}(K, X)$ is a fibration, and the fiber of a vertex $\alpha \in \mathbf{Hom}(K, X)_0$ consists of all $f \in \mathbf{Hom}(L, X)$ such that $f|_L = \alpha$ (recall $\mathbf{Hom}(K, X)_0 = \text{hom}_{\text{Set}}(k, X)$). These fibers are all fibrant by the above lemma.

Now we claim that $f \xrightarrow{\sim} g \text{ (rel } L)$ iff f, g are homotopic as vertices of the fiber of i^* . Let $\alpha = f|_L = g|_L$. A homotopy $h: f \xrightarrow{\sim} g \text{ (rel } L)$ corresponds to a map $h: \Delta^1 \times K \rightarrow X$, which by the exponential law corresponds to $\tilde{h}: \Delta^1 \rightarrow \mathbf{Hom}(K, X)$.

Note that a homotopy diagram of the form

$$\begin{array}{ccc}
 K \times \Delta^0 = K & & \\
 \downarrow 1 \times \delta^1 & \searrow f & \\
 K \times \Delta^1 & \xrightarrow{h} & X \\
 \uparrow 1 \times \delta^0 & \nearrow g & \\
 K \times \Delta^0 = K & &
 \end{array}$$

Corresponds to a homotopy of vertices

$$\begin{array}{ccc}
 \Delta^0 & & \\
 \downarrow \delta^1 & \searrow \iota_f & \\
 \Delta^1 & \xrightarrow{\tilde{h}} & \mathbf{Hom}(K, X) \\
 \uparrow \delta^0 & \nearrow \iota_g & \\
 \Delta^0 & &
 \end{array}$$

and vice versa.

Homotopy relative to L is the same as homotopy of vertices in the fibers of i^* , which are fibrant. By the above lemma this is an equivalence. ■

1.8 Simplicial homotopy groups

1.8.1 Definition

Let X be a fibrant simplicial set and $v \in X_0$ a vertex. For $n \geq 1$, define $\pi_n(X, v)$ to be the set of homotopy classes of maps $\alpha: \Delta^n \rightarrow X \text{ (rel } \partial\Delta^n)$ such that α fits into a diagram

$$\begin{array}{ccc}
 \Delta^n & \xrightarrow{\alpha} & X \\
 \uparrow & & \uparrow v \\
 \partial\Delta^n & \xrightarrow{\quad} & *
 \end{array}$$

Equivalently, α restricts to the constant map v on $\partial\Delta^n$.

For $n = 0$ define $\pi_0(X)$ to be the set of homotopy classes of vertices of X . This is called the **path components** of X . X is said to be **connected** if $\pi_0(X)$ is trivial (a singleton).

We write $[\alpha]$ for the homotopy class of α .

Note that α restricting to v on $\partial\Delta^n$ just means that $d_i\alpha = v$ for all i . Indeed, $d_i\alpha$ is just $\alpha \circ \delta^i$ and since δ^i is in $\partial\Delta^n$, this is mapped to v .

Let $\alpha, \beta: \Delta^n \rightarrow X$ represent elements of $\pi_n(X, v)$. Then define the simplices

$$v_i = v \text{ for } 0 \leq i \leq n-2, \quad v_{n-1} = \alpha, \quad v_{n+1} = \beta.$$

Then $d_i v_j = d_{j-1} v_i$ for $i < j$ and $i, j \neq n$, as all the faces are just v . Thus the v_i define a simplicial map $(v_0, \dots, v_{n-1}, v_n): \Lambda_n^{n+1} \rightarrow X$ and as X is fibrant, there is an extension

$$\begin{array}{ccc} \Lambda_n^{n+1} & \xrightarrow{(v_0, \dots, v_{n-1}, v_n)} & X \\ \downarrow & \nearrow \omega & \\ \Delta^{n+1} & & \end{array}$$

Notice then that

$$\partial(d_n \omega) = (d_0 d_n \omega, \dots, d_{n-1} d_n \omega, d_n d_n \omega) = (d_{n-1} d_0 \omega, \dots, d_{n-1} d_{n-1} \omega, d_n d_{n+1} \omega) = (v, \dots, v).$$

Thus $d_n \omega$ represents an element of $\pi_n(X, v)$.

1.8.2 Lemma

The homotopy class of $d_n \omega$ (rel $\partial \Delta^n$) is independent on choice of representatives of $[\alpha]$ and $[\beta]$ and choice of extension ω .

Proof

Suppose that $h_{n-1}: \alpha \xrightarrow{\sim} \alpha'$ (rel $\partial \Delta^n$) and $h_{n+1}: \beta \xrightarrow{\sim} \beta'$ (rel $\partial \Delta^n$) are homotopies and

$$\partial \omega = (v, \dots, v, \alpha, d_n \omega, \beta), \quad \partial \omega' = (v, \dots, v, \alpha', d_n \omega', \beta')$$

are extensions.

There is a map

$$(\Delta^{n+1} \times \partial \Delta^n) \cup (\Lambda_n^{n+1} \times \Delta^1) \xrightarrow{(\omega, \omega', (v, \dots, v, h_{n-1}, h_{n+1}))} X$$

given by this data. We can then extend this to $\theta: \Delta^{n+1} \times \Delta^1 \rightarrow X$. The composite

$$\Delta^n \times \Delta^1 \xrightarrow{\delta^n \times 1} \Delta^{n+1} \times \Delta^1 \xrightarrow{\theta} X$$

is a homotopy $d_n \omega \xrightarrow{\sim} d_n \omega'$ (rel $\partial \Delta^n$). ■

TODO: What?

From this lemma, we can define multiplication on $\pi_n(X, v)$ by $[\alpha] \cdot [\beta] = [d_n \omega]$, where $\partial \omega = (v, \dots, v, \alpha, d_n \omega, \beta)$. Let $e \in \pi_n(X, v)$ denote the homotopy class of v , i.e. the constant map $\Delta^n \rightarrow * \xrightarrow{v} X$.

1.8.3 Theorem

For a simplicial set X and a vertex $v \in X_0$, $\pi_n(X, v)$ forms a group for all $n \geq 1$.

Proof

First we show that $\alpha \cdot e = \alpha$. To do this, it is sufficient to find an $\omega: \Delta^{n+1} \rightarrow X$ such that $d_{n-1} \omega = d_n \omega = \alpha$ and $d_i \omega = v$ for all other i . Then $\partial \omega = (v, \dots, v, \alpha, \alpha, v)$ showing $\alpha \cdot e = \alpha$. We can take $\omega = s_{n-1} \alpha$ and this is sufficient.

Now we want to show that α has an inverse, so there is a β where $\alpha \cdot \beta = e$. We want an ω with $d_n \omega = v$,

$d_{n-1}\omega = \alpha$, and $d_i\omega = v$ for all other i . That is, we want to complete $(v, \dots, v, \alpha, v)$. This is possible as X is fibrant.

Now we show associativity. Let α, β, γ represent elements of $\pi_n(X, v)$. Then we have $(n+1)$ -simplices $\omega_{n-1}, \omega_{n+1}, \omega_{n+2}$ representing $\alpha\beta$, $(\alpha\beta)\gamma$, and $\beta\gamma$ respectively. That is,

$$\begin{aligned}\partial\omega_{n-1} &= (v, \dots, v, \alpha, d_n\omega_{n-1}, \beta), \\ \partial\omega_{n+1} &= (v, \dots, v, d_n\omega_{n-1}, d_n\omega_{n+1}, \gamma), \\ \partial\omega_{n+2} &= (v, \dots, v, \beta, d_n\omega_{n+2}, \gamma).\end{aligned}$$

These form a map

$$\Lambda_n^{n+2} \xrightarrow{(v, \dots, v, \omega_{n-1}, \omega_{n+1}, \omega_{n+2})} X$$

which extends to a map $\theta: \Delta^{n+2} \rightarrow X$. But then

$$\partial(d_n\theta) = (v, \dots, v, \alpha, d_n\omega_{n+1}, d_n\omega_{n+2})$$

so that $\alpha \cdot d_n\omega_{n+2} = d_n\omega_{n+1}$.

Unrolling definitions, we have

$$(\alpha\beta)\gamma = d_n\omega_{n-1} \cdot \gamma = d_n\omega_{n+1} = \alpha \cdot d_n\omega_{n+2} = \alpha(\beta\gamma)$$

as desired. ■

We will show that $\pi_n(X, v)$ is an Abelian group for $n \geq 2$. To do this, we will construct the *loop space* ΩX where $\pi_n(X, v) \cong \pi_{n-1}(\Omega X, v)$ and show that $\pi_k(\Omega X, v)$ is Abelian for all $k \geq 1$.

Note:

We digress a bit here for the sake of some background. The constructions thus far all have a topological analogue, which you should be familiar with. The loop space does as well, which we describe here.

First of all, let us construct the **function space** of topological spaces. Let X and Y be topological spaces, then their function space is the space $C(X, Y)$ of continuous functions $X \rightarrow Y$. We define a subbase as follows: for $K \subseteq X$ compact and $U \subseteq Y$ open, define

$$V(K, U) = \{f \in C(X, Y) \mid f(K) \subseteq U\}.$$

These define our subbase as K, U range. This is called the **compact-open topology**.

An important result is that, if X is locally compact Hausdorff, then $C(X, Y)$ forms an internal hom. That is, there is an adjunction $\bullet \times X \dashv C(X, \bullet)$. The adjunction is given by currying: map $\alpha: Y \times X \rightarrow Z$ to $\tilde{\alpha}: Y \rightarrow C(X, Z)$ given by $\tilde{\alpha}(y)(x) = \alpha(y, x)$. And similarly map $\beta: Y \rightarrow C(X, Z)$ to $\hat{\beta}: Y \times X \rightarrow Z$ by $\hat{\beta}(y, x) = \beta(y)(x)$. It is standard to see that these are well-defined (i.e. $\tilde{\alpha}, \hat{\beta}$ are continuous); this is dependent on X being locally compact Hausdorff.

The adjunction defines natural isomorphisms

$$C(Y \times X, Z) \cong C(Y, C(X, Z)),$$

and it is again standard to show that these isomorphisms are continuous. Thus $C(Y \times X, Z)$ is homeomorphic to $C(Y, C(X, Z))$. We have then enriched **Top** over itself, and shown that the internal hom is an internal hom in this enrichment.

We are a bit more concerned with **Top**_{*}, the category of pointed topological spaces (i.e. topological spaces with a distinguished point). If X, Y are pointed topological spaces, then $C_*(X, Y)$ is the subspace of $C(X, Y)$ consisting of basepoint-preserving maps $X \rightarrow Y$. On the other hand, $X \times Y$ has no natural basepoint: any choice of $X \times *$ or $* \times Y$ would be valid. Thus we define their **smash product** to be $X \wedge Y = (X \times Y)/(X \times * \cup * \times Y)$. It is then standard to show that the above adjunction restricts to

$$C_*(Y \wedge X, Z) \cong C_*(Y, C_*(X, Z)).$$

We now define the **loop space** of X to be $\Omega X = C_*(S^1, X)$, for any choice of basepoint of S^1 . Similarly we define the **suspension** of X to be $\Sigma X = X \wedge S^1$. The adjunction then shows that

$$C_*(\Sigma X, Y) \cong C_*(X, \Omega Y).$$

Now we claim that

$$\pi_0 C(X, Y) \cong [X, Y]$$

where the right-hand side denotes the collection of maps $X \rightarrow Y$ quotiented by homotopy. Indeed, a path $I \rightarrow C(X, Y)$ corresponds under the adjunction to a homotopy $I \times X \rightarrow Y$ and vice versa. The endpoints of the path are the edges of the homotopy, giving the desired result. Similarly if we let $[X, Y]_*$ denote the class of basepoint-preserving homotopies, i.e. maps $h: I \times X \rightarrow Y$ such that $h(t, *) = *$, then

$$\pi_0 C_*(X, Y) \cong [X, Y]_*.$$

Now we know that by definition, $\pi_n X = [S^n, X]_*$. Thus

$$\pi_1 X \cong \pi_0 C_*(S^1, X) = \pi_0 \Omega(X),$$

as desired. It can be shown that $S^{n+m} \cong S^n \wedge S^m$, so that by induction $\pi_n \Omega(X) = \pi_{n+1} X$.

Further note that

$$[\Sigma X, Y]_* \cong \pi_0 C_*(\Sigma X, Y) \cong \pi_0 C_*(X, \Omega Y) \cong [X, \Omega Y]_*.$$

This is called **Eckmann-Hilton duality**.

First we note that the homotopy groups are functorial. That is, for each $n \geq 1$, π_n is functorial $\mathbf{sSet}_* \rightarrow \mathbf{Grp}$ where $\mathbf{sSet}$ is the category of simplicial sets with a distinguished vertex. Indeed suppose $f: (X, v) \rightarrow (Y, fv)$ is a simplicial map, then we define $f_*: \pi_n(X, v) \rightarrow \pi_n(Y, fv)$ by mapping $[\alpha] \in \pi_n(X, v)$ to $f \circ \alpha$.

This being well-defined is due to the following result.

1.8.4 Lemma

Suppose we have maps $f, g: K \rightarrow X$ and $f', g': X \rightarrow Y$, and a simplicial subset $L \subseteq K$ such that $f \simeq g$ (rel L) and $f' \simeq g'$ (rel $f(L)$). Then $f' \circ f \simeq g' \circ g$ (rel L).

Proof

Let $h: f \simeq g$ (rel L) and $h': f' \simeq g'$ (rel $f(L)$) be homotopies. Then we can compose the homotopies as in the following diagram, which is easy to verify as commuting.

$$\begin{array}{ccccc}
 K \times \Delta^0 & \xrightarrow{f} & X \times \Delta^0 & & \\
 \downarrow 1 \times \delta^1 & & \downarrow 1 \times \delta^1 & \searrow f' & \\
 K \times \Delta^1 & \xrightarrow{(h, 1)} & X \times \Delta^1 & \xrightarrow{h'} & Y \\
 \uparrow 1 \times \delta^0 & & \uparrow 1 \times \delta^0 & \nearrow g' & \\
 K \times \Delta^0 & \xrightarrow{g} & X \times \Delta^0 & &
 \end{array}$$

So $h' \circ (h, 1)$ defines a homotopy $f' \circ f \simeq g' \circ g$ as desired. It is clear that this is relative to L . ■

Now that we have shown $f_*: \pi_n(X, v) \rightarrow \pi_n(Y, fv)$ is well-defined, we show that it is a group morphism. Suppose $\alpha, \beta: \Delta^n \rightarrow X$ are representatives of elements from $\pi_n(X, v)$, so that $\alpha\beta = \gamma$ and ω witnesses this:

$\partial\omega = (v, \dots, v, \alpha, \gamma, \beta)$. Notice then that $f \circ \omega: \Delta^{n+1} \rightarrow Y$ has

$$d_k(f \circ \omega) = f(d_k\omega) \implies \partial(f \circ \omega) = (f(v), \dots, f(v), f \circ \alpha, f \circ \gamma, f \circ \beta)$$

so that $[f\alpha][f\beta] = [f\gamma]$ as desired.

Note that the above lemma also shows that $\pi_0: \mathbf{sSet} \rightarrow \mathbf{Set}$ is functorial. Indeed, if $f: X \rightarrow Y$ is a simplicial map, then we define $f_*: \pi_0 X \rightarrow \pi_0 Y$ by mapping the homotopy class of $[v]$ to $[fv]$. This corresponds to mapping $\iota_v: \Delta^0 \rightarrow X$ to $f \circ \iota_v$.

Thus we have a sequence of functors $\{\pi_n\}_{n \geq 0}$. We would expect them to be related somehow, and indeed they are. But before we continue, we make the following observation.

1.8.5 Lemma

A simplex $\alpha: \Delta^n \rightarrow X$ represents the identity of $\pi_n(X, v)$ iff there is an $(n+1)$ -simplex ω of X such that $\partial\omega = (v, \dots, v, \alpha)$.

Proof

Note that such an ω exists iff $e \cdot \alpha = e$. This is iff $\alpha = e$. ■

Now suppose $p: X \rightarrow Y$ is a Kan fibration, and let F be the fiber over a vertex $*$ in Y . By lemma 1.7.4 F is fibrant. Now let $v \in F$ be a vertex, and suppose $\alpha: \Delta^n \rightarrow Y$ represents an element of $\pi_n(Y, *)$. Then in the following diagram, we have a lift θ :

$$\begin{array}{ccc} \Lambda_0^n & \xrightarrow{(\cdot, v, \dots, v)} & X \\ \downarrow & \nearrow \theta & \downarrow p \\ \Delta^n & \xrightarrow{\alpha} & Y \end{array}$$

Now, $[d_0\theta] \in \pi_{n-1}(F, v)$ is independent on choice of θ and representative of α . Firstly, $p \circ (d_0\theta) = d_0(p\theta) = d_0\alpha = *$ (because $\alpha \in \pi_n(Y, *)$ meaning $\partial\alpha = (*, \dots, *)$) so that $d_0\theta \in F$.

TODO: Why

This then defines a function

$$\partial: \pi_n(Y, *) \rightarrow \pi_{n-1}(F, v)$$

called the **boundary map**.

1.8.6 Lemma

- (1) For $n > 1$, the boundary map is a group morphism $\pi_n(Y, *) \rightarrow \pi_{n-1}(F, v)$.
- (2) The sequence

$$\begin{aligned} \cdots \rightarrow \pi_n(F, v) &\xrightarrow{i_*} \pi_n(X, v) \xrightarrow{p_*} \pi_n(Y, *) \xrightarrow{\partial} \pi_{n-1}(F, v) \rightarrow \\ &\rightarrow \cdots \xrightarrow{p_*} \pi_1(Y, *) \xrightarrow{\partial} \pi_0(F) \xrightarrow{i_*} \pi_0(X) \xrightarrow{p_*} \pi_0(Y) \end{aligned}$$

is exact in the sense that the kernel equals the image everywhere (for $n = 0$ this means e.g. that $\text{im } i_* = (p_*)^{-1}*$). Moreover, there is an action of $\pi_1(Y, *)$ on $\pi_0(F)$ such that two elements of $\pi_0(F)$ have the same image under i_* iff they are in the same orbit for the $\pi_1(Y, *)$ -action.

Proof

First we show (1), that ∂ is a group morphism. Suppose for $i = n - 1, n, n + 1$ we have diagrams

$$\begin{array}{ccc} \Lambda_0^n & \xrightarrow{(\cdot, v, \dots, v)} & X \\ \downarrow & \nearrow \theta_i & \downarrow p \\ \Delta^n & \xrightarrow{\alpha_i} & Y \end{array}$$

where α_i represent elements of $\pi_n(Y, *)$, as well as an $(n + 1)$ -simplex ω witnessing $[\alpha_{n-1}][\alpha_{n+1}] = [\alpha_n]$:

$$\partial\omega = (*, \dots, *, \alpha_{n-1}, \alpha_n, \alpha_{n+1}).$$

This then defines a map

$$\Lambda_0^{n+1} \xrightarrow{(\cdot, v, \dots, v, \theta_{n-1}, \theta_n, \theta_{n+1})} X$$

because $d_j\theta_i = v$ for $j > 0$. Furthermore it fits into the following diagram, which then lifts to give γ :

$$\begin{array}{ccc} \Lambda_0^{n+1} & \xrightarrow{(\cdot, v, \dots, v, \theta_{n-1}, \theta_n, \theta_{n+1})} & X \\ \downarrow & \nearrow \gamma & \downarrow p \\ \Delta^n & \xrightarrow{\omega} & Y \end{array}$$

Then

$$\begin{aligned} \partial(d_0\gamma) &= (d_0d_0\gamma, d_1d_0\gamma, \dots, d_nd_0\gamma) = (d_0d_1\gamma, d_0d_2\gamma, \dots, d_0d_{n-1}\gamma, d_0d_n\gamma, d_0d_{n+1}\gamma) \\ &= (v, \dots, v, d_0\theta_{n-1}, d_0\theta_n, d_0\theta_{n+1}). \end{aligned}$$

Thus $d_0\gamma$ witnesses the fact that $[d_0\theta_{n-1}][d_0\theta_{n+1}] = [d_0\theta_n]$. Because $\partial(\alpha_i) = [d_0\theta_i]$ this means that we have $\partial(\alpha_{n-1})\partial(\alpha_{n+1}) = \partial(\alpha_n)$ as desired.

For (2), we first show that

$$\pi_n(X, v) \xrightarrow{p_*} \pi_n(Y, *) \xrightarrow{\partial} \pi_{n-1}(F, v)$$

is exact. The composition is trivial, as we consider the following diagram for $[\alpha] \in \pi_n(X, v)$:

$$\begin{array}{ccc} \Lambda_0^n & \xrightarrow{(\cdot, v, \dots, v)} & X \\ \downarrow & \nearrow \alpha & \downarrow p \\ \Delta^n & \xrightarrow{p\alpha} & Y \end{array}$$

So that $\partial \circ p(\alpha) = d_0\alpha = v$ and so $\text{im } p_* \subseteq \ker \partial$. Conversely suppose $\gamma: \Delta^n \rightarrow Y$ represents $[\gamma] \in \pi_n(Y, *)$ such that $\partial[\gamma] = e$. Take a diagram

$$\begin{array}{ccc} \Lambda_0^n & \xrightarrow{(\cdot, v, \dots, v)} & X \\ \downarrow & \nearrow \theta & \downarrow p \\ \Delta^n & \xrightarrow{\gamma} & Y \end{array}$$

so that $[d_0\theta] = \partial[\gamma] = [e]$. There is then a homotopy $h_0: \Delta^{n-1} \times \Delta^1 \rightarrow F$ where $d_0\theta \cong v \pmod{\partial\Delta^n}$. This defines a map

$$(\Delta^n \times *) \cup (\partial\Delta^n \times \Delta^1) \xrightarrow{(\theta, (h_0, v, \dots, v))} X$$

whose component on $\Delta^n \times *$ is θ , and $(h_0, v, \dots, v)$ defines the map out of $\partial\Delta^n \times \Delta^1$ (which maps (δ^0, t) to $h_0(\text{id}, t)$ and (δ^i, t) to v for all other i). On the common intersection $\partial\Delta^n \times *$, both of these maps agree as they are equal to v , so this map is well-defined.

Because X is fibrant, it extends to $h: \Delta^n \times \Delta^1 \rightarrow X$. Then the homotopy $p \circ h$ is a homotopy $\gamma \cong v \pmod{\partial\Delta^n}$ as desired.

All other instances of exactness are proven similarly. ■

Let X be fibrant, and consider $\mathbf{Hom}(\Delta^1, X)$. We have the maps $(\delta^\varepsilon)^*: \mathbf{Hom}(\Delta^1, X) \rightarrow \mathbf{Hom}(\Delta^0, X) \cong X$ which maps $f: \Delta^1 \times \Delta^n \rightarrow X$ to $f(\neg\varepsilon, \bullet)$ (where $\neg 0 = 1$ and $\neg 1 = 0$). We can thus view these as the endpoints of f : if $n = 0$ then these are just $f(0), f(1)$.

Because X is fibrant and δ^ε are monic, by corollary 1.6.4 so are $(\delta^\varepsilon)^*$. Thus their fibers are fibrant as well. So for a choice of vertex $* \in X$, define the **path space** of X to be the fiber of $*$ under $(\delta^0)^*$. Denote it by PX .

The path space can be viewed as the set of paths ending at $* \in X$. We define $\pi: PX \rightarrow X$ via the composite:

$$\pi = PX \hookrightarrow \mathbf{Hom}(\Delta^1, X) \xrightarrow{(\delta^1)^*} \mathbf{Hom}(\Delta^0, X) \cong X,$$

which maps a path to its other endpoint.

1.8.7 Lemma

Let X be fibrant and PX its path space. Then for all vertices $v \in PX$ and $i \geq 0$, $\pi_i(PX, v)$ is trivial. Furthermore $\pi: PX \rightarrow X$ is a fibration.

Proof

By lemma 1.6.5 $(\delta^0)^*$ has the right-lifting property relative to all $\partial\Delta^n \subseteq \Delta^n$. The following commutative diagram demonstrates that the restriction of $(\delta^0)^*$ to PX , which is just $PX \rightarrow *$ also has this right-lifting property.

$$\begin{array}{ccccc} \partial\Delta^n & \longrightarrow & PX & \hookrightarrow & \mathbf{Hom}(\Delta^1, X) \\ \downarrow & & \downarrow & \nearrow & \downarrow (\delta^0)^* \\ \Delta^n & \longrightarrow & * & \longrightarrow & X \end{array}$$

Because this diagram commutes, composing the dashed arrow with $(\delta^0)^*$ is constant. Thus the dashed arrow lies in PX , giving the desired lift.

If $v, u \in PX$ are vertices then d_0v and d_0u are both $*$ by definition of PX , and thus there is a map $(v, u): \partial\Delta^1 \rightarrow PX$. By the above paragraph, this extends to $\omega: \Delta^1 \rightarrow PX$ where $\partial\omega = (v, u)$ so that v and u are homotopic. Thus $\pi_0(PX)$ is trivial.

Similarly if $\alpha: \Delta^n \rightarrow PX$ represents an element of $\pi_n(PX, v)$ then there is a lift

$$\begin{array}{ccc} \partial\Delta^{n+1} & \xrightarrow{(v, \dots, v, \alpha)} & PX \\ \downarrow & \nearrow \omega & \\ \Delta^{n+1} & & \end{array}$$

Thus $\partial\omega = (v, \dots, v, \alpha)$ which means by lemma 1.8.5 $\alpha = [e]$.

To show that π is fibrant, notice that we have a pullback

$$\begin{array}{ccc} PX & \longrightarrow & \mathbf{Hom}(\Delta^1, X) \\ \pi \downarrow & & \downarrow i_* \\ X & \xrightarrow{(*, 1_X)} & \mathbf{Hom}(\partial\Delta^1, X) \cong X \times X \end{array}$$

Because i_* is a fibration, and the pullback of fibrations are fibrations, π is a fibrant. ■

1.8.8 Definition

Let X be a fibrant simplicial set, with a distinguished vertex $* \in X$. Let PX be its path space relative to this vertex, and $\pi: PX \rightarrow X$ its endpoint map. Define X 's **loop space**, ΩX , to be the fiber of $* \in X$ under π .

Because π is a fibration, ΩX is fibrant.

1.8.9 Lemma

$\pi_n(\Omega X, *)$ is Abelian for $n \geq 1$.

Proof

By exponentiation, as a set, $\pi_n(\Omega X, *)$ consists of homotopy classes of maps α fitting into diagrams of the form

$$\begin{array}{ccc} \Delta^n \times \Delta^1 & \xrightarrow{\alpha} & X \\ \uparrow & & \searrow * \\ (\partial\Delta^n \times \Delta^1) \cup (\Delta^n \times \partial\Delta^1) & & \end{array}$$

relative the boundary $(\partial\Delta^n \times \Delta^1) \cup (\Delta^n \times \partial\Delta^1)$.

We can define a second multiplication on $\pi_n(\Omega X, *)$ by multiplying α, β along $\alpha(x, \bullet)$ and $\beta(x, \bullet)$ (as 1-simplices). Denote this multiplication by $\star$. It can then be shown that $[*]$ is the identity for $\star$, and this new multiplication and the usual multiplication on $\pi_n(\Omega X, *)$ satisfy the following identity:

$$([\alpha] \star [\beta])([\gamma] \star [\delta]) = ([\alpha][\gamma]) \star ([\beta][\delta]).$$

By a standard argument (given below), this implies that $\star$ and the usual multiplication coincide and are Abelian. ■

1.8.10 Proposition (Eckmann-Hilton)

Let X be a set equipped with two binary operators, $*$ and $\star$. Further suppose that

- (1) Both $*$ and $\star$ have units, 1_* and $1_\star$ respectively.
- (2) The operators satisfy the following identity:

$$(a \star b) * (c \star d) = (a * c) \star (b * d), \quad \text{for all } a, b, c, d \in X.$$

Then $*$ and $\star$ are the same operator, and are commutative and associative.

Proof

We give a visual proof, courtesy of Wikipedia. Let us write $\begin{smallmatrix} a \\ b \end{smallmatrix}$ for $a * b$ and $a \star b$ for $a \star b$. Notice that (2) tells us that

$$\begin{pmatrix} a \star b \\ c \star d \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} \star \begin{pmatrix} b \\ d \end{pmatrix}$$

so we can write

$$\begin{smallmatrix} a & b \\ c & d \end{smallmatrix} = \begin{pmatrix} a \star b \\ c \star d \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} \star \begin{pmatrix} b \\ d \end{pmatrix}$$

unambiguously.

Let us write $\bullet, \circ$ for the identities of $\ast, \star$ respectively. Then we have

$$\bullet = \begin{array}{c} \bullet \\ \bullet \end{array} = \begin{array}{c} \bullet \\ \circ \end{array} \circ = \circ \circ = \circ$$

showing that both units are equal.

Now for $a, b \in X$,

$$a \ b = \begin{array}{c} a \\ \bullet \end{array} \begin{array}{c} \bullet \\ b \end{array} = \begin{array}{c} a \\ b \end{array} = \begin{array}{c} \bullet \\ b \end{array} \begin{array}{c} a \\ \bullet \end{array} = b \ a = \begin{array}{c} b \\ \bullet \end{array} \begin{array}{c} \bullet \\ a \end{array} = \begin{array}{c} b \\ a \end{array}$$

showing that horizontal and vertical composition ($\ast$ and $\star$) are the same, and commutative.

Finally for associativity, we have

$$a \ (b \ c) = a \ \begin{pmatrix} b \\ c \end{pmatrix} = \begin{array}{c} a \\ \bullet \end{array} \begin{array}{c} b \\ c \end{array} = \begin{pmatrix} a \ b \\ c \end{pmatrix} = (a \ b) \ c. \quad \blacksquare$$

Note:

The Eckmann-Hilton argument is also called the “sliding boxes argument” in lieu of how the above proof relies on sliding around the elements.

Because we can show that $\pi_{n+1}(X, \ast) \cong \pi_n(\Omega X, \ast)$, it follows that $\pi_n(X, \ast)$ is Abelian for $n \geq 2$.

1.8.11 Corollary

Let X be fibrant. Then for $n \geq 2$, $\pi_n(X, \ast)$ is Abelian.

1.8.12 Definition

A simplicial map $f: X \rightarrow Y$ is a **weak equivalence** if it induces an isomorphism between all of the homotopy groups. Explicitly,

- (1) $f_*: \pi_0 X \rightarrow \pi_0 Y$ is a bijection,
- (2) for all $x \in X$, $f_*: \pi_n(X, x) \rightarrow \pi_n(Y, fx)$ is an isomorphism.

1.8.13 Theorem

A simplicial map $f: X \rightarrow Y$ is a fibration and a weak equivalence iff it has the right-lifting property with respect to all $\partial \Delta^n \subseteq \Delta^n$ for $n \geq 0$.

Proof

Tedious.

1.8.14 Example

By the above theorem and lemma 1.6.5, we see that $(\delta^k)^*$ is a weak equivalence between the internal-homs

$\mathbf{Hom}(\Delta^n, X) \rightarrow \mathbf{Hom}(\Delta^{n-1}, X)$. Inductively, this shows that

$$(\delta^{k_1})^* \circ \dots \circ (\delta^{k_n})^* = (\delta^{k_n} \circ \dots \circ \delta^{k_1})^*: \mathbf{Hom}(\Delta^n, X) \rightarrow \mathbf{Hom}(\Delta^0, X) \cong X$$

is a weak equivalence. Note that $\delta = \delta^{k_n} \circ \dots \circ \delta^{k_1}: \Delta^0 \rightarrow \Delta^n$ just corresponds to a choice of vertex of Δ^n , i.e. a number in $[n]$. Then $\delta^*: \mathbf{Hom}(\Delta^n, X) \rightarrow X$ corresponds to evaluating $f: \Delta^n \times \Delta^k \rightarrow X$ at $f(\delta, 1)$.

Regardless, this shows that the evaluation map is a weak equivalence and a fibration $\mathbf{Hom}(\Delta^n, X) \rightarrow X$.

2 Enriched Category Theory

2.1 Monoidal categories

For a given commutative ring R , there is a functor $\otimes: \mathbf{Mod}_R \times \mathbf{Mod}_R \rightarrow \mathbf{Mod}_R$ (the tensor product) which maps M, N to $M \otimes N$ defined up to unique isomorphism as representing bilinear maps out of $M \times N$. That is,

$$\mathrm{hom}_R(M \otimes N, \bullet) \cong \{\text{bilinear maps } M \times N \rightarrow \bullet\}.$$

The tensor product is of much importance in algebra, and we are concerned with it here for certain properties it exhibits.

For one, $M \otimes R \cong R \otimes M \cong M$, so that R acts as a sort of unit for the tensor. Furthermore $M \otimes (N \otimes W) \cong (M \otimes N) \otimes W$, so the tensor is sort of associative. By “sort of” we mean that R is a unit and the tensor is associative, up to a natural isomorphism.

We are interested in categories equipped with a similar structure. For the time being, we focus on the associativity of $\otimes$.

The simplest such attempt of a generalization is a *strict monoidal category*, which we describe here.

2.1.1 Definition (Strict nonunital monoidal categories)

A **strict nonunital monoidal category** is a category $\mathcal{C}$ equipped with a bifunctor $\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ which is strictly associative in the following sense:

- (1) For all $X, Y, Z \in \mathcal{C}$ we have an *equality*:

$$X \otimes (Y \otimes Z) = (X \otimes Y) \otimes Z.$$

- (2) For a triple of morphisms $f: X \rightarrow X', g: Y \rightarrow Y', h: Z \rightarrow Z'$, we have an *equality*:

$$\begin{array}{ccccc} f \otimes (g \otimes h) & X \otimes (Y \otimes Z) & X' \otimes (Y' \otimes Z') \\ = & : & = & \longrightarrow & = \\ (f \otimes g) \otimes h & (X \otimes Y) \otimes Z & (X' \otimes Y') \otimes Z' \end{array}.$$

Such a definition is neat and simple, but betrays one of the core tenets of category theory: things should be defined *up to isomorphism*, not up to equality. For example, $\mathbf{Mod}_R$ equipped with the tensor product is not a strict monoidal category.

A simple example of a strict nonunital monoidal category are semigroups. Recall that a semigroup is a set M equipped with an associative binary operation. A set M can be viewed as a discrete category, and then any semigroup structure on M corresponds to a strict nonunital monoidal category structure on M and vice versa. A more involved example are endofunctor categories. If $\mathcal{C}$ is an (ordinary) category, then the functor category $\mathbf{End}(\mathcal{C}) = \mathbf{Fun}(\mathcal{C}, \mathcal{C})$ is a strict monoidal category whose monoidal structure is given by composition:

$$\circ: \mathbf{End}(\mathcal{C}) \times \mathbf{End}(\mathcal{C}) \rightarrow \mathbf{End}(\mathcal{C}), \quad (F, G) \mapsto F \circ G.$$

If we want to generalize the tensor product structure on $\mathbf{Mod}_R$ we need to encode information about the isomorphisms $X \otimes (Y \otimes Z) \xrightarrow{\sim} (X \otimes Y) \otimes Z$.

2.1.2 Definition (Nonunital monoidal categories)

A **nonunital monoidal category** is a category $\mathcal{C}$ equipped with the following:

- (1) a bifunctor $\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$,
- (2) for every $X, Y, Z \in \mathcal{C}$ an isomorphism $\alpha_{X,Y,Z}: X \otimes (Y \otimes Z) \xrightarrow{\sim} (X \otimes Y) \otimes Z$, called the **associator**.

We further require the following from the associators.

- (1) The associator forms a natural isomorphism between the following two functors:

$$\begin{aligned} \mathcal{C} \times \mathcal{C} \times \mathcal{C} &\xrightarrow{(X,Y,Z) \mapsto X \otimes (Y \otimes Z)} \mathcal{C} \\ \mathcal{C} \times \mathcal{C} \times \mathcal{C} &\xrightarrow{(X,Y,Z) \mapsto (X \otimes Y) \otimes Z} \mathcal{C}. \end{aligned}$$

Explicitly, this means that for all $f: X \rightarrow X'$, $g: Y \rightarrow Y'$, $h: Z \rightarrow Z'$, the following diagram commutes:

$$\begin{array}{ccc} X \otimes (Y \otimes Z) & \xrightarrow{\alpha_{X,Y,Z}} & (X \otimes Y) \otimes Z \\ \downarrow f \otimes (g \otimes h) & & \downarrow (f \otimes g) \otimes h \\ X' \otimes (Y' \otimes Z') & \xrightarrow{\alpha_{X',Y',Z'}} & (X' \otimes Y') \otimes Z' \end{array}$$

- (2) The following **pentagon identity** is satisfied: for $W, X, Y, Z \in \mathcal{C}$ the following diagram of isomorphism commutes:

$$\begin{array}{ccc} W \otimes ((X \otimes Y) \otimes Z) & \xrightarrow{\alpha_{W,X \otimes Y, Z}} & (W \otimes (X \otimes Y)) \otimes Z \\ \uparrow 1_W \otimes \alpha_{X,Y,Z} & & \downarrow \alpha_{W,X,Y} \otimes 1_Z \\ W \otimes (X \otimes (Y \otimes Z)) & & ((W \otimes X) \otimes Y) \otimes Z \\ \searrow \alpha_{W,X,Y \otimes Z} & & \nearrow \alpha_{W \otimes X, Y, Z} \\ & (W \otimes X) \otimes (Y \otimes Z) & \end{array}$$

If $(\mathcal{C}, \otimes)$ is a strict nonunital monoidal category, then taking $\alpha = \text{id}$ makes it into a (non-strict) nonunital monoidal category. So this definition is a generalization of the previous one.

The tensor product on Mod_R is an example of a nonunital monoidal category.

2.1.3 Example

If $\mathcal{D}$ is a nonunital monoidal category, then for any category $\mathcal{C}$ we can form $\text{Fun}(\mathcal{C}, \mathcal{D})$ into a nonunital monoidal category. The monoidal product is given by postcomposition with $\otimes$ in the following

$$\text{Fun}(\mathcal{C}, \mathcal{D}) \times \text{Fun}(\mathcal{C}, \mathcal{D}) \cong \text{Fun}(\mathcal{C}, \mathcal{D} \times \mathcal{D}) \xrightarrow{\otimes \circ -} \text{Fun}(\mathcal{C}, \mathcal{D}).$$

The associator for $F, G, H: \mathcal{C} \rightarrow \mathcal{D}$ is given by the natural isomorphism

$$F \otimes (G \otimes H) \xrightarrow{\sim} (F \otimes G) \otimes H, \quad X \mapsto \alpha_{F(X), G(X), H(X)}.$$

Now we extend our definitions to consider units.

2.1.4 Definition (Strict monoidal categories)

A **strict monoidal category** is a nonunital strict monoidal category $\mathcal{C}$ equipped with an object $\mathbf{1} \in \mathcal{C}$ (the unit) such that

- (1) $X \otimes \mathbf{1} = X = \mathbf{1} \otimes X$ for all objects $X \in \mathcal{C}$,
- (2) For every $f: X \rightarrow X'$, $f \otimes \text{id}_{\mathbf{1}} = f = \text{id}_{\mathbf{1}} \otimes f$.

For example for any category $\mathcal{C}$, $\text{End}(\mathcal{C})$ is a strict monoidal category whose unit is the identity functor $1_{\mathcal{C}}$.

For a set M viewed as a discrete category, strict monoidal structures on M correspond to monoids defined on M . Recall that a monoid is a set equipped with a binary operator which is associative and unital.

Note that the unit of a strict monoidal category is not invariant under isomorphism. Indeed, if $\mathbf{1}, \mathbf{1}'$ are two units then they must be equal: $\mathbf{1} \otimes \mathbf{1}' = \mathbf{1}$ because $\mathbf{1}'$ is the unit, but it is also equal to $\mathbf{1}'$ as $\mathbf{1}$ is the unit.

2.1.5 Definition (Monoidal categories)

A **monoidal category** is a nonunital monoidal category $\mathcal{C}$ equipped with a **unit**: a pair $(\mathbf{1}, v)$ where $\mathbf{1} \in \mathcal{C}$ is an object and $v: \mathbf{1} \otimes \mathbf{1} \xrightarrow{\sim} \mathbf{1}$ is an isomorphism. We further require that the two functors

$$\mathcal{C} \rightarrow \mathcal{C}, \quad X \mapsto \mathbf{1} \otimes X, \quad X \mapsto X \otimes \mathbf{1}$$

are fully faithful.

If $\mathcal{C}$ is a strict monoidal category, then $(\mathbf{1}, \text{id}_{\mathbf{1}})$ forms a unit of $\mathcal{C}$ (in the non-strict sense) so that $\mathcal{C}$ is also a non-strict monoidal category.

It is not obvious a priori that the unit of a monoidal category is unique up to unique isomorphism, but we will show that it is.

If $\mathcal{C}$ is a monoidal category, then we have an isomorphism

$$\mathbf{1} \otimes (\mathbf{1} \otimes X) \xrightarrow{\alpha_{\mathbf{1}, \mathbf{1}, X}} (\mathbf{1} \otimes \mathbf{1}) \otimes X \xrightarrow{v \otimes \text{id}_X} \mathbf{1} \otimes X.$$

Because $Y \mapsto \mathbf{1} \otimes Y$ is fully faithful, there is a unique isomorphism $\lambda_X: \mathbf{1} \otimes X \xrightarrow{\sim} X$ making the following diagram commute:

$$\begin{array}{ccc} \mathbf{1} \otimes (\mathbf{1} \otimes X) & \xrightarrow{\alpha_{\mathbf{1}, \mathbf{1}, X}} & (\mathbf{1} \otimes \mathbf{1}) \otimes X \\ & \searrow \text{id}_{\mathbf{1}} \otimes \lambda_X & \swarrow v \otimes \text{id}_X \\ & \mathbf{1} \otimes X & \end{array}$$

We call λ_X the **left unitor**.

Similarly there is a unique isomorphism $\rho_X: X \otimes \mathbf{1} \xrightarrow{\sim} X$ making the following diagram commute:

$$\begin{array}{ccc} X \otimes (\mathbf{1} \otimes \mathbf{1}) & \xrightarrow{\alpha_{X, \mathbf{1}, \mathbf{1}}} & (X \otimes \mathbf{1}) \otimes \mathbf{1} \\ & \searrow \text{id}_X \otimes v & \swarrow \rho_X \otimes \text{id}_{\mathbf{1}} \\ & X \otimes \mathbf{1} & \end{array}$$

And ρ_X is called the **right unitor**.

By their uniqueness, it is easy to see that the unitors are natural. That is, λ and ρ form natural isomorphisms

$$\lambda: (X \mapsto \mathbf{1} \otimes X) \xrightarrow{\sim} 1_{\mathcal{C}}, \quad \rho: (X \mapsto X \otimes \mathbf{1}) \xrightarrow{\sim} 1_{\mathcal{C}}.$$

2.1.6 Proposition (The triangle identity)

Let $\mathcal{C}$ be a monoidal category. Then for any $X, Y \in \mathcal{C}$, the following diagram of isomorphisms commutes:

$$\begin{array}{ccc}
 X \otimes (\mathbf{1} \otimes Y) & \xrightarrow{\alpha_{X, \mathbf{1}, Y}} & (X \otimes \mathbf{1}) \otimes Y \\
 \text{id}_X \otimes \lambda_Y \searrow & & \nearrow \rho_X \otimes \text{id}_Y \\
 & X \otimes Y &
 \end{array}$$

Proof

This follows from the pentagon identity and the functoriality of α, λ, ρ . ■

2.1.7 Corollary

Let $\mathcal{C}$ be a monoidal category. Then the left and right unitors $\lambda_1, \rho_1: \mathbf{1} \otimes \mathbf{1} \xrightarrow{\sim} \mathbf{1}$ are equal to the unit constraint v .

Proof

The left unitor is characterized by the following commutative diagram

$$\begin{array}{ccc}
 \mathbf{1} \otimes (\mathbf{1} \otimes X) & \xrightarrow{\alpha_{\mathbf{1}, \mathbf{1}, X}} & (\mathbf{1} \otimes \mathbf{1}) \otimes X \\
 \text{id}_{\mathbf{1}} \otimes \lambda_X \searrow & & \nearrow v \otimes \text{id}_X \\
 & \mathbf{1} \otimes X &
 \end{array}$$

so by the triangle identity we have that $v \otimes \text{id}_X = \rho_1 \otimes \text{id}_X$. Because $Y \mapsto Y \otimes \mathbf{1}$ is fully faithful, this means that $v = \rho_1$. Similarly we have that $v = \lambda_1$. ■

Note:

In lieu of this result, it should not be surprising that we can axiomatize monoidal categories in terms of the unitors λ and ρ instead of v . To do so, we just require the pentagon identity and the triangle identity. That is, we require the existence of the natural isomorphisms α, λ, ρ , and that they satisfy the pentagon identity and the triangle identity. Because λ, ρ are natural isomorphisms of $X \mapsto \mathbf{1} \otimes X, X \otimes \mathbf{1}$ to $\text{id}_{\mathcal{C}}$ respectively, it follows immediately that these functors are fully faithful. The above proof relies only on these functors being fully faithful and the triangle identity, so we can reconstruct v from $\lambda_1 = \rho_1$.

2.1.8 Proposition

If $\mathcal{C}$ is a nonunital monoidal category equipped with units $(\mathbf{1}, v)$ and $(\mathbf{1}', v')$, then there is a unique isomorphism $u: \mathbf{1} \xrightarrow{\sim} \mathbf{1}'$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathbf{1} \otimes \mathbf{1} & \xrightarrow{v} & \mathbf{1} \\ u \otimes u \downarrow & & \downarrow u \\ \mathbf{1}' \otimes \mathbf{1}' & \xrightarrow{v'} & \mathbf{1}' \end{array}$$

Proof

Let λ, ρ be the unitors relative to $(\mathbf{1}, v)$. Because $v = \lambda_1$, this claim is equivalent to the outer rectangle in the below diagram commuting:

$$\begin{array}{ccc} \mathbf{1} \otimes \mathbf{1} & \xrightarrow{\lambda_1} & \mathbf{1} \\ \text{id}_1 \otimes u \downarrow & & \downarrow u \\ \mathbf{1} \otimes \mathbf{1}' & \xrightarrow{\lambda_1} & \mathbf{1}' \\ u \otimes \text{id}_{1'} \downarrow & & \downarrow \text{id}_{1'} \\ \mathbf{1}' \otimes \mathbf{1}' & \xrightarrow{\rho'} & \mathbf{1}' \end{array}$$

The upper square commutes by naturality of λ . So we want to show the existence and uniqueness of u making the bottom square commute. This follows immediately from $X \mapsto X \otimes \mathbf{1}'$ being fully faithful ($((v')^{-1} \circ \lambda_{1'})$ has a unique origin). ■

Note:

The conditions for $\mathcal{C}$ being monoidal are actually quite strong. A powerful theorem due to Mac Lane states that in a monoidal category, all diagrams whose vertices are obtained via tensoring with objects of $\mathcal{C}$ and $\mathbf{1}$ and whose edges are tensors of the associators, unitors, and identities, and their inverses, commutes.

This is called **Mac Lane's coherence theorem**.

Note that this result is restricted only to so-called “formal diagrams” whose vertices and edges are composed of data from the monoidal structure on $\mathcal{C}$.

2.1.9 Example

If $\mathcal{C}$ is a category admitting binary products, then it forms a nonunital monoidal category. Indeed, $X \otimes (Y \otimes Z)$ and $(X \otimes Y) \otimes Z$ both satisfy the same universal properties, so there is a unique isomorphism between them. A standard diagram chasing argument shows that these isomorphisms satisfy the pentagon identity.

If $\mathcal{C}$ has a terminal object $\mathbf{1}$, then there is a unique isomorphism $v: \mathbf{1} \times \mathbf{1} \cong \mathbf{1}$, and it is standard to see that this makes $\mathcal{C}$ into a monoidal category.

2.1.10 Example

If $\mathcal{C}$ is a monoidal category, then there is a canonical way to turn its opposite category $\mathcal{C}^{\text{op}}$ into one. This follows from the commutativity of $\bullet^{\text{op}}$ with product categories, so

$$\text{Fun}(\mathcal{C} \times \mathcal{C}, \mathcal{C})^{\text{op}} \cong \text{Fun}(\mathcal{C}^{\text{op}} \times \mathcal{C}^{\text{op}}, \mathcal{C}^{\text{op}})$$

meaning there is a canonical choice for $\otimes^{\text{op}}: \mathcal{C}^{\text{op}} \times \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$.

If we denote the associated object of $X \in \mathcal{C}$ in the opposite category by $X^{\text{op}} \in \mathcal{C}^{\text{op}}$, then the associator $\alpha_{X^{\text{op}}, Y^{\text{op}}, Z^{\text{op}}}$ is defined to be the inverse $\alpha_{X, Y, Z}^{-1}$.

And if $(1, v)$ is the unit for $\mathcal{C}$, then $(1^{\text{op}}, v^{-1})$ is the unit for $\mathcal{C}^{\text{op}}$.

2.1.11 Example

If $\mathcal{C}$ is a monoidal category, we can “reverse” the monoidal product to give another monoidal structure on $\mathcal{C}$. Denote the reverse monoidal category by $\mathcal{C}^{\text{rev}}$. Then $\otimes^{\text{rev}}$ acts on objects and morphisms by

$$X \otimes^{\text{rev}} Y = Y \otimes X, \quad f \otimes^{\text{rev}} g = g \otimes^{\text{rev}} f.$$

The associator is then

$$\alpha_{X, Y, Z}^{\text{rev}} = \alpha_{Z, Y, X}^{-1}$$

and the unit remains the same: $(1^{\text{rev}}, v^{\text{rev}}) = (1, v)$.

2.2 Monoidal functors

2.2.1 Nonunital monoidal functors

Now that we have a notion of monoidal categories, it is natural to consider functors between them. We are of course interested in functors which preserve the monoidal structure, which not all functors will. A first approach is the following.

2.2.1 Definition (Nonunital strict monoidal functors)

Let $\mathcal{C}, \mathcal{D}$ be nonunital monoidal categories. A **nonunital strict monoidal functor** from $\mathcal{C}$ to $\mathcal{D}$ is a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ such that

- (1) The following diagram of functors commutes:

$$\begin{array}{ccc} \mathcal{C} \times \mathcal{C} & \xrightarrow{\quad} & \mathcal{C} \\ \downarrow F \times F & \circlearrowleft \otimes & \downarrow F \\ \mathcal{D} \times \mathcal{D} & \xrightarrow{\quad} & \mathcal{D} \end{array}$$

- (2) F carries the associator of $\mathcal{C}$ to the associator of $\mathcal{D}$. That is, $F(\alpha_{X, Y, Z}) = \alpha_{FX, FY, FZ}$. Alternatively we can write this as the whiskering: $F\alpha = \alpha_{F \times F \times F}$.

For example, the identity of $\mathcal{C}$ is a nonunital strict monoidal functor $\mathcal{C} \rightarrow \mathcal{C}$.

But this approach is clearly too restrictive. Many constructions of a monoidal product are defined only up to isomorphism, so requiring a strict equality of $F(X \otimes Y) = F(X) \otimes F(Y)$ is too restrictive. For example, if $\mathcal{C}$ has finite products, then there are different ways of choosing $X \times Y$ although they are all uniquely isomorphic. Then the identity functor $1_{\mathcal{C}}$ will not be a strict monoidal functor from $\mathcal{C}$ equipped with one choice of products to another.

Thus we will want some sort of isomorphism $\mu_{X,Y}: F(X) \otimes F(Y) \xrightarrow{\sim} F(X \otimes Y)$. Of course we will want these μ to be natural. We start developing our theory by not requiring μ to be an isomorphism.

2.2.2 Definition (Nonunital lax monoidal functors)

Let $\mathcal{C}, \mathcal{D}$ be nonunital monoidal categories. A **nonunital lax monoidal functor** from $\mathcal{C}$ to $\mathcal{D}$ is a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ along with morphisms $\mu_{X,Y}: F(X) \otimes F(Y) \rightarrow F(X \otimes Y)$ for $X, Y \in \mathcal{C}$ such that

- (1) μ forms a natural transformation $\otimes \circ (F \times F) \Rightarrow F \circ \otimes$:

$$\begin{array}{ccc} \mathcal{C} \times \mathcal{C} & \xrightarrow{\otimes} & \mathcal{C} \\ F \times F \downarrow & \nearrow \mu & \downarrow F \\ \mathcal{D} \times \mathcal{D} & \xrightarrow{\otimes} & \mathcal{D} \end{array}$$

- (2) μ is compatible with the associators on $\mathcal{C}$ and $\mathcal{D}$: the following diagram commutes

$$\begin{array}{ccc} F(X) \otimes (F(Y) \otimes F(Z)) & \xrightarrow{\alpha_{FX,FY,FZ}} & (F(X) \otimes F(Y)) \otimes F(Z) \\ \downarrow \text{id}_{FX} \otimes \mu_{Y,Z} & & \downarrow \mu_{X,Y} \otimes \text{id}_{FZ} \\ F(X) \otimes F(Y \otimes Z) & & F(X \otimes Y) \otimes F(Z) \\ \downarrow \mu_{X,Y \otimes Z} & & \downarrow \mu_{X \otimes Y, Z} \\ F(X \otimes (Y \otimes Z)) & \xrightarrow{F\alpha_{X,Y,Z}} & F((X \otimes Y) \otimes Z) \end{array}$$

We call μ the **tensoror** of F .

2.2.3 Definition (Nonunital monoidal functors)

A **nonunital monoidal functor** $F: \mathcal{C} \rightarrow \mathcal{D}$ is a nonunital lax monoidal functor whose tensoror μ is a natural isomorphism.

Note that a nonunital strict monoidal functor is a nonunital monoidal functor whose tensoror μ is the identity.

2.2.4 Example

The tensor $M \otimes_R N$ is characterized by an R -bilinear map $\mu_{X,Y}: M \times N \rightarrow M \otimes_R N$. Then the forgetful functor $\text{Mod}_R \rightarrow \text{Set}$ is a nonunital lax monoidal functor whose tensoror is μ . Here Set is made into a monoidal category by direct products.

This functor is not monoidal, as $U(M \otimes N)$ is not isomorphic in general to $U(M) \times U(N)$ (U denoting the forgetful functor).

2.2.5 Example

Let $\mathcal{C}$ be a monoidal category, and endow $\text{End}(\mathcal{C})$ with the induced monoidal structure. For $X \in \mathcal{C}$ define $\ell_X: \mathcal{C} \rightarrow \mathcal{C}$ by $\ell_X(Y) = X \otimes Y$ and $\ell_X(f) = \text{id}_X \otimes f$. This then defines a functor $\ell: \mathcal{C} \rightarrow \text{End}(\mathcal{C})$ (this is just the currying of $\otimes$).

For $X, Y \in \mathcal{C}$, there is a natural isomorphism $\mu_{X,Y} \cong \ell_X \circ \ell_Y \xrightarrow{\sim} \ell_{X \otimes Y}$ whose value on $Z \in \mathcal{C}$ is the associator:

$$(\ell_X \circ \ell_Y)(Z) = X \otimes (Y \otimes Z) \xrightarrow{\alpha_{X,Y,Z}} (X \otimes Y) \otimes Z = \ell_{X \otimes Y}(Z).$$

Then (ℓ, μ) forms a nonunital monoidal functor $\mathcal{C} \rightarrow \text{End}(\mathcal{C})$.

2.2.6 Definition (Nonunital monoidal natural transformations)

Let $F, F': \mathcal{C} \rightarrow \mathcal{D}$ be nonunital lax monoidal functors with tensorors μ, μ' respectively. A natural transformation $\gamma: F \Rightarrow F'$ is **nonunital monoidal** if for all $X, Y \in \mathcal{C}$ the following diagram commutes:

$$\begin{array}{ccc} F(X) \otimes F(Y) & \xrightarrow{\mu_{X,Y}} & F(X \otimes Y) \\ \gamma_X \otimes \gamma_Y \downarrow & & \downarrow \gamma_{X \otimes Y} \\ F'(X) \otimes F'(Y) & \xrightarrow{\mu'_{X,Y}} & F'(X \otimes Y) \end{array}$$

Define $\text{Fun}_{\text{nu}}^{\text{lax}}(\mathcal{C}, \mathcal{D})$ to be the category whose objects are nonunital lax monoidal functors $\mathcal{C} \rightarrow \mathcal{D}$ and whose morphisms are nonunital monoidal natural transformations. And let $\text{Fun}_{\text{nu}}^{\otimes}(\mathcal{C}, \mathcal{D})$ be the full subcategory of $\text{Fun}_{\text{nu}}^{\text{lax}}(\mathcal{C}, \mathcal{D})$ whose objects are the nonunital monoidal functors $\mathcal{C} \rightarrow \mathcal{D}$.

2.2.7 Example

Let $\mathcal{C}$ be a nonunital monoidal category and $A \in \mathcal{C}$ an object. A **nonunital monoid structure** on A is a map $m: A \otimes A \rightarrow A$ (multiplication) which is associative in the sense that the following diagram commutes

$$\begin{array}{ccc} A \otimes (A \otimes A) & \xrightarrow{\alpha} & (A \otimes A) \otimes A \\ \text{id} \otimes m \downarrow & & \downarrow m \otimes \text{id} \\ A \otimes A & & A \otimes A \\ & \searrow m & \swarrow m \\ & A & \end{array}$$

Then (A, m) is a **nonunital monoid object** of $\mathcal{C}$. If (A, m) and (A', m') are nonunital monoid objects of $\mathcal{C}$, a **nonunital monoid homomorphism** is a morphism $f: A \rightarrow A'$ such that the following commutes

$$\begin{array}{ccc} A \otimes A & \xrightarrow{m} & A \\ f \otimes f \downarrow & & \downarrow f \\ A' \otimes A' & \xrightarrow{m'} & A' \end{array}$$

Let $\text{Mon}^{\text{nu}}(\mathcal{C})$ denote the category of nonunital monoid objects of $\mathcal{C}$ whose morphisms are nonunital monoid homomorphisms. For example, if $\mathcal{C} = \text{Set}$ with the cartesian product, then nonunital monoids in Set are simply ordinary nonunital monoids (semigroups).

Let $\{e\}$ be the trivial monoid, which can be regarded as a discrete (strict) monoidal category. Objects of $\mathcal{C}$ can be identified with functors $F: \{e\} \rightarrow \mathcal{C}$ (A is identified with $F(e) = A$). Then nonunital lax monoidal structures on $F: \{e\} \rightarrow \mathcal{C}$ can be identified with nonunital monoid structures on $A = F(e)$.

Indeed the tensoror is a map $\mu: A \otimes A \rightarrow A$ (since $e \otimes e = e$), and compatibility with the associators means that it forms multiplication on A . Naturality is trivially satisfied.

Thus there is an isomorphism of categories $\text{Fun}_{\text{nu}}^{\text{lax}}(\{e\}, \mathcal{C}) \cong \text{Mon}^{\text{nu}}(\mathcal{C})$.

Note:

The unit of a monoid object is unique. That is, if $M \in \mathcal{C}$ is equipped with multiplication $m: M \otimes M \rightarrow M$, then any two units $\varepsilon, \varepsilon': \mathbf{1} \rightarrow M$ must be equal.

Indeed, consider the following diagram:

$$\begin{array}{ccccc}
 \mathbf{1} \otimes \mathbf{1} & \xrightarrow{\text{id} \otimes \varepsilon} & \mathbf{1} \otimes M & \xrightarrow{\varepsilon' \otimes \text{id}} & M \otimes M \\
 \downarrow v = \lambda_1 & & \searrow \lambda & & \downarrow m \\
 \mathbf{1} & \xrightarrow{\varepsilon} & & & M
 \end{array}$$

Both regions of this diagram (the left due to λ 's naturality, the right by definition of units), so we have

$$\varepsilon \circ v = m \circ (\varepsilon' \otimes \varepsilon).$$

A similar diagram with ρ in place of λ yields

$$\varepsilon \circ v = m \circ (\varepsilon \otimes \varepsilon').$$

By symmetry we can swap $\varepsilon, \varepsilon'$ so that we get

$$\varepsilon \circ v = m \circ (\varepsilon \otimes \varepsilon') = m \circ (\varepsilon' \otimes \varepsilon) = \varepsilon' \circ v$$

and since v is an isomorphism, $\varepsilon = \varepsilon'$ as desired.

2.2.8 Example

Let $\mathcal{C}, \mathcal{D}$ be nonunital monoidal categories whose reverses are $\mathcal{C}^{\text{rev}}, \mathcal{D}^{\text{rev}}$. Because $\mathcal{C}, \mathcal{C}^{\text{rev}}$ have the same underlying categorical structure, a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ determines a functor $F^{\text{rev}}: \mathcal{C}^{\text{rev}} \rightarrow \mathcal{D}^{\text{rev}}$ and vice versa. This bijection extends to nonunital lax monoidal functors: if $F: \mathcal{C} \rightarrow \mathcal{D}$ is a nonunital lax monoidal functor, then F^{rev} can be made into one with $\mu_{X,Y}^{\text{rev}} = \mu_{Y,X}$.

Thus we have an isomorphism of categories $\text{Fun}_{\text{nu}}^{\text{lax}}(\mathcal{C}, \mathcal{D}) \cong \text{Fun}_{\text{nu}}^{\text{lax}}(\mathcal{C}^{\text{rev}}, \mathcal{D}^{\text{rev}})$. Clearly this restricts to an isomorphism of subcategories $\text{Fun}_{\text{nu}}^{\otimes}(\mathcal{C}, \mathcal{D}) \cong \text{Fun}_{\text{nu}}^{\otimes}(\mathcal{C}^{\text{rev}}, \mathcal{D}^{\text{rev}})$.

2.2.9 Example

The previous example then raises the question regarding the relation between lax monoidal functors $\mathcal{C} \rightarrow \mathcal{D}$ and lax monoidal functors $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$. Unwinding definitions, we see that the obvious bijection would require us defining $\mu_{X,Y}^{\text{op}} = \mu_{X,Y}^{-1}$. This defines an isomorphism $\text{Fun}_{\text{nu}}^{\otimes}(\mathcal{C}, \mathcal{D})^{\text{op}} \cong \text{Fun}_{\text{nu}}^{\otimes}(\mathcal{C}^{\text{op}}, \mathcal{D}^{\text{op}})$. But this does not extend to an isomorphism of *lax* monoidal functors!

In lieu of this example, we give the following definition.

2.2.10 Definition (Nonunital colax monoidal functors)

Let $\mathcal{C}, \mathcal{D}$ be nonunital monoidal categories. A **nonunital colax monoidal functor** is a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ which forms a nonunital lax monoidal functor $F^{\text{op}}: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$. Explicitly, it comes equipped with morphisms $\mu = \{\mu_{X,Y}: F(X \otimes Y) \rightarrow F(X) \otimes F(Y)\}_{X,Y \in \mathcal{C}}$ satisfying the following:

- (1) μ is a natural transformation $F \circ \otimes \Rightarrow \otimes \circ (F \times F)$:

$$\begin{array}{ccc} \mathcal{C} \times \mathcal{C} & \xrightarrow{\otimes} & \mathcal{C} \\ \downarrow F \times F & \searrow \mu & \downarrow F \\ \mathcal{D} \times \mathcal{D} & \xrightarrow{\otimes} & \mathcal{D} \end{array}$$

Or explicitly, the following diagram commutes for all morphisms f, g in $\mathcal{C}$:

$$\begin{array}{ccc} F(X \otimes Y) & \xrightarrow{\mu_{X,Y}} & F(X) \otimes F(Y) \\ \downarrow F(f \otimes g) & & \downarrow F(f) \otimes F(g) \\ F(X' \otimes Y') & \xrightarrow{\mu_{X',Y'}} & F(X') \otimes F(Y') \end{array}$$

- (2) For every $X, Y, Z \in \mathcal{C}$ the following diagram commutes:

$$\begin{array}{ccc} F(X \otimes (Y \otimes Z)) & \xrightarrow{F\alpha_{X,Y,Z}} & F((X \otimes Y) \otimes Z) \\ \uparrow \mu_{X,Y \otimes Z} & & \uparrow \mu_{X \otimes Y, Z} \\ F(X) \otimes F(Y \otimes Z) & & F(X \otimes Y) \otimes F(Z) \\ \uparrow \text{id}_{FX} \otimes \mu_{Y,Z} & & \uparrow \mu_{X,Y} \otimes \text{id}_{FZ} \\ F(X) \otimes (F(Y) \otimes F(Z)) & \xrightarrow{\alpha_{FX,FY,FZ}} & (F(X) \otimes F(Y)) \otimes F(Z) \end{array}$$

Suppose we have nonunital lax monoidal functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{E}$ with respective tensorors μ, ν . We want to then define a tensoror on $G \circ F$. We have the following situation:

$$\begin{array}{ccc} \mathcal{C} \times \mathcal{C} & \xrightarrow{\otimes} & \mathcal{C} \\ \downarrow F \times F & \searrow \mu & \downarrow F \\ \mathcal{D} \times \mathcal{D} & \xrightarrow{\otimes} & \mathcal{D} \\ \downarrow G \times G & \searrow \nu & \downarrow G \\ \mathcal{E} \times \mathcal{E} & \xrightarrow{\otimes} & \mathcal{E} \end{array}$$

So we see that we can construct a natural transformation $\otimes \circ (G \times G) \circ (F \times F) \Rightarrow G \circ F \circ \otimes$ as follows. First, $\nu_{F \times F}$ is a natural transformation $\otimes \circ (G \times G) \circ (F \times F) \Rightarrow G \circ \otimes \circ (F \times F)$. Second, $G\mu$ is a natural transformation $G \circ \otimes \circ (F \times F) \Rightarrow G \circ F \circ \otimes$. Thus composing them gives the desired natural transformation:

$$\begin{aligned} G\mu \circ \nu_{F \times F}: \otimes \circ (G \times G) \circ (F \times F) &\Rightarrow G \circ F \circ \otimes, \\ (G \circ F)(X) \otimes (G \circ F)(Y) &\xrightarrow{\nu_{FX,FY}} G(F(X) \otimes F(Y)) \xrightarrow{G\mu_{X,Y}} G(F(X \otimes Y)). \end{aligned}$$

It is then routine to check that this is compatible with the associators of $\mathcal{C}$ and $\mathcal{E}$.

Thus we can form a category of nonunital monoidal categories with nonunital lax monoidal functors as morphisms. It is also clear that in the above expression, if μ and ν are isomorphisms, so is their composition. Thus this restricts to the wide (meaning containing all objects) subcategory whose morphisms are nonunital monoidal functors.

2.2.11 Definition (Oriented fiber products)

Let $\mathcal{C}, \mathcal{D}, \mathcal{E}$ be categories along with functors $F: \mathcal{C} \rightarrow \mathcal{E}$ and $G: \mathcal{D} \rightarrow \mathcal{E}$. Recall that $[n]$ is the poset $0 < \dots < n$ which we can view as a category. The **oriented fiber product** is the iterated pullback

$$\mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{E}} \mathcal{D} = \mathcal{C} \times_{\text{Fun}(\{0\}, \mathcal{E})} \text{Fun}([1], \mathcal{E}) \times_{\text{Fun}(\{1\}, \mathcal{E})} \mathcal{D}$$

where the functors $\mathcal{C}, \mathcal{D} \rightarrow \text{Fun}(\{x\}, \mathcal{E})$ are obtained via F, G and the isomorphisms $\text{Fun}(\{x\}, \mathcal{E}) \cong \mathcal{E}$.

We explicitly describe this category.

- (1) Objects of $\mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{E}} \mathcal{D}$ are triples (C, D, η) where $C \in \mathcal{C}$ and $D \in \mathcal{D}$ are objects and $\eta: F(C) \rightarrow G(D)$ is a morphism in $\mathcal{E}$.
- (2) If (C, D, η) and (C', D', η') are objects of $\mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{E}} \mathcal{D}$, a morphism $(C, D, \eta) \rightarrow (C', D', \eta')$ is a pair (u, v) where $u: C \rightarrow C'$ is a morphism in $\mathcal{C}$, $v: D \rightarrow D'$ a morphism in $\mathcal{D}$, and the following diagram commutes (in $\mathcal{E}$):

$$\begin{array}{ccc} FC & \xrightarrow{\quad \eta \quad} & GD \\ F(u) \downarrow & & \downarrow G(v) \\ FC' & \xrightarrow{\quad \eta' \quad} & GD' \end{array}$$

Note:

The oriented fiber product is often called the **comma category** $F \downarrow G$.

2.2.12 Proposition

Let $\mathcal{C}, \mathcal{D}$ be nonunital monoidal categories and $G: \mathcal{D} \rightarrow \mathcal{C}$ a functor. Let $\mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{C}} \mathcal{D}$ be the oriented fiber product relative to $1_{\mathcal{C}}$ and G .

- (1) Let μ be a nonunital lax monoidal structure on G . Then there is a unique nonunital monoidal structure $\otimes_{\mu}$ on $\mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{C}} \mathcal{D}$ with the following properties:

- (i) The forgetful functor

$$U: \mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{C}} \mathcal{D} \rightarrow \mathcal{C} \times \mathcal{D}, \quad (C, D, \eta) \mapsto (C, D)$$

is a strict monoidal functor.

- (ii) On objects, the tensor product $\otimes_{\mu}$ is given by

$$(C, D, \eta) \otimes_{\mu} (C', D', \eta') = (C \otimes C', D \otimes D', t(\eta, \eta'))$$

where $t(\eta, \eta')$ is the composition

$$t(\eta, \eta') = C \otimes C' \xrightarrow{\eta \otimes \eta'} G(D) \otimes G(D') \xrightarrow{\mu_{D, D'}} G(D \otimes D').$$

- (2) Conversely every nonunital monoidal structure on $\mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{C}} \mathcal{D}$ defines a nonunital lax monoidal structure on G so that $\mu \mapsto \otimes_{\mu}$ is a bijection.

Proof

For (1) clearly the two conditions uniquely determine $\otimes_\mu$. Indeed, U maps morphisms to themselves (the oriented fiber product is a subcategory of the direct product). For U to be strictly monoidal we must have that for morphisms (u, v) and (u', v') in the oriented fiber product,

$$(u, v) \otimes_\mu (u', v') = U((u, v) \otimes_\mu (u', v')) = U((u, v)) \otimes U((u', v')) = (u, v) \otimes (u', v') = (u \otimes u, v \otimes v').$$

The associators are uniquely determined again because U is strictly monoidal: the isomorphism

$$(C, D, \eta) \otimes_\mu ((C', D', \eta') \otimes_\mu (C'', D'', \eta'')) \cong ((C, D, \eta) \otimes_\mu (C', D', \eta')) \otimes_\mu (C'', D'', \eta'')$$

maps to $(\alpha_{C, C', C'', \alpha_{D, D', D''}})$ so this must be the associator in the oriented fiber product.

To show that this does form a well-defined monoidal structure on the oriented fiber product, we must show two things. First we must show that $(u \otimes u', v \otimes v')$ is a morphism $(C, D, \eta) \otimes_\mu (C', D', \eta') \rightarrow (\bar{C}, \bar{D}, \bar{\eta}) \otimes_\mu (\bar{C}', \bar{D}', \bar{\eta}')$. That is if the following diagrams commute:

$$\begin{array}{ccc} C & \xrightarrow{\eta} & G(D) \\ u \downarrow & & \downarrow G(v) \\ \bar{C} & \xrightarrow{\bar{\eta}} & G(\bar{D}) \end{array} \quad \begin{array}{ccc} C' & \xrightarrow{\eta'} & G(D') \\ u' \downarrow & & \downarrow G(v') \\ \bar{C}' & \xrightarrow{\bar{\eta}'} & G(\bar{D}') \end{array}$$

Then the following diagram commutes

$$\begin{array}{ccc} C \otimes C' & \xrightarrow{t(\eta, \eta')} & G(D \otimes D') \\ u \otimes u' \downarrow & & \downarrow G(v \otimes v') \\ \bar{C} \otimes \bar{C}' & \xrightarrow{t(\bar{\eta}, \bar{\eta}')} & G(\bar{D} \otimes \bar{D}') \end{array}$$

Now we must show that the associators are morphisms. This is sufficient, because the naturality of the associator and it satisfying the pentagon identity follow from them doing so in $\mathcal{C} \times \mathcal{D}$. To show that they are morphisms, we must have that the following diagram commutes

$$\begin{array}{ccc} C \otimes (C' \otimes C'') & \xrightarrow{\alpha_{C, C', C''}} & (C \otimes C') \otimes C'' \\ t(\eta, t(\eta', \eta'')) \downarrow & & \downarrow t(t(\eta, \eta'), \eta'') \\ G(D \otimes (D' \otimes D'')) & \xrightarrow{G(\alpha_{D, D', D''})} & G((D \otimes D') \otimes D'') \end{array}$$

Conversely any structure satisfying these two conditions defines $\mu_{D, D'}$ by $\mu_{D, D'} = t(\text{id}_{G(D)}, \text{id}_{G(D')})$. Notice that if (C, D, η) is an object in the oriented fiber product, then there is a morphism

$$(\eta, \text{id}_D): (C, D, \eta) \rightarrow (G(D), D, \text{id}_{G(D)}).$$

Thus the first condition implies the following diagram commutes

$$\begin{array}{ccc} C \otimes C' & \xrightarrow{t(\eta, \eta')} & G(D \otimes D') \\ \eta \otimes \eta' \downarrow & & \downarrow \text{id} \\ G(D) \otimes G(D') & \xrightarrow{t(\text{id}, \text{id})} & G(D \otimes D') \end{array}$$

meaning that $t(\eta, \eta') = \mu_{D, D'} \circ (\eta \otimes \eta')$. Thus t can be recovered from μ .

Thus it is sufficient for (1) and (2) to show that if $\mu = \{\mu_{D, D'}\}$ is any system of morphisms, it defines a nonunital lax monoidal structure on G iff the two conditions above are satisfied.

The first condition requires that outer rectangle in the following diagram commutes:

$$\begin{array}{ccccc}
C \otimes C' & \xrightarrow{\eta \otimes \eta'} & G(D) \otimes G(D') & \xrightarrow{\mu_{D,D'}} & G(D \otimes D') \\
\downarrow u \otimes u' & & \downarrow G(v) \otimes G(v') & & \downarrow G(v \otimes v') \\
\bar{C} \otimes \bar{C}' & \xrightarrow{\bar{\eta} \otimes \bar{\eta}'} & G(\bar{D}) \otimes G(\bar{D}') & \xrightarrow{\mu_{\bar{D},\bar{D}'}} & G(\bar{D} \otimes \bar{D}')
\end{array}$$

The left square commutes automatically by functoriality of $\otimes, G$. This then commutes iff the right square commutes (taking the η to be the identities for the converse). Thus the first condition holds iff μ is natural.

The second condition requires that the outer rectangle in the following diagram commutes:

$$\begin{array}{ccc}
C \otimes (C' \otimes C'') & \xrightarrow{\alpha_{C,C',C''}} & (C \otimes C') \otimes C'' \\
\downarrow \eta \otimes (\eta' \otimes \eta'') & & \downarrow (\eta \otimes \eta') \otimes \eta'' \\
G(C) \otimes (G(C') \otimes G(C'')) & \xrightarrow{\alpha_{G(C),G(C'),G(C'')}} & (G(C) \otimes G(C')) \otimes G(C'') \\
\downarrow \text{id}_{G(C)} \otimes \mu_{C',C''} & & \downarrow \mu_{C,C'} \otimes \text{id}_{G(C'')} \\
G(C) \otimes G(C' \otimes C'') & & G(C \otimes C') \otimes G(C'') \\
\downarrow \mu_{C,C' \otimes C''} & & \downarrow \mu_{C \otimes C',C''} \\
G(C \otimes (C' \otimes C'')) & \xrightarrow{G\alpha_{C,C',C''}} & G((C \otimes C') \otimes C'')
\end{array}$$

The top square automatically commutes by naturality of the associators of $\mathcal{C}$. Thus the outer rectangle commutes iff the lower rectangle does (for the converse, taking η to be the identities). And this is iff μ commutes with the associators of $\mathcal{C}, \mathcal{D}$, i.e. forms a lax monoidal structure on G . ■

Note that a pair of functors $F: \mathcal{C} \rightleftarrows \mathcal{D}: G$ form an adjoint pair $F \dashv G$ iff $1_{\mathcal{C}} \downarrow G$ and $F \downarrow 1_{\mathcal{D}}$ are isomorphic. Indeed, an object of the former is (C, D, η) where $\eta: C \rightarrow G(D)$. Applying the adjunction gives $\eta^\sharp: F(C) \rightarrow D$, so that $(C, D, \eta^\sharp)$ is an object of the latter category. Naturality of the adjunction tells us this is functorial. And the inverse, $\eta: F(C) \rightarrow D$ mapping to $\eta^\flat: C \rightarrow G(D)$ is its inverse.

Thus the following are equivalent:

- (1) The datum of a nonunital lax monoidal structure on $G: \mathcal{D} \rightarrow \mathcal{C}$.
- (2) The datum of a nonunital colax monoidal structure on $F: \mathcal{C} \rightarrow \mathcal{D}$.
- (3) The datum of a nonunital monoidal structure on $\mathcal{C} \xrightarrow{\vec{\cdot}}_{\mathcal{C}} \mathcal{D} \cong \mathcal{C} \xrightarrow{\vec{\cdot}}_{\mathcal{D}} \mathcal{D}$ such that the forgetful functor to $\mathcal{C} \times \mathcal{D}$ is strict monoidal.

The above proposition showed the equivalence of (1) and (3). Then (2) is equivalent to the datum of a nonunital lax monoidal structure on F^{op} , equivalent to a nonunital monoidal structure on $1_{\mathcal{D}^{\text{op}}} \downarrow F^{\text{op}}$ compatible with $\mathcal{C}^{\text{op}}$ and $\mathcal{D}^{\text{op}}$. Because this is the opposite category of $F \downarrow 1_{\mathcal{D}} \cong 1_{\mathcal{C}} \downarrow G$, this is then equivalent to (3).

2.2.2 Unital lax monoidal functors

We now consider monoidal functors between monoidal categories, that is we consider units.

2.2.13 Definition

Let $\mathcal{C}, \mathcal{D}$ be monoidal categories with units $1_{\mathcal{C}}, 1_{\mathcal{D}}$ respectively. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a nonunital lax monoidal functor with tensoror μ . A **left unit** for F is a morphism $\varepsilon: 1_{\mathcal{D}} \rightarrow F(1_{\mathcal{C}})$ in $\mathcal{D}$ such that $\mathcal{D}$'s left unitor $\lambda_{F(X)}: 1_{\mathcal{D}} \otimes F(X) \xrightarrow{\sim} F(X)$ is equal to

$$\lambda_{F(X)} = 1_{\mathcal{D}} \otimes F(X) \xrightarrow{\varepsilon \otimes \text{id}_{F(X)}} F(1_{\mathcal{C}}) \otimes F(X) \xrightarrow{\mu_{1_{\mathcal{C}}, X}} F(1_{\mathcal{C}} \otimes X) \xrightarrow{F(\lambda_X)} F(X),$$

where $\lambda_X: \mathbf{1}_{\mathcal{C}} \otimes X \rightarrow X$ is the left unitor of $\mathcal{C}$.

Similarly ε is a **right unit** for F if $\mathcal{D}$'s right unitor $\rho_{F(X)}: F(X) \otimes \mathbf{1}_{\mathcal{D}} \xrightarrow{\sim} F(X)$ is the composition

$$\rho_{F(X)} = F(X) \otimes \mathbf{1}_{\mathcal{D}} \xrightarrow{\text{id}_{F(X)} \otimes \varepsilon} F(X) \otimes F(\mathbf{1}_{\mathcal{C}}) \xrightarrow{\mu_{X, \mathbf{1}_{\mathcal{C}}}} F(X \otimes \mathbf{1}_{\mathcal{C}}) \xrightarrow{F(\rho_X)} F(X)$$

where $\rho_X: X \otimes \mathbf{1}_{\mathcal{C}} \rightarrow X$ is the right unitor of $\mathcal{C}$.

If ε is both a left and right unit, it is just called a **unit**.

It should not be surprising that the unit of a lax monoidal functor is unique.

2.2.14 Proposition

Suppose that $F: \mathcal{C} \rightarrow \mathcal{D}$ is a nonunital lax monoidal functor with left unit ε_L and right unit ε_R . Then $\varepsilon_L = \varepsilon_R$.

Proof

Consider the following diagram

$$\begin{array}{ccccc}
 \mathbf{1}_{\mathcal{D}} \otimes \mathbf{1}_{\mathcal{D}} & \xrightarrow{\text{id} \otimes \varepsilon_R} & \mathbf{1}_{\mathcal{D}} \otimes F(\mathbf{1}_{\mathcal{C}}) & \xrightarrow{\varepsilon_L \otimes \text{id}} & F(\mathbf{1}_{\mathcal{C}}) \otimes F(\mathbf{1}_{\mathcal{C}}) \\
 \downarrow \lambda_{\mathbf{1}_{\mathcal{D}}} & & \searrow \lambda_{F(\mathbf{1}_{\mathcal{C}})} & & \downarrow \mu_{\mathbf{1}_{\mathcal{C}}, \mathbf{1}_{\mathcal{C}}} \\
 & & & & F(\mathbf{1}_{\mathcal{C}} \otimes \mathbf{1}_{\mathcal{C}}) \\
 & & & & \downarrow F(\lambda_{\mathbf{1}_{\mathcal{C}}}) \\
 \mathbf{1}_{\mathcal{D}} & \xrightarrow{\varepsilon_R} & & & F(\mathbf{1}_{\mathcal{C}})
 \end{array}$$

The left square commutes by naturality of the left unitor, while the right square commutes because ε_L is a left unit for F . Recall that the units of $\mathcal{C}, \mathcal{D}$ are equal to $v_{\mathcal{C}} = \lambda_{\mathbf{1}_{\mathcal{C}}}$ and $v_{\mathcal{D}} = \lambda_{\mathbf{1}_{\mathcal{D}}}$ respectively. Thus the above commutative diagram tells us that

$$\varepsilon_R \circ v_{\mathcal{D}} = \mathbf{1}_{\mathcal{D}} \otimes \mathbf{1}_{\mathcal{D}} \xrightarrow{\varepsilon_L \otimes \varepsilon_R} F(\mathbf{1}_{\mathcal{C}}) \otimes F(\mathbf{1}_{\mathcal{C}}) \xrightarrow{\mu_{\mathbf{1}_{\mathcal{C}}, \mathbf{1}_{\mathcal{C}}}} F(\mathbf{1}_{\mathcal{C}} \otimes \mathbf{1}_{\mathcal{C}}) \xrightarrow{F(v_{\mathcal{C}})} F(\mathbf{1}_{\mathcal{C}}).$$

By symmetry, we see that $\varepsilon_L \circ v_{\mathcal{D}}$ is equal to this as well. Because $v_{\mathcal{D}}$ is an isomorphism, it follows that $\varepsilon_L = \varepsilon_R$. ■

2.2.15 Corollary

Let $\mathcal{C}, \mathcal{D}$ be monoidal categories and $F: \mathcal{C} \rightarrow \mathcal{D}$ a nonunital lax monoidal functor. Then F admits a unit $\varepsilon: \mathbf{1}_{\mathcal{D}} \rightarrow F(\mathbf{1}_{\mathcal{C}})$ iff it has both a left and right unit, and in such a case the unit is unique.

2.2.16 Proposition

Let $\mathcal{C}, \mathcal{D}$ be monoidal categories with respective units $\mathbf{1}_{\mathcal{C}}, \mathbf{1}_{\mathcal{D}}$. Let $G: \mathcal{D} \rightarrow \mathcal{C}$ be a functor with a nonunital lax monoidal structure, identified with a nonunital monoidal structure on the oriented fiber product $\mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D}$. Let $\varepsilon: \mathbf{1}_{\mathcal{C}} \rightarrow G(\mathbf{1}_{\mathcal{D}})$ be a morphism in $\mathcal{C}$, and let $\mathbf{1} = (\mathbf{1}_{\mathcal{C}}, \mathbf{1}_{\mathcal{D}}, \varepsilon)$ be an object of $\mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D}$. Then

- (1) ε is a left unit for G iff for every object $(C, D, \eta) \in \mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D}$, the left unitors $\lambda_C: \mathbf{1}_{\mathcal{C}} \otimes C \cong C$ and $\lambda_D: \mathbf{1}_{\mathcal{D}} \otimes D \cong D$ determine an isomorphism $(\lambda_C, \lambda_D): \mathbf{1} \otimes (C, D, \eta) \cong (C, D, \eta)$ in $\mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D}$.
- (2) Dually, ε is a right unit for G iff for every object $(C, D, \eta) \in \mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D}$, the right unitors $\rho_C: \mathbf{1}_{\mathcal{C}} \otimes C \cong C$ and $\rho_D: \mathbf{1}_{\mathcal{D}} \otimes D \cong D$ determine an isomorphism $(\rho_C, \rho_D): \mathbf{1} \otimes (C, D, \eta) \cong (C, D, \eta)$ in $\mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D}$.

Proof

The proof of (2) is similar to the proof of (1). Note that (λ_C, λ_D) forms a morphism $\mathbf{1} \otimes (C, D, \eta) \rightarrow (C, D, \eta)$ iff the outer rectangle in the following diagram commutes

$$\begin{array}{ccc}
 \mathbf{1}_{\mathcal{C}} \otimes C & \xrightarrow{\lambda_C} & C \\
 \text{id} \otimes \eta \downarrow & & \downarrow \eta \\
 \mathbf{1}_{\mathcal{C}} \otimes G(D) & \xrightarrow{\lambda_{GD}} & G(D) \\
 \varepsilon \otimes \text{id} \downarrow & & \downarrow \text{id} \\
 G(\mathbf{1}_{\mathcal{D}}) \otimes G(D) & & \\
 \mu \downarrow & & \\
 G(\mathbf{1}_{\mathcal{D}} \otimes D) & \xrightarrow{G(\lambda_D)} & G(D)
 \end{array}$$

In this diagram, the upper square commutes by λ 's naturality. The lower rectangle commutes because ε is a left unit for G .

Conversely if this outer rectangle commutes then specializing to $C = G(D)$ and $\eta = \text{id}$ shows that the lower rectangle commutes. This is true for all D so that ε is a left unit for G .

Note that (λ_C, λ_D) is a morphism iff $(\lambda_C^{-1}, \lambda_D^{-1})$ is, so that it is an isomorphism as desired. $\blacksquare$

2.2.17 Corollary

Let $\mathcal{C}, \mathcal{D}$ be monoidal categories with respective units $(\mathbf{1}_{\mathcal{C}}, v_{\mathcal{C}})$ and $(\mathbf{1}_{\mathcal{D}}, v_{\mathcal{D}})$. Let $G: \mathcal{D} \rightarrow \mathcal{C}$ be a nonunital lax monoidal functor and $\varepsilon: \mathbf{1}_{\mathcal{C}} \rightarrow G(\mathbf{1}_{\mathcal{D}})$ a morphism in $\mathcal{C}$. Let $\mathbf{1} = (\mathbf{1}_{\mathcal{C}}, \mathbf{1}_{\mathcal{D}}, \varepsilon)$ be an object of $\mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D}$. Then the following are equivalent:

- (1) ε is a unit for G ,
- (2) The pair $v = (v_{\mathcal{C}}, v_{\mathcal{D}})$ is a morphism $\mathbf{1} \otimes \mathbf{1} \rightarrow \mathbf{1}$ in $\mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D}$, and $(\mathbf{1}, v)$ is a unit relative to $\otimes_{\mu}$.

Proof

Assume (1), then the previous proposition demonstrated that the functors

$$\mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D} \rightarrow \mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D}, \quad X \mapsto \mathbf{1} \otimes X, \quad X \mapsto X \otimes \mathbf{1}$$

are naturally isomorphic to the identity (via the left and right unitors), and thus are fully faithful. Thus to show (2) it is sufficient to show that $v = (v_{\mathcal{C}}, v_{\mathcal{D}})$ is a morphism $\mathbf{1} \otimes \mathbf{1} \rightarrow \mathbf{1}$ (because then it is an isomorphism). This is again due to the previous proposition as $v_{\mathcal{C}} = \lambda_{\mathbf{1}_{\mathcal{C}}}$ and $v_{\mathcal{D}} = \lambda_{\mathbf{1}_{\mathcal{D}}}$.

Now assume (2) so that $\mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D}$ is a monoidal category with unit $(\mathbf{1}, v)$. Because the forgetful functor $U: \mathcal{C} \vec{\times}_{\mathcal{C}} \mathcal{D} \rightarrow \mathcal{C} \times \mathcal{D}$ is strict monoidal, it carries the left and right unitors of the oriented fiber product to

the left and right unitors of $\mathcal{C}, \mathcal{D}$. Indeed, this is immediate because $v = (v_{\mathcal{C}}, v_{\mathcal{D}})$ is a unit for $\otimes_{\mu}$ and by the definition of the left/right unitors.

This means that $(\lambda_{\mathcal{C}}, \lambda_{\mathcal{D}})$ and $(\rho_{\mathcal{C}}, \rho_{\mathcal{D}})$ are isomorphisms between the necessary objects, so by the above proposition ε is both a left and right unit for G . ■

2.2.18 Definition

Let $\mathcal{C}, \mathcal{D}$ be monoidal categories. A **lax monoidal functor** $F: \mathcal{C} \rightarrow \mathcal{D}$ is a nonunital lax monoidal functor for which there exists a unit $\varepsilon: \mathbf{1}_{\mathcal{D}} \rightarrow F(\mathbf{1}_{\mathcal{C}})$.

2.2.19 Corollary

Let $\mathcal{C}, \mathcal{D}$ be monoidal categories and $G: \mathcal{D} \rightarrow \mathcal{C}$ a functor. Then there is a bijection between lax monoidal structures on G and monoidal structures on $\mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{C}} \mathcal{D}$ such that the forgetful functor is strictly monoidal. This is given by $\mu \mapsto \otimes_{\mu}$.

2.2.20 Definition (Colax monoidal functors)

Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a functor between monoidal categories. It is a **colax monoidal functor** if its opposite $F^{\text{op}}: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$ is a lax monoidal functor. That is F is a nonunital colax monoidal functor with a **counit** $\varepsilon: F(\mathbf{1}_{\mathcal{C}}) \rightarrow \mathbf{1}_{\mathcal{D}}$ such that the left and right unitors of $F(X)$ are

$$\begin{aligned}\lambda_{FX} &= F(X) \xrightarrow{F(\lambda_X)} F(\mathbf{1}_{\mathcal{C}} \otimes X) \xrightarrow{\mu_{\mathbf{1}_{\mathcal{C}}, X}} F(\mathbf{1}_{\mathcal{C}}) \otimes F(X) \xrightarrow{\varepsilon \otimes \text{id}} \mathbf{1}_{\mathcal{D}} \otimes F(X) \\ \rho_{FX} &= F(X) \xrightarrow{F(\rho_X)} F(X \otimes \mathbf{1}_{\mathcal{C}}) \xrightarrow{\mu_{X, \mathbf{1}_{\mathcal{C}}}} F(X) \otimes F(\mathbf{1}_{\mathcal{C}}) \xrightarrow{\text{id} \otimes \varepsilon} F(X) \otimes \mathbf{1}_{\mathcal{D}}.\end{aligned}$$

2.2.21 Corollary

Suppose that $\mathcal{C}, \mathcal{D}$ are monoidal categories and $F: \mathcal{C} \rightleftarrows \mathcal{D}: G$ forms an adjunction, giving an isomorphism $\mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{C}} \mathcal{D} \cong \mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{D}} \mathcal{D}$. Then the following are equivalent:

- (1) The datum of a lax monoidal structure on $G: \mathcal{D} \rightarrow \mathcal{C}$,
- (2) The datum of a colax monoidal structure on $F: \mathcal{C} \rightarrow \mathcal{D}$,
- (3) The datum of a monoidal structure on $\mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{C}} \mathcal{D} \cong \mathcal{C} \overset{\rightarrow}{\times}_{\mathcal{D}} \mathcal{D}$ which is compatible with the monoidal structures of $\mathcal{C}, \mathcal{D}$.

We can give a more direct equivalent characterization of the unit of a lax monoidal functor in terms of the units of $\mathcal{C}, \mathcal{D}$, instead of the left and right unitors.

2.2.22 Proposition

Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a nonunital lax monoidal functor between monoidal categories with units $\mathbf{1}_{\mathcal{C}}, \mathbf{1}_{\mathcal{D}}$, and let $\varepsilon: \mathbf{1}_{\mathcal{D}} \rightarrow F(\mathbf{1}_{\mathcal{C}})$ be a morphism in $\mathcal{D}$. Then ε is a left unit for F iff it satisfies the following two conditions:

- (1) The following diagram commutes in $\mathcal{D}$:

$$\begin{array}{ccc}
\mathbf{1}_{\mathcal{D}} \otimes \mathbf{1}_{\mathcal{D}} & \xrightarrow{\varepsilon \otimes \varepsilon} & F(\mathbf{1}_{\mathcal{C}}) \otimes F(\mathbf{1}_{\mathcal{C}}) \\
\downarrow v_{\mathcal{D}} & & \downarrow \mu_{\mathbf{1}_{\mathcal{C}}, \mathbf{1}_{\mathcal{C}}} \\
\mathbf{1}_{\mathcal{D}} & \xrightarrow{\varepsilon} & F(\mathbf{1}_{\mathcal{C}} \otimes \mathbf{1}_{\mathcal{C}}) \\
& & \downarrow F(v_{\mathcal{C}}) \\
& & F(\mathbf{1}_{\mathcal{C}})
\end{array}$$

(2) For all $X \in \mathcal{C}$ the following composite is a monomorphism in $\mathcal{D}$:

$$\mathbf{1}_{\mathcal{D}} \otimes F(X) \xrightarrow{\varepsilon \otimes \text{id}_{FX}} F(\mathbf{1}_{\mathcal{C}}) \otimes F(X) \xrightarrow{\mu_{\mathbf{1}_{\mathcal{C}}, X}} F(\mathbf{1}_{\mathcal{C}} \otimes X).$$

Moreover if these conditions are satisfied then the composite in (2) is an isomorphism.

Proof

Let us write $(\mathbf{1}, v)$ for the units of both $\mathcal{C}$ and $\mathcal{D}$, and λ for the left unitor of both $\mathcal{C}$ and $\mathcal{D}$. If ε is a left unit then consider the following diagram:

$$\begin{array}{ccccc}
\mathbf{1} \otimes \mathbf{1} & \xrightarrow{\text{id} \otimes \varepsilon} & \mathbf{1} \otimes F(\mathbf{1}) & \xrightarrow{\varepsilon \otimes \text{id}} & F(\mathbf{1}) \otimes F(\mathbf{1}) \\
\downarrow \lambda_1 & & \searrow \lambda_{F\mathbf{1}} & & \downarrow \mu_{\mathbf{1}, \mathbf{1}} \\
& & & & F(\mathbf{1} \otimes \mathbf{1}) \\
& & & & \downarrow F(v) \\
\mathbf{1} & \xrightarrow{\varepsilon} & & & F(\mathbf{1})
\end{array}$$

This diagram commutes: the left square commutes by naturality of the left unitors, the right square commutes because ε is a left unit of F . Thus the outer rectangle commutes showing (1).

Now for $X \in \mathcal{C}$, the left unitor λ_{FX} is

$$\lambda_{FX} = \mathbf{1} \otimes F(X) \xrightarrow{\varepsilon \otimes \text{id}} F(\mathbf{1}) \otimes F(X) \xrightarrow{\mu_{\mathbf{1}, X}} F(\mathbf{1} \otimes X) \xrightarrow{F(\lambda_X)} F(X).$$

This is an isomorphism and since $F(\lambda_X)$ is an isomorphism as well, so too is $\mu_{\mathbf{1}, X} \circ (\varepsilon \otimes \text{id})$, showing (2).

Now suppose ε satisfies (1) and (2), we want to show that it is the left unit of F . Let $f: \mathbf{1} \otimes F(X) \rightarrow F(X)$ be

$$f = \mathbf{1} \otimes F(X) \xrightarrow{\varepsilon \otimes \text{id}} F(\mathbf{1}) \otimes F(X) \xrightarrow{\mu_{\mathbf{1}, X}} F(\mathbf{1} \otimes X) \xrightarrow{F(\lambda_X)} F(X),$$

we want to show that it is λ_{FX} . By definition of the left unitor, this is equivalent to saying that $\text{id}_{\mathbf{1}} \otimes f$ is equal to

$$\text{id}_{\mathbf{1}} \otimes f = \mathbf{1} \otimes (\mathbf{1} \otimes F(X)) \xrightarrow{\alpha_{\mathbf{1}, \mathbf{1}, X}} (\mathbf{1} \otimes \mathbf{1}) \otimes F(X) \xrightarrow{v \otimes \text{id}} \mathbf{1} \otimes F(X).$$

By virtue of (2), it is sufficient to show that both sides are equal after postcomposition with the monomorphism

$$\mathbf{1} \otimes F(X) \xrightarrow{\varepsilon \otimes \text{id}} F(\mathbf{1}) \otimes F(X) \xrightarrow{\mu_{\mathbf{1}, X}} F(\mathbf{1} \otimes X).$$

This can be shown by filling in the necessary commutative diagram. ■

2.2.23 Example

When $\mathcal{C} = \{e\}$ is the trivial monoid, then we can identify nonunital lax monoidal functors $F: \mathcal{C} \rightarrow \mathcal{D}$ with a nonunital monoid object $A \in \mathcal{D}$. Then the above proposition shows that $\varepsilon: \mathbf{1}_{\mathcal{D}} \rightarrow A$ is a left unit iff the diagram

$$\begin{array}{ccc} \mathbf{1} \otimes \mathbf{1} & \xrightarrow{\varepsilon \otimes \varepsilon} & A \otimes A \\ v \downarrow & & \downarrow m \\ \mathbf{1} & \xrightarrow{\varepsilon} & A \end{array}$$

is commutative (i.e. ε is idempotent) and the map

$$\mathbf{1} \otimes A \xrightarrow{\varepsilon \otimes \text{id}} A \otimes A \xrightarrow{m} A$$

is monic (i.e. ε is left-cancelative).

If $\mathcal{D}$ is the category of sets, this means that if M is a nonunital monoid (in the classical sense) then $e \in M$ is a left unit iff it is idempotent ($e^2 = e$) and left multiplication with e is monic ($e \cdot x = e \cdot y$ implies $x = y$).

2.2.24 Example

Let Mod_R be the category of R -modules for a commutative ring R . Consider then the forgetful functor $U: \text{Mod}_R \rightarrow \text{Set}$ equipped with the nonunital lax monoidal structure given by the canonical maps $b_{M,N}: M \times N \rightarrow M \otimes N$. The function $\varepsilon: \{*\} \rightarrow U(R)$ mapping $*$ to $1 \in R$ is a unit for U .

This is clear from the above proposition.

2.2.25 Example

Let $\mathcal{C}$ be a monoidal category and let $h: \mathcal{C} \rightarrow \text{Set}$ be the functor corepresented by $\mathbf{1}$: $h = \text{hom}_{\mathcal{C}}(\mathbf{1}, \bullet)$. Then for every $X, Y \in \mathcal{C}$ we have a canonical map

$$\mu_{X,Y}: h(X) \times h(Y) \rightarrow h(X \otimes Y)$$

which maps $x \in h(X)$ and $y \in h(Y)$ to the composite

$$h(x, y) = \mathbf{1} \xrightarrow{v^{-1}} \mathbf{1} \otimes \mathbf{1} \xrightarrow{x \otimes y} X \otimes Y.$$

This defines a lax monoidal structure on h with a unit given by

$$\varepsilon: \{*\} \rightarrow h(\mathbf{1}) = \text{hom}_{\mathcal{C}}(\mathbf{1}, \mathbf{1}), \quad \varepsilon(*) = \text{id}_{\mathbf{1}}.$$

2.2.26 Definition (Monoidal natural transformations)

Let $F, F': \mathcal{C} \rightarrow \mathcal{D}$ be lax monoidal functors. A natural transformation $\gamma: F \Rightarrow F'$ is **monoidal** if it satisfies the following conditions:

- (1) It is monoidal in the sense of a nonunital lax monoidal functors. Explicitly the following diagram

commutes

$$\begin{array}{ccc}
 F(X) \otimes F(Y) & \xrightarrow{\mu_{X,Y}} & F(X \otimes Y) \\
 \gamma_X \otimes \gamma_Y \downarrow & & \downarrow \gamma_{X \otimes Y} \\
 F'(X) \otimes F'(Y) & \xrightarrow{\mu'_{X,Y}} & F'(X \otimes Y)
 \end{array}$$

(2) The unit of F' is the composition

$$\varepsilon' = \mathbf{1}_{\mathcal{D}} \xrightarrow{\varepsilon} F(\mathbf{1}_{\mathcal{C}}) \xrightarrow{\gamma_{\mathbf{1}_{\mathcal{C}}}} F'(\mathbf{1}_{\mathcal{C}})$$

where ε is F 's unit.

Let $\text{Fun}^{\text{lax}}(\mathcal{C}, \mathcal{D})$ be the category of lax monoidal functors $\mathcal{C} \rightarrow \mathcal{D}$ whose morphisms are monoidal natural transformations. This is a (non-full) subcategory of $\text{Fun}_{\text{nu}}^{\text{lax}}(\mathcal{C}, \mathcal{D})$.

Note:

Let $\mathcal{C}, \mathcal{D}$ be monoidal categories and $F: \mathcal{C} \rightarrow \mathcal{D}$ a functor. Then F is a lax monoidal functor iff F^{rev} is. Similarly γ is a monoidal natural transformation iff γ^{rev} is. Thus we obtain an isomorphism of categories $\text{Fun}^{\text{lax}}(\mathcal{C}, \mathcal{D}) \cong \text{Fun}^{\text{lax}}(\mathcal{C}^{\text{rev}}, \mathcal{D}^{\text{rev}})$.

Recall that we can compose nonunital lax monoidal functors. Explicitly if we have nonunital lax monoidal functors $\mathcal{C} \xrightarrow{F} \mathcal{D} \xrightarrow{G} \mathcal{E}$ with tensorors μ, ν , then their composition has a nonunital lax monoidal structure given by the whiskering $\xi = G\mu \circ \nu_{F \times F}$:

$$\xi_{X,Y} = (G \circ F)(X) \otimes (G \circ F)(Y) \xrightarrow{\nu_{FX,FY}} G(F(X) \otimes F(Y)) \xrightarrow{G\mu_{X,Y}} (G \circ F)(X \otimes Y).$$

If F, G have units

$$\delta: \mathbf{1}_{\mathcal{D}} \rightarrow F(\mathbf{1}_{\mathcal{C}}), \quad \varepsilon: \mathbf{1}_{\mathcal{E}} \rightarrow G(\mathbf{1}_{\mathcal{D}})$$

then the composite

$$\mathbf{1}_{\mathcal{E}} \xrightarrow{\varepsilon} G(\mathbf{1}_{\mathcal{D}}) \xrightarrow{G\delta} (G \circ F)(\mathbf{1}_{\mathcal{C}})$$

is a unit for $G \circ F$.

Thus composition restricts to a functor

$$\circ: \text{Fun}^{\text{lax}}(\mathcal{D}, \mathcal{E}) \times \text{Fun}^{\text{lax}}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}^{\text{lax}}(\mathcal{C}, \mathcal{E}).$$

2.2.27 Example

Let $\mathcal{C}$ be a monoidal category. A **monoid object** in $\mathcal{C}$ is an object $M \in \mathcal{C}$ equipped with *multiplication* $m: A \otimes A \rightarrow A$ and a *unit* $\varepsilon: \mathbf{1} \rightarrow A$ such that the following diagrams commute

$$\begin{array}{ccc}
 M \otimes (M \otimes M) & \xrightarrow{\text{id} \otimes m} & M \otimes M \\
 \alpha \downarrow & & \downarrow m \\
 (M \otimes M) \otimes M & \xrightarrow{m \otimes \text{id}} & M \otimes M \\
 & & \uparrow m \\
 & & M
 \end{array}
 \qquad
 \begin{array}{ccccc}
 \mathbf{1} \otimes M & \xrightarrow{\varepsilon \otimes \text{id}} & M \otimes M & \xleftarrow{\text{id} \otimes \varepsilon} & M \otimes \mathbf{1} \\
 & \searrow \lambda & \downarrow m & \swarrow \rho & \\
 & & M & &
 \end{array}$$

If M, M' are monoid objects of $\mathcal{C}$ with multiplications m, m' and units $\varepsilon, \varepsilon'$, then a morphism $f: M \rightarrow M'$ is a **monoid homomorphism** if the following diagram commutes:

$$\begin{array}{ccc} M \otimes M & \xrightarrow{m} & M \\ f \otimes f \downarrow & & \downarrow f \\ M' \otimes M' & \xrightarrow{m'} & M' \end{array}$$

and f carries the unit of M to the unit of M' : $\varepsilon' = f \circ \varepsilon$.

It is clear that the monoid objects of $\mathcal{C}$ form a category, denoted $\text{Mon}(\mathcal{C})$ (this is a non-full subcategory of $\text{Mon}^{\text{nu}}(\mathcal{C})$).

If we let $\{e\}$ denote the trivial monoid, then monoid objects of $\mathcal{C}$ can be identified with lax monoidal functors $\{e\} \rightarrow \mathcal{C}$. Thus we obtain an isomorphism of categories $\text{Fun}^{\text{lax}}(\{e\}, \mathcal{C}) \cong \text{Mon}(\mathcal{C})$.

2.2.3 Unital monoidal functors

2.2.28 Definition (Monoidal functors)

A functor $F: \mathcal{C} \rightarrow \mathcal{D}$ between monoidal categories is a **monoidal functor** if it is a lax monoidal functor and

- (1) its tensoror is an isomorphism: $\mu_{X,Y}: F(X) \otimes F(Y) \rightarrow F(X \otimes Y)$ is an isomorphism for all X, Y ;
- (2) its unit $\varepsilon: \mathbf{1}_{\mathcal{D}} \rightarrow F(\mathbf{1}_{\mathcal{C}})$ is an isomorphism.

Note:

Note that a monoidal functor is both a nonunital monoidal functor (i.e. μ is an isomorphism) and a lax monoidal functor (i.e. has a unit). But the converse is not true: a functor which is both a nonunital monoidal functor and a lax monoidal functor is a monoidal functor iff its unit is an isomorphism.

Note:

Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a nonunital monoidal functor and $\varepsilon: \mathbf{1}_{\mathcal{D}} \rightarrow F(\mathbf{1}_{\mathcal{C}})$ be an isomorphism in $\mathcal{D}$. Then ε satisfies condition (2) in proposition 2.2.22: both the maps in

$$\mathbf{1}_{\mathcal{D}} \otimes F(X) \xrightarrow{\varepsilon \otimes \text{id}} F(\mathbf{1}_{\mathcal{C}}) \otimes F(X) \xrightarrow{\mu} F(\mathbf{1}_{\mathcal{C}} \otimes X)$$

are isomorphisms. Thus ε is a unit of F iff condition (1) holds: the following diagram commutes

$$\begin{array}{ccc} \mathbf{1}_{\mathcal{D}} \otimes \mathbf{1}_{\mathcal{D}} & \xrightarrow{\varepsilon \otimes \varepsilon} & F(\mathbf{1}_{\mathcal{C}}) \otimes F(\mathbf{1}_{\mathcal{C}}) \\ \downarrow v_{\mathcal{D}} & & \downarrow \mu_{\mathbf{1}_{\mathcal{C}}, \mathbf{1}_{\mathcal{C}}} \\ & & F(\mathbf{1}_{\mathcal{C}} \otimes \mathbf{1}_{\mathcal{C}}) \\ & & \downarrow F(v_{\mathcal{C}}) \\ \mathbf{1}_{\mathcal{D}} & \xrightarrow{\varepsilon} & F(\mathbf{1}_{\mathcal{C}}) \end{array}$$

This is iff $(F(\mathbf{1}_{\mathcal{C}}), F(v_{\mathcal{C}}) \circ \mu_{\mathbf{1}_{\mathcal{C}}, \mathbf{1}_{\mathcal{C}}})$ is a unit of $\mathcal{C}$. Thus a nonunital monoidal functor $F: \mathcal{C} \rightarrow \mathcal{D}$ is monoidal

iff the functors

$$\mathcal{D} \rightarrow \mathcal{D}, \quad X \mapsto F(\mathbf{1}_{\mathcal{C}}) \otimes X, \quad X \mapsto X \otimes F(\mathbf{1}_{\mathcal{C}})$$

are fully faithful (and then naturally isomorphic to the identity functor).

2.2.29 Definition

If $F, F': \mathcal{C} \rightarrow \mathcal{D}$ are monoidal functors, a natural transformation $\gamma: F \Rightarrow F'$ is **monoidal** if it is monoidal in the sense of a monoidal natural transformation between unital lax monoidal functors (definition 2.2.26).

For monoidal categories $\mathcal{C}, \mathcal{D}$ we let $\text{Fun}^{\otimes}(\mathcal{C}, \mathcal{D})$ be the full subcategory of $\text{Fun}^{\text{lax}}(\mathcal{C}, \mathcal{D})$ consisting of monoidal functors $\mathcal{C} \rightarrow \mathcal{D}$ with monoidal natural transformations between them.

2.2.30 Example

As before, we have isomorphisms

$$\begin{aligned} \text{Fun}^{\otimes}(\mathcal{C}, \mathcal{D}) &\cong \text{Fun}^{\otimes}(\mathcal{C}^{\text{rev}}, \mathcal{D}^{\text{rev}}), & F &\mapsto F^{\text{rev}} \\ \text{Fun}^{\otimes}(\mathcal{C}, \mathcal{D})^{\text{op}} &\cong \text{Fun}^{\otimes}(\mathcal{C}^{\text{op}}, \mathcal{D}^{\text{op}}), & F &\mapsto F^{\text{op}}. \end{aligned}$$

I.e. the opposite and reverse functors are monoidal iff F is.

2.2.31 Example

If F, G are monoidal functors so too is their composite $G \circ F$. This is because the tensoror $G\mu \circ \nu_{F \times F}$ is the composition of isomorphisms and thus an isomorphism, as is the unit $G(\delta) \circ \varepsilon$.

2.2.32 Example

Suppose $\mathcal{C}, \mathcal{D}$ are categories with finite products, which we view as a monoidal structure. Any functor $F: \mathcal{C} \rightarrow \mathcal{D}$ is lax monoidal as there is a canonical map $F(X \times Y) \rightarrow F(X) \times F(Y)$ which forms its tensoror, and there is as well a unique morphism $F(\mathbf{1}_{\mathcal{C}}) \rightarrow \mathbf{1}_{\mathcal{D}}$. This is a monoidal functor iff F preserves limits; this is the same as saying that the canonical maps $F(X \times Y) \rightarrow F(X) \times F(Y)$ and $F(\mathbf{1}_{\mathcal{C}}) \rightarrow \mathbf{1}_{\mathcal{D}}$ are isomorphisms.

2.3 Enriched categories

2.3.1 Definition ($\mathcal{A}$ -enriched category)

Let $\mathcal{A}$ be a monoidal category with unit $\mathbf{1}$. An **$\mathcal{A}$ -enriched category** $\mathcal{C}$ consists of the following data:

- (1) A collection $\text{Ob}(\mathcal{C})$, whose elements we refer to as **objects** of $\mathcal{C}$. We abuse notation by writing $X \in \mathcal{C}$ instead of $X \in \text{Ob}(\mathcal{C})$.
- (2) For every $X, Y \in \mathcal{C}$ an object of $\mathcal{A}$, $\underline{\text{hom}}_{\mathcal{C}}(X, Y)$.
- (3) For every $X, Y, Z \in \mathcal{C}$ an $\mathcal{A}$ -morphism

$$c_{Z,Y,X}: \underline{\text{hom}}_{\mathcal{C}}(Y, Z) \otimes \underline{\text{hom}}_{\mathcal{C}}(X, Y) \rightarrow \underline{\text{hom}}_{\mathcal{C}}(X, Z)$$

which we call **composition**.

- (4) For every $X \in \mathcal{C}$ a morphism $e_X: \mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{C}}(X, X)$ called the **identity** of X .

The data must satisfy the following conditions:

- (1) Associativity: for $W, X, Y, Z \in \mathcal{C}$:

$$\begin{array}{ccc}
 \underline{\text{hom}}(Y, Z) \otimes (\underline{\text{hom}}(X, Y) \otimes \underline{\text{hom}}(W, X)) & \xrightarrow{\text{id} \otimes c_{Y,X,W}} & \underline{\text{hom}}(Y, Z) \otimes \underline{\text{hom}}(W, Y) \\
 \downarrow \alpha & & \downarrow c_{Z,Y,W} \\
 & & \underline{\text{hom}}(W, Z) \\
 & & \uparrow c_{Z,X,W} \\
 (\underline{\text{hom}}(Y, Z) \otimes \underline{\text{hom}}(X, Y)) \otimes \underline{\text{hom}}(W, X) & \xrightarrow{c_{Z,Y,X} \otimes \text{id}} & \underline{\text{hom}}(X, Z) \otimes \underline{\text{hom}}(W, X)
 \end{array}$$

Here α is the associator of $\mathcal{A}$.

- (2) Unit: for every $X, Y \in \mathcal{C}$ the two diagrams below must commute:

$$\begin{array}{ccc}
 \mathbf{1} \otimes \underline{\text{hom}}(X, Y) & \xrightarrow{e_Y \otimes \text{id}} & \underline{\text{hom}}(Y, Y) \otimes \underline{\text{hom}}(X, Y) \\
 & \searrow \lambda & \downarrow c_{Y,Y,X} \\
 & & \underline{\text{hom}}(X, Y)
 \end{array}$$

$$\begin{array}{ccc}
 \underline{\text{hom}}(X, Y) \otimes \mathbf{1} & \xrightarrow{\text{id} \otimes e_X} & \underline{\text{hom}}(X, Y) \otimes \underline{\text{hom}}(X, X) \\
 & \searrow \rho & \downarrow c_{Y,X,X} \\
 & & \underline{\text{hom}}(X, Y)
 \end{array}$$

Here λ, ρ are the unitors of $\mathcal{A}$.

2.3.2 Example

Set-enriched categories are precisely the locally small categories.

2.3.3 Example

Let $\mathcal{A}$ be a monoidal category and $\mathcal{C}$ an $\mathcal{A}$ -enriched category. For every object $X \in \mathcal{C}$, the composition law and unit

$$c_{X,X,X}: \underline{\text{hom}}(X, X) \otimes \underline{\text{hom}}(X, X) \rightarrow \underline{\text{hom}}(X, X), \quad e_X: \mathbf{1} \rightarrow \underline{\text{hom}}(X, X)$$

turn $\underline{\text{hom}}(X, X)$ into a monoid object of $\mathcal{A}$. This induces a bijection between $\text{Mon}(\mathcal{A})$ and $\mathcal{A}$ -enriched categories with a single object.

Thus enriched category theory can be seen as a generalization of the theory of monoid objects.

2.3.4 Remark (Functorality)

Let $\mathcal{A}, \mathcal{A}'$ be monoidal categories and $F: \mathcal{A} \rightarrow \mathcal{A}'$ a lax monoidal functor, with tensoror μ and unit ε . Then every $\mathcal{A}$ -enriched category $\mathcal{C}$ defines an $\mathcal{A}'$ -enriched category $\mathcal{C}'$ as follows:

- (1) The objects of $\mathcal{C}'$ are the same as those of $\mathcal{C}$: $\text{Ob}(\mathcal{C}') = \text{Ob}(\mathcal{C})$.
- (2) For $X, Y \in \mathcal{C}'$ set $\underline{\text{hom}}_{\mathcal{C}'}(X, Y) = F(\underline{\text{hom}}_{\mathcal{C}}(X, Y))$.
- (3) For $X, Y, Z \in \mathcal{C}'$ define the composition law $c'_{X,Y,Z}$ in $\mathcal{C}'$ as the composition

$$\begin{aligned} \underline{\text{hom}}_{\mathcal{C}'}(Y, Z) \otimes \underline{\text{hom}}_{\mathcal{C}'}(X, Y) &= F(\underline{\text{hom}}_{\mathcal{C}}(Y, Z)) \otimes F(\underline{\text{hom}}_{\mathcal{C}}(X, Y)) \\ &\xrightarrow{\mu} F(\underline{\text{hom}}_{\mathcal{C}}(Y, Z) \otimes \underline{\text{hom}}_{\mathcal{C}}(X, Y)) \\ &\xrightarrow{F c_{Z,Y,X}} F(\underline{\text{hom}}_{\mathcal{C}}(X, Z)) \\ &= \underline{\text{hom}}_{\mathcal{C}'}(X, Z) \end{aligned}$$

- (4) For every $X \in \mathcal{C}'$ the identity e'_X is the composition

$$\mathbf{1}_{\mathcal{A}'} \xrightarrow{\varepsilon} F(\mathbf{1}_{\mathcal{A}}) \xrightarrow{F(e_X)} F(\underline{\text{hom}}_{\mathcal{C}}(X, X)) = \underline{\text{hom}}_{\mathcal{C}'}(X, X).$$

2.3.5 Definition (The underlying category)

Let $\mathcal{A}$ be a monoidal category and $\mathcal{C}$ an $\mathcal{A}$ -enriched category. Consider then the monoidal functor $F: \mathcal{A} \rightarrow \text{Set}$ given by $F = \text{hom}_{\mathcal{A}}(\mathbf{1}, \bullet)$ (example 2.2.25). By the above remark, this gives rise to a **Set**-enriched category, which we can identify with an ordinary category. We call this the **underlying category** of $\mathcal{C}$, and we abuse notation and denote it by $\mathcal{C}$.

Concretely, this category has the same objects as $\mathcal{C}$, and its morphisms are

$$\text{hom}_{\mathcal{C}}(X, Y) = \text{hom}_{\mathcal{A}}(\mathbf{1}, \underline{\text{hom}}_{\mathcal{C}}(X, Y)).$$

Note:

Let $\mathcal{A}$ be a monoidal category and $\mathcal{C}$ an ordinary category. An $\mathcal{A}$ -**enrichment** of $\mathcal{C}$ is an $\mathcal{A}$ -enriched category $\hat{\mathcal{C}}$ such that the underlying category of $\hat{\mathcal{C}}$ is isomorphic to $\mathcal{C}$.

2.3.6 Example

Let **Top** denote the category of topological spaces, endowed with the monoidal structure given by the cartesian product. A **Top**-enriched category is called a **topologically enriched category**. Note that the corepresentable functor $\text{hom}_{\text{Top}}(\mathbf{1}, \bullet)$ (where $\mathbf{1} = \{*\}$ is the unit/terminal object of **Top**) is naturally isomorphic to the forgetful functor $\text{Top} \rightarrow \text{Set}$.

Thus if $\mathcal{C}$ is a topologically enriched category, its underlying category $\mathcal{C}_0$ can be described as follows:

- (1) Its objects are the same as the objects of $\mathcal{C}$.
- (2) For $X, Y \in \mathcal{C}_0$, morphisms $X \rightarrow Y$ in $\mathcal{C}_0$ are the same as points in the space $\underline{\text{hom}}_{\mathcal{C}}(X, Y)$.
- (3) Given $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ in $\mathcal{C}_0$, the composition $g \circ f$ is the image of (g, f) under the continuous map

$$c_{Z,Y,X}: \underline{\text{hom}}_{\mathcal{C}}(Y, Z) \times \underline{\text{hom}}_{\mathcal{C}}(X, Y) \rightarrow \underline{\text{hom}}_{\mathcal{C}}(X, Z).$$

- (4) For $X \in \mathcal{C}_0$, id_X is the image of $*$ in $\{*\}$ under $e_X: \{*\} \rightarrow \underline{\text{hom}}_{\mathcal{C}}(X, X)$.

Conversely, enriching an ordinary category $\mathcal{C}_0$ over $\mathbf{Top}$ is the same as endowing each hom-set $\text{hom}_{\mathcal{C}_0}(X, Y)$ with a topology such that composition is continuous $\text{hom}_{\mathcal{C}_0}(Y, Z) \times \text{hom}_{\mathcal{C}_0}(X, Y) \rightarrow \text{hom}_{\mathcal{C}_0}(X, Z)$.

Note:

The identities of an enriched category are unique. That is, if $\mathcal{C}, \mathcal{C}'$ are two $\mathcal{A}$ -enriched categories with the same objects, morphisms, and compositions, then $e_X = e'_X$ for all $X \in \mathcal{C}$. Indeed, e_X is simply the unit of the $\mathcal{A}$ -monoid $\underline{\text{hom}}_{\mathcal{C}}(X, X)$ (with $c_{X,X,X}$ as multiplication) which we showed is unique.

2.3.7 Definition (Enriched functors)

Let $\mathcal{A}$ be a monoidal category and $\mathcal{C}, \mathcal{D}$ be $\mathcal{A}$ -enriched categories. An $\mathcal{A}$ -enriched functor $F: \mathcal{C} \rightarrow \mathcal{D}$ consists of the following data:

- (1) For every $X \in \mathcal{C}$ an associated object $F(X) \in \mathcal{D}$.
- (2) For every $X, Y \in \mathcal{C}$ a morphism in $\mathcal{A}$

$$F_{X,Y}: \underline{\text{hom}}_{\mathcal{C}}(X, Y) \rightarrow \underline{\text{hom}}_{\mathcal{D}}(F(X), F(Y)).$$

This data is subject to the following conditions:

- (1) For every $X \in \mathcal{C}$, the morphism $e_{FX}: \mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{D}}(X, X)$ factors as

$$\mathbf{1} \xrightarrow{e_X} \underline{\text{hom}}_{\mathcal{C}}(X, X) \xrightarrow{F_{X,X}} \underline{\text{hom}}_{\mathcal{D}}(X, X).$$

- (2) For every $X, Y, Z \in \mathcal{C}$ the following diagram commutes

$$\begin{array}{ccc} \underline{\text{hom}}_{\mathcal{C}}(Y, Z) \otimes \underline{\text{hom}}_{\mathcal{C}}(X, Y) & \xrightarrow{c_{Z,Y,X}} & \underline{\text{hom}}_{\mathcal{C}}(X, Z) \\ \downarrow F_{Y,Z} \otimes F_{X,Y} & & \downarrow F_{X,Z} \\ \underline{\text{hom}}_{\mathcal{D}}(F(Y), F(Z)) \otimes \underline{\text{hom}}_{\mathcal{D}}(F(X), F(Y)) & \xrightarrow{c_{FZ,FY,FX}} & \underline{\text{hom}}_{\mathcal{D}}(F(X), F(Z)) \end{array}$$

If $\mathcal{C} \xrightarrow{F} \mathcal{D} \xrightarrow{G} \mathcal{E}$ are $\mathcal{A}$ -enriched functors, then we can define their composition $G \circ F: \mathcal{C} \rightarrow \mathcal{E}$ by

$$(G \circ F)(X) = G(F(X)), \quad X \in \mathcal{C}, \quad (G \circ F)_{X,Y} = G_{FX,FY} \circ F_{X,Y}.$$

Thus we can form the (ordinary) category $\text{Cat}(\mathcal{A})$ of small (meaning $\text{Ob}(\mathcal{C})$ is small) $\mathcal{A}$ -enriched categories, whose morphisms are $\mathcal{A}$ -enriched functors.

2.3.8 Example

A $\mathbf{Set}$ -enriched functor is easily seen to be an ordinary functor. Thus there is an isomorphism of categories $\text{Cat} \cong \text{Cat}(\mathbf{Set})$.

2.3.9 Example

If $F: \mathcal{A} \rightarrow \mathcal{A}'$ is a lax monoidal functor between monoidal categories, then we showed that it maps $\mathcal{A}$ -enriched categories $\mathcal{C}$ to $\mathcal{A}'$ -enriched categories $\mathcal{C}'$. Similarly it maps an $\mathcal{A}$ -enriched functor $f: \mathcal{C} \rightarrow \mathcal{D}$ to the $\mathcal{A}'$ -enriched functor $f': \mathcal{C}' \rightarrow \mathcal{D}'$ which acts on objects as f does, and $f'_{X,Y} = F(f_{X,Y})$.

This defines a functor of ordinary categories $\text{Cat}(\mathcal{A}) \rightarrow \text{Cat}(\mathcal{A}')$. In particular for the functor $F = \underline{\text{hom}}_{\mathcal{A}}(\mathbf{1}, \bullet)$, we obtain a forgetful functor

$$\text{Cat}(\mathcal{A}) \rightarrow \text{Cat}(\text{Set}) \cong \text{Cat}$$

which assigns to each (small) $\mathcal{A}$ -enriched category its underlying ordinary category.

We can also define enriched natural transformations. Recall that an ordinary natural transformation $\alpha: F \Rightarrow G: \mathcal{C} \rightarrow \mathcal{D}$ is given by a $\mathcal{C}$ -indexed family of $\mathcal{D}$ -morphisms $\alpha_X: FX \rightarrow GX$ such that the usual diagram commutes. We can write this naturality condition as requiring that the following square commutes for all $X, Y \in \mathcal{C}$:

$$\begin{array}{ccc} \text{hom}_{\mathcal{C}}(X, Y) & \xrightarrow{F} & \text{hom}_{\mathcal{D}}(FX, FY) \\ G \downarrow & & \downarrow \alpha_Y \circ \\ \text{hom}_{\mathcal{D}}(GX, GY) & \xrightarrow{\circ \alpha_X} & \text{hom}_{\mathcal{D}}(FX, GY) \end{array}$$

In enriched category theory in order to talk about “picking out” a morphism α_X , we talk instead about a morphism $\alpha_X: \mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{D}}(FX, FY)$. This motivates the following.

2.3.10 Definition (Enriched natural transformations)

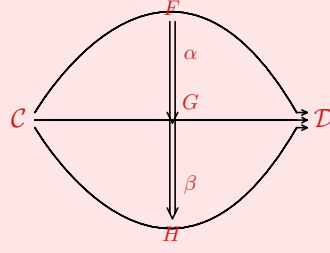
Let $F, G: \mathcal{C} \rightarrow \mathcal{D}$ be $\mathcal{A}$ -enriched functors. An **$\mathcal{A}$ -enriched natural transformation** $\alpha: F \Rightarrow G$ is an $\text{Ob}(\mathcal{C})$ indexed family of $\mathcal{A}$ -morphisms $\{\alpha_X: \mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{D}}(FX, GX)\}_{X \in \mathcal{C}}$ such that the following diagram commutes (in $\mathcal{A}$) for all $X, Y \in \mathcal{C}$:

$$\begin{array}{ccc} \mathbf{1} \otimes \underline{\text{hom}}_{\mathcal{C}}(X, Y) & \xrightarrow{\alpha_Y \otimes F} & \underline{\text{hom}}_{\mathcal{D}}(FY, GY) \otimes \underline{\text{hom}}_{\mathcal{D}}(FX, FY) \\ \lambda^{-1} \uparrow & & \downarrow c \\ \underline{\text{hom}}_{\mathcal{C}}(X, Y) & & \underline{\text{hom}}_{\mathcal{D}}(FX, GY) \\ \rho^{-1} \downarrow & & \uparrow c \\ \underline{\text{hom}}_{\mathcal{C}}(X, Y) \otimes \mathbf{1} & \xrightarrow{G \otimes \alpha_X} & \underline{\text{hom}}_{\mathcal{D}}(GX, GY) \otimes \underline{\text{hom}}_{\mathcal{D}}(FX, GX) \end{array}$$

As with ordinary natural transformations, we can compose enriched natural transformations both vertically and horizontally.

2.3.11 Definition

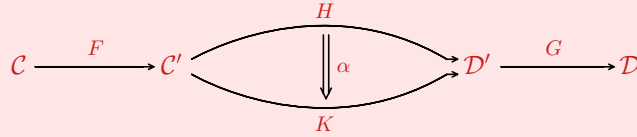
Suppose $\mathcal{C}, \mathcal{D}$ are $\mathcal{A}$ -enriched categories and we have enriched functors and natural transformations as in the following diagram:



Then we define the **vertical composition** $\beta \circ \alpha: F \Rightarrow H$ to be

$$(\beta \circ \alpha)_X = \mathbf{1} \xrightarrow{v^{-1}} \mathbf{1} \otimes \mathbf{1} \xrightarrow{\beta_X \otimes \alpha_X} \underline{\text{hom}}_{\mathcal{D}}(GX, HX) \otimes \underline{\text{hom}}_{\mathcal{D}}(FX, GX) \xrightarrow{c} \underline{\text{hom}}_{\mathcal{D}}(FX, HX).$$

And if we have a diagram of the form

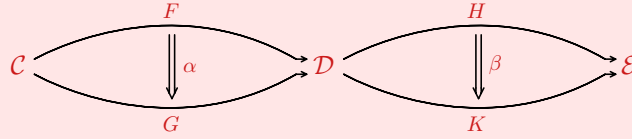


We define the **whiskering** of α to be

$$(G\alpha)_X = \mathbf{1} \xrightarrow{\alpha_X} \underline{\text{hom}}_{\mathcal{D}'}(HX, KX) \xrightarrow{G} \underline{\text{hom}}_{\mathcal{D}}(GHX, GKX): G \circ H \Rightarrow G \circ K$$

$$(\alpha F)_X = \alpha_{FX}: H \circ F \Rightarrow K \circ F$$

Finally if we have a diagram of the form



We define the **horizontal composition** $\beta \star \alpha: H \circ F \Rightarrow K \circ G$ to be

$$\beta \star \alpha = (\beta G) \circ (H\alpha) = (K\alpha) \circ (\beta F). \quad (1)$$

It is straightforward to show that vertical composition, whiskering, and horizontal composition are all natural transformations. It is furthermore straightforward to show the identity (1). Furthermore it is straightforward to show that $\text{Cat}(\mathcal{A})$ forms a strict 2-category (i.e. a category enriched over Cat).

2.3.12 Remark

Recall the construction of an underlying category for an $\mathcal{A}$ -enriched category $\mathcal{C}$, denoted $\mathcal{C}_0$. As it turns out this construction is functorial, 2-functorial even.

Let $\mathcal{I}$ be the **unit $\mathcal{A}$ -category** with a single object $*$ and $\mathcal{I}(*, *) = \mathbf{1}$; composition is given by the unit v . Then the underlying category functor is just the representable 2-functor

$$\bullet_0: \text{Cat}(\mathcal{A}) \rightarrow \text{Cat}, \quad \bullet_0 = \text{hom}_{\text{Cat}(\mathcal{A})}(\mathcal{I}, \bullet).$$

Indeed, $\mathcal{A}$ -functors $\mathcal{I} \rightarrow \mathcal{C}$ correspond to a choice of object $A \in \mathcal{C}$. A natural transformation $A \Rightarrow B: \mathcal{I} \rightarrow \mathcal{C}$ corresponds to a morphism $\mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{C}}(A, B)$. Thus the category $\text{hom}_{\text{Cat}(\mathcal{A})}(\mathcal{I}, \mathcal{C})$ has as objects the objects of $\mathcal{C}$ and as morphisms the morphisms $\mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{C}}(A, B)$ for $A, B \in \mathcal{C}$. This is precisely $\mathcal{C}_0$.

Note that we can write this as

$$\text{hom}_{\mathcal{C}_0}(A, B) = \mathcal{A}_1(\underline{\text{hom}}_{\mathcal{C}}(A, B)),$$

where $\mathcal{A}_1: \mathcal{A} \rightarrow \text{Set}$ is the representable functor $\mathcal{A}_1 = \text{hom}_{\mathcal{A}}(\mathbf{1}, \bullet)$. It then follows that the relation of $\mathcal{C}_0$ and $\mathcal{C}$ depends heavily on how faithful the functor $\mathcal{A}_1$ is.

For example if $\mathcal{A} = \text{Cat}$ then Cat_1 is not faithful: $\mathcal{C}$ is a 2-category while $\mathcal{C}_0$ is the category obtained by forgetting the 2-cells. When $\mathcal{A} = \text{Top}$ then $\mathcal{C}_0$ only loses the topology on the hom-sets. And when $\mathcal{A} = \text{Ab}$ or Mod_R then $\mathcal{A}_1$ is faithful and conservative (reflects isomorphisms) so that $\mathcal{C}_0$ is relatively similar to $\mathcal{C}$. An enriched functor $F: \mathcal{C} \rightarrow \mathcal{D}$ then corresponds to the ordinary functor $F_0: \mathcal{C}_0 \rightarrow \mathcal{D}_0$ which maps $A: \mathcal{I} \rightarrow \mathcal{C}$ to $F \circ A$ and $f: A \rightarrow B$ to the whiskering Ff , which is the composite

$$\mathbf{1} \xrightarrow{f} \underline{\text{hom}}_{\mathcal{C}}(A, B) \xrightarrow{F} \underline{\text{hom}}_{\mathcal{D}}(FA, FB).$$

As we identify $A: \mathcal{I} \rightarrow \mathcal{C}$ with $A \in \mathcal{C}$, and $Ff: \mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{D}}(FA, FB)$ with morphisms $FA \rightarrow FB$ this tells us that

$$F_0(A) = FA, \quad F_0(f) = Ff.$$

In other words,

$$(F_0)_{A,B}: \text{hom}_{\mathcal{C}_0}(A, B) \rightarrow \text{hom}_{\mathcal{D}_0}(FA, FB) \text{ is } \mathcal{A}_1 F_{A,B}: \mathcal{A}_1 \underline{\text{hom}}_{\mathcal{C}}(A, B) \rightarrow \mathcal{A}_1 \underline{\text{hom}}_{\mathcal{D}}(FA, FB).$$

For $\mathcal{A}$ -natural transformations $\alpha: F \Rightarrow G$, these become ordinary natural transformations $\alpha_0: F_0 \Rightarrow G_0$ whose components $(\alpha_0)_A \in \text{hom}_{\mathcal{D}_0}(FA, GA)$ is precisely $\alpha_A: \mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{D}}(FA, GA)$.

2.4 Symmetric monoidal categories

In the sequel, monoidal categories will in general be denoted by $\mathcal{V}$ (instead of $\mathcal{A}, \mathcal{B}, \mathcal{C}$, etc.).

2.4.1 Definition

Let $\mathcal{V}$ be a monoidal category. A **braiding** for $\mathcal{V}$ is a natural isomorphism $\beta: \otimes \Rightarrow \otimes^{\text{rev}}$ which satisfies the following two coherence conditions:

$$\begin{array}{ccccc} X \otimes (Y \otimes Z) & \xrightarrow{\alpha} & (X \otimes Y) \otimes Z & \xrightarrow{\beta_{X \otimes Y, Z}} & Z \otimes (X \otimes Y) \\ \downarrow \text{id} \otimes \beta_{Y, Z} & & & & \downarrow \alpha \\ X \otimes (Z \otimes Y) & \xrightarrow{\alpha} & (X \otimes Z) \otimes Y & \xrightarrow{\beta_{X, Z} \otimes \text{id}} & (Z \otimes X) \otimes Y \\ & & & & \\ (X \otimes Y) \otimes Z & \xrightarrow{\alpha^{-1}} & X \otimes (Y \otimes Z) & \xrightarrow{\beta_{X, Y \otimes Z}} & (Y \otimes Z) \otimes X \\ \downarrow \beta_{X, Y} \otimes \text{id} & & & & \downarrow \alpha^{-1} \\ (Y \otimes X) \otimes Z & \xrightarrow{\alpha^{-1}} & Y \otimes (X \otimes Z) & \xrightarrow{\text{id} \otimes \beta_{X, Z}} & Y \otimes (Z \otimes X) \end{array}$$

A monoidal category equipped with a braiding is called a **braided monoidal category**.

It follows from the two coherence conditions above that in a braided monoidal category, the following diagram commutes

$$\begin{array}{ccc} \mathbf{1} \otimes X & \xrightarrow{\beta_{\mathbf{1}, X}} & X \otimes \mathbf{1} \\ & \searrow \lambda & \swarrow \rho \\ & X & \end{array}$$

2.4.2 Definition

A **symmetric monoidal category** is a braided monoidal category $\mathcal{V}$ such that $\beta_{Y,X} = \beta_{X,Y}^{-1}$ for all X, Y .

That is, the braiding forms a natural isomorphism $X \otimes \bullet \cong \bullet \otimes X$.

Symmetric monoidal categories are important because they allow us to extend many of the categorical constructions on ordinary categories to $\mathcal{V}$ -enriched categories.

2.4.3 Example

If $\mathcal{C}$ is a category with finite products, viewed as a monoidal category it is symmetric monoidal. The braiding is given by the unique isomorphism $X \times Y \xrightarrow{\sim} Y \times X$ commuting with projections.

In particular **Set** and **Cat** are symmetric monoidal categories.

2.4.4 Example

The tensor product turns $\mathbf{Mod}_R$ into a symmetric monoidal category.

2.4.5 Remark

Let $\mathcal{V}$ be a symmetric monoidal category, then we can form a 2-functor

$$\otimes: \mathbf{Cat}(\mathcal{V}) \times \mathbf{Cat}(\mathcal{V}) \rightarrow \mathbf{Cat}(\mathcal{V}).$$

We define it on objects, functors, and natural transformations as follows.

- (1) For $\mathcal{V}$ -categories $\mathcal{A}, \mathcal{B}$ we define the $\mathcal{V}$ -category $\mathcal{A} \otimes \mathcal{B}$ whose objects are pairs (A, B) for $A \in \mathcal{A}, B \in \mathcal{B}$. And we define the hom-sets by

$$\underline{\mathrm{hom}}_{\mathcal{A} \otimes \mathcal{B}}((A, B), (A', B')) = \underline{\mathrm{hom}}_{\mathcal{A}}(A, A') \otimes \underline{\mathrm{hom}}_{\mathcal{B}}(B, B').$$

Composition is defined as

$$\begin{aligned} & (\underline{\mathrm{hom}}_{\mathcal{A}}(A', A'') \otimes \underline{\mathrm{hom}}_{\mathcal{B}}(B', B'')) \otimes (\underline{\mathrm{hom}}_{\mathcal{A}}(A, A') \otimes \underline{\mathrm{hom}}_{\mathcal{B}}(B, B')) \\ & \xrightarrow{\sim} (\underline{\mathrm{hom}}_{\mathcal{A}}(A', A'') \otimes \underline{\mathrm{hom}}_{\mathcal{A}}(A, A')) \otimes (\underline{\mathrm{hom}}_{\mathcal{B}}(B, B'') \otimes \underline{\mathrm{hom}}_{\mathcal{B}}(B, B')) \\ & \xrightarrow{c \otimes c} \underline{\mathrm{hom}}_{\mathcal{A}}(A, A'') \otimes \underline{\mathrm{hom}}_{\mathcal{B}}(B, B''). \end{aligned}$$

The first isomorphism here is any composition of α, β ; the coherence conditions ensure that any composition will give the same isomorphism.

The identity is the composite

$$\mathbf{1} \xrightarrow{v^{-1}} \mathbf{1} \otimes \mathbf{1} \xrightarrow{e_A \otimes e_B} \underline{\mathrm{hom}}_{\mathcal{A}}(A, A) \otimes \underline{\mathrm{hom}}_{\mathcal{B}}(B, B).$$

It is straightforward to check that $\mathcal{A} \otimes \mathcal{B}$ forms a $\mathcal{V}$ -category.

- (2) Now for $F: \mathcal{A} \rightarrow \mathcal{A}'$ and $G: \mathcal{B} \rightarrow \mathcal{B}'$ we define $F \otimes G: \mathcal{A} \otimes \mathcal{B} \rightarrow \mathcal{A}' \otimes \mathcal{B}'$ which maps (A, B) to (FA, GB) and

$$\begin{aligned} & (F \otimes G)_{(A,B),(A',B')} \\ & = \underline{\mathrm{hom}}_{\mathcal{A}}(A, A') \otimes \underline{\mathrm{hom}}_{\mathcal{B}}(B, B') \xrightarrow{F_{A,A'} \otimes G_{B,B'}} \underline{\mathrm{hom}}_{\mathcal{A}'}(FA, FA') \otimes \underline{\mathrm{hom}}_{\mathcal{B}'}(FB, FB'). \end{aligned}$$

(3) And for $\alpha: F \Rightarrow F'$ and $\beta: G \Rightarrow G'$ we define $\alpha \otimes \beta: F \otimes G \Rightarrow F' \otimes G'$ where

$$(\alpha \otimes \beta)_{A,B}: \mathbf{1} \xrightarrow{v^{-1}} \mathbf{1} \otimes \mathbf{1} \xrightarrow{\alpha_A \otimes \beta_B} \underline{\text{hom}}_{\mathcal{A}'}(FA, F'A) \otimes \underline{\text{hom}}_{\mathcal{B}'}(GB, G'B).$$

It is straightforward to check that this is all well-defined. Furthermore there are evident isomorphisms

$$\mathcal{A} \otimes (\mathcal{B} \otimes \mathcal{C}) \cong (\mathcal{A} \otimes \mathcal{B}) \otimes \mathcal{C}, \quad \mathcal{I} \otimes \mathcal{A} \cong \mathcal{A} \otimes \mathcal{A} \otimes \mathcal{I}, \quad \mathcal{A} \otimes \mathcal{B} \cong \mathcal{B} \otimes \mathcal{A}$$

where $\mathcal{I}$ is the unit $\mathcal{V}$ -category (with a single object and $\mathcal{I}(*, *) = \mathbf{1}$). It is again straightforward to check that these isomorphism satisfy the necessary coherence conditions to form $\text{Cat}(\mathcal{V})$ into a **symmetric monoidal 2-category**.

2.4.6 Remark (Opposite categories)

We can also form the 2-functor

$$\bullet^{\text{op}}: \text{Cat}(\mathcal{V}) \rightarrow \text{Cat}(\mathcal{V})$$

which maps a $\mathcal{V}$ -category $\mathcal{A}$ to its **opposite**: $\mathcal{A}^{\text{op}}$. $\mathcal{A}^{\text{op}}$ has the same objects as $\mathcal{A}$, and $\underline{\text{hom}}_{\mathcal{A}^{\text{op}}}(A, B) = \underline{\text{hom}}_{\mathcal{A}}(B, A)$, with composition given by

$$\underline{\text{hom}}_{\mathcal{A}}(C, B) \otimes \underline{\text{hom}}_{\mathcal{A}}(B, A) \xrightarrow{\beta} \underline{\text{hom}}_{\mathcal{A}}(B, A) \otimes \underline{\text{hom}}_{\mathcal{A}}(C, B) \xrightarrow{c} \underline{\text{hom}}_{\mathcal{A}}(C, A).$$

And the unit is just that of $\mathcal{A}$'s: $\mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{A}^{\text{op}}}(A, A) = \underline{\text{hom}}_{\mathcal{A}}(A, A)$.

A functor $F: \mathcal{A} \rightarrow \mathcal{B}$ is mapped to $F^{\text{op}}: \mathcal{A}^{\text{op}} \rightarrow \mathcal{B}^{\text{op}}$ which acts as F does on objects and

$$F_{A,B}^{\text{op}} = F_{B,A}: \underline{\text{hom}}_{\mathcal{A}}(B, A) \rightarrow \underline{\text{hom}}_{\mathcal{B}}(FB, FA).$$

A natural transformation $\alpha: F \Rightarrow G$ maps to $\alpha^{\text{op}}: G^{\text{op}} \Rightarrow F^{\text{op}}$ where

$$\alpha_A^{\text{op}} = \alpha_A: \mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{B}^{\text{op}}}(GA, FA) = \underline{\text{hom}}_{\mathcal{B}}(FA, GA).$$

Enriched functors $\mathcal{A}^{\text{op}} \rightarrow \mathcal{B}$ are called **contravariant functors** from $\mathcal{A}$ to $\mathcal{B}$.

It is easy to see that in terms of the underlying categories, we have

$$(\mathcal{A}^{\text{op}})_0 = (\mathcal{A}_0)^{\text{op}}, \quad (F^{\text{op}})_0 = (F_0)^{\text{op}}$$

for $\mathcal{V}$ -categories $\mathcal{A}$ and $\mathcal{V}$ -functors F .

2.4.7 Remark (Two-variable functors)

Enriched functors $\mathcal{A} \otimes \mathcal{B} \rightarrow \mathcal{C}$ can be viewed as two-variable functors. A two-variable functor $F: \mathcal{A} \otimes \mathcal{B} \rightarrow \mathcal{C}$ gives rise two single-variable functors $F(A, \bullet): \mathcal{B} \rightarrow \mathcal{C}$ and $F(\bullet, B): \mathcal{A} \rightarrow \mathcal{C}$ for $A \in \mathcal{A}$ and $B \in \mathcal{B}$:

$$F(A, \bullet) = \mathcal{B} \cong \mathcal{I} \otimes \mathcal{B} \xrightarrow{A \otimes \text{id}} \mathcal{A} \otimes \mathcal{B} \xrightarrow{F} \mathcal{C}, \quad F(\bullet, B) = \mathcal{A} \cong \mathcal{A} \otimes \mathcal{I} \xrightarrow{\text{id} \otimes B} \mathcal{A} \otimes \mathcal{B} \xrightarrow{F} \mathcal{C}.$$

Explicitly for $B, B' \in \mathcal{B}$ we have that

$$\begin{aligned} F(A, \bullet)_{B, B'} &= \underline{\text{hom}}_{\mathcal{B}}(B, B') \xrightarrow{\lambda} \mathbf{1} \otimes \underline{\text{hom}}_{\mathcal{B}}(B, B') \xrightarrow{e_A} \underline{\text{hom}}_{\mathcal{A}}(A, A) \otimes \underline{\text{hom}}_{\mathcal{B}}(B, B') \\ &\xrightarrow{F_{(A, B), (A, B')}} \underline{\text{hom}}_{\mathcal{C}}(F(A, B), F(A, B')) \end{aligned}$$

and similar for $F(\bullet, B)$. The functoriality of these functors is due to the functoriality of the composition of functors.

Conversely suppose we have two families of functors

$$F(A, \bullet): \mathcal{B} \rightarrow \mathcal{C}, \quad F(\bullet, B): \mathcal{A} \rightarrow \mathcal{C}$$

the first indexed by $\text{Ob}(\mathcal{A})$, the second by $\text{Ob}(\mathcal{B})$. Further suppose that for all $A \in \mathcal{A}, B \in \mathcal{B}$ we have $F(A, \bullet)B = F(\bullet, B)A$, and let us denote this common object by $F(A, B)$. Then there is a functor $F: \mathcal{A} \otimes \mathcal{B} \rightarrow \mathcal{C}$ for which these are its partial functors iff the following diagram commutes

$$\begin{array}{ccc} \underline{\text{hom}}_{\mathcal{A}}(A, A') \otimes \underline{\text{hom}}_{\mathcal{B}}(B, B') & \xrightarrow{F(\bullet, B') \otimes F(A, \bullet)} & \underline{\text{hom}}_{\mathcal{C}}(F(A, B'), F(A', B')) \otimes \underline{\text{hom}}_{\mathcal{C}}(F(A, B), F(A, B')) \\ \downarrow \beta & & \downarrow c \\ & & \underline{\text{hom}}_{\mathcal{C}}(F(A, B), F(A', B')) \\ & & \uparrow c \\ \underline{\text{hom}}_{\mathcal{B}}(B, B') \otimes \underline{\text{hom}}_{\mathcal{A}}(A, A') & \xrightarrow{F(A', \bullet) \otimes F(\bullet, B)} & \underline{\text{hom}}_{\mathcal{C}}(F(A', B), F(A', B')) \otimes \underline{\text{hom}}_{\mathcal{C}}(F(A, B), F(A, B')) \end{array}$$

Moreover such a functor F is unique, and $F_{(A, B), (A', B')}$ is the diagonal of the above diagram.

Furthermore, if $F, G: \mathcal{A} \otimes \mathcal{B} \rightarrow \mathcal{C}$ are two functors, a family $\alpha_{A, B}: \mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{C}}(F(A, B), G(A, B))$ forms an enriched natural transformation iff for each fixed $A \in \mathcal{A}$, $\alpha_{A, \bullet}$ forms a natural transformation $F(A, \bullet) \Rightarrow G(A, \bullet)$ and for each fixed $B \in \mathcal{B}$, $\alpha_{\bullet, B}$ a natural transformation $F(\bullet, B) \Rightarrow G(\bullet, B)$. Thus naturality may be verified variable-by-variable.

Note:

In general it is not true that $(\mathcal{A} \otimes \mathcal{B})_0$ is $\mathcal{A}_0 \times \mathcal{B}_0$. Their objects are the same, both being $\text{Ob}(\mathcal{A}) \times \text{Ob}(\mathcal{B})$, but a morphism in the former is a morphism (in $\mathcal{V}$) $\mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{A}}(A, A') \otimes \underline{\text{hom}}_{\mathcal{B}}(B, B')$, while a morphism in the latter is a pair of morphisms $\mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{A}}(A, A')$ and $\mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{B}}(B, B')$. There is then a canonical functor $\mathcal{A}_0 \times \mathcal{B}_0 \rightarrow (\mathcal{A} \otimes \mathcal{B})_0$ which is the identity on objects and maps the pair $f: \mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{A}}(A, A')$ and $g: \mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{B}}(B, B')$ to

$$\mathbf{1} \cong \mathbf{1} \otimes \mathbf{1} \xrightarrow{f \otimes g} \underline{\text{hom}}_{\mathcal{A}}(A, A') \otimes \underline{\text{hom}}_{\mathcal{B}}(B, B').$$

2.5 Closed monoidal categories

2.5.1 Definition

A monoidal category $\mathcal{V}$ is **left** (resp. **right**) **closed** if each functor $Y \otimes \bullet: \mathcal{V} \rightarrow \mathcal{V}$ (resp. $\bullet \otimes Y: \mathcal{V} \rightarrow \mathcal{V}$) has a right adjoint $[Y, \bullet]$. If a monoidal category is both left and right closed, it is called **biclosed**.

When $\mathcal{V}$ is a symmetric monoidal category, there is no differentiation between being left or right closed, and we just say that $\mathcal{V}$ is **closed**.

Suppose $\mathcal{V}$ is right closed, then consider the unit and counit of the adjunction for each Y :

$$\text{coev}: X \rightarrow [Y, X \otimes Y], \quad \text{ev}: [Y, X] \otimes Y \rightarrow X$$

called **coevaluation** and **evaluation** respectively.

It is a classical result that for a functor $F: \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$, if $F(\bullet, B)$ has a right adjoint $G(B, \bullet)$ for all $B \in \mathcal{B}$, then G assembles into a functor $G: \mathcal{B}^{\text{op}} \times \mathcal{C} \rightarrow \mathcal{A}$. In this case we have a functor

$$[\bullet, \bullet]: \mathcal{V}^{\text{op}} \times \mathcal{V} \rightarrow \mathcal{V}.$$

Consider the adjunction:

$$\text{hom}_{\mathcal{V}}(X \otimes Y, Z) \cong \text{hom}_{\mathcal{V}}(X, [Y, Z]),$$

natural in all variables. Letting $X = \mathbf{1}$ we get

$$\text{hom}_{\mathcal{V}}(Y, Z) \cong \text{hom}_{\mathcal{V}}(\mathbf{1} \otimes Y, Z) \cong \text{hom}_{\mathcal{V}}(\mathbf{1}, [Y, Z]) = \mathcal{V}_1[Y, Z]$$

where $\mathcal{V}_1$ is the corepresentable functor $\mathcal{V}_1 = \text{hom}_{\mathcal{V}}(\mathbf{1}, \bullet)$. Due to this relation between $\text{hom}_{\mathcal{V}}$ and $[\bullet, \bullet]$ (and a deeper relation we will see soon), $[\bullet, \bullet]$ is called the **internal hom** of $\mathcal{V}$.

Consider the adjunction again, letting $Y = \mathbf{1}$ we get

$$\text{hom}_{\mathcal{V}}(X, Z) \cong \text{hom}_{\mathcal{V}}(X \otimes \mathbf{1}, Z) \cong \text{hom}_{\mathcal{V}}(X, [\mathbf{1}, Z])$$

which by Yoneda means that there is an isomorphism $Z \cong [\mathbf{1}, Z]$. Again in the adjunction, replace X by $W \otimes X$ so that we get

$$\text{hom}_{\mathcal{V}}(W, [X, [Y, Z]]) \cong \text{hom}_{\mathcal{V}}((W \otimes X) \otimes Y, Z) \cong \text{hom}_{\mathcal{V}}(W \otimes (X \otimes Y), Z) \cong \text{hom}_{\mathcal{V}}(W, [X \otimes Y, Z])$$

so that again by Yoneda, $[X, [Y, Z]] \cong [X \otimes Y, Z]$.

Thus we see that the adjunction $\bullet \otimes Y \dashv [Y, \bullet]$ becomes *internal* to $\mathcal{V}$. That is, the natural isomorphism $[X, [Y, Z]] \cong [X \otimes Y, Z]$ gives rise to the usual isomorphism $\text{hom}_{\mathcal{V}}(X \otimes Y, Z) \cong \text{hom}_{\mathcal{V}}(X, [Y, Z])$ via $\mathcal{V}_1$.

Note:

A cartesian monoidal category $\mathcal{V}$ (cartesian meaning its monoidal product is given by the direct product) is automatically symmetric. When it is closed we call it **cartesian closed**.

Set is an example of a cartesian closed category, its hom-sets are internal homs.

2.5.2 Example

The category **Top** of topological spaces is a cartesian monoidal category, but it is not cartesian closed. This is because in general $\bullet \times Y$ does not preserve equalizers.

But if we limit ourselves to **CGHaus**, the full subcategory of **Top** consisting of compactly generated Hausdorff spaces, then **CGHaus** is cartesian closed. A compactly generated Hausdorff space is a Hausdorff space X where $U \subseteq X$ is open iff $U \cap K \subseteq K$ is open for all $K \subseteq X$ compact.

We can give each hom-set $C(X, Y)$ a topology, called the **compact-open topology** consisting of a subbase

given by subsets of the form

$$V(K, U) = \{f: X \rightarrow Y \mid f(K) \subseteq U\}, \quad K \subseteq X \text{ compact}, \quad U \subseteq Y \text{ open}.$$

It is then straightforward to verify that the obvious maps (currying)

$$C(X \times Y, Z) \cong C(X, C(Y, Z))$$

are well-defined and homeomorphisms.

2.5.3 Example

An **lattice** is a poset for which every two elements a, b has a **join** and **meet**: $a \vee b = \sup\{a, b\}$ and $a \wedge b = \inf\{a, b\}$. Viewing posets as categories, this is the same as requiring that the category have all finite limits and colimits, as supremums and infimums are coproducts and products respectively in a poset category.

A lattice is then a cartesian monoidal category, and it is called a **Heyting algebra** if it is closed.

The reason we care about symmetric closed monoidal categories is because the inner hom enriches the category over itself. That is, we can form a $\mathcal{V}$ -category, which we will call $\mathcal{V}$ as well, such that its underlying category is isomorphic to the original ordinary monoidal category.

The construction is straightforward: for $X, Y \in \mathcal{V}$ let the enriched hom be the internal hom $\underline{\text{hom}}_{\mathcal{V}}(X, Y) = [X, Y]$. Then composition $c: [Y, Z] \otimes [X, Y] \rightarrow [X, Z]$ corresponds under the adjunction to the composite

$$([Y, Z] \otimes [X, Y]) \otimes X \xrightarrow{\alpha_{Y,Z}} \otimes([X, Y] \otimes X) \xrightarrow{\text{id} \otimes \text{ev}} [Y, Z] \otimes Y \xrightarrow{\text{ev}} Z.$$

That is, c is the unique morphism satisfying

$$\text{ev} \circ (\text{id} \otimes c) = \text{ev} \circ (\text{id} \otimes \text{ev}) \circ \alpha. \quad (1)$$

(Recall that if ε is the counit of an adjunction, the isomorphism $\text{hom}(x, Gy) \rightarrow \text{hom}(Fx, y)$ is given by $\varepsilon \circ F(\bullet)$.)

The identity $e_X: \mathbf{1} \rightarrow [X, X]$ corresponds under the adjunction to $\lambda_X: \mathbf{1} \otimes X \rightarrow X$. That is, e is the unique morphism satisfying

$$\text{ev} \circ (\text{id} \otimes e) = \lambda. \quad (2)$$

With equations (1) and (2) in mind, it is straightforward to verify that this forms a $\mathcal{V}$ -category.

It is furthermore straightforward (albeit tedious) to verify that the underlying category of this $\mathcal{V}$ -category is $\mathcal{V}$ itself. This follows from the natural isomorphism

$$\mathcal{V}_1[X, Y] \cong \text{hom}_{\mathcal{V}}(X, Y).$$

Note:

In lieu of this, and in order to distinguish (when necessary) between the ordinary category $\mathcal{V}$ and the self-enriched category $\mathcal{V}$, we will denote the ordinary category by $\mathcal{V}_0$. We will sometimes do this for monoidal categories which are not necessarily closed, or symmetric.

Now we note that if $\mathcal{A}$ is a $\mathcal{V}$ -enriched category, then we can form a $\mathcal{V}$ -enriched functor for each $A \in \mathcal{A}$,

$$\underline{\text{hom}}_{\mathcal{A}}(A, \bullet): \mathcal{A} \rightarrow \mathcal{V}, \quad B \mapsto \underline{\text{hom}}_{\mathcal{A}}(A, B)$$

and

$$\underline{\text{hom}}_{\mathcal{A}}(A, \bullet)_{B,C}: \underline{\text{hom}}_{\mathcal{A}}(B, C) \rightarrow [\underline{\text{hom}}_{\mathcal{A}}(A, B), \underline{\text{hom}}_{\mathcal{A}}(A, C)]$$

corresponds under the adjunction to composition:

$$\underline{\text{hom}}_{\mathcal{A}}(B, C) \otimes \underline{\text{hom}}_{\mathcal{A}}(A, B) \xrightarrow{c} \underline{\text{hom}}_{\mathcal{A}}(A, C).$$

It is again straightforward to verify that this is indeed a $\mathcal{V}$ -functor. Replacing $\mathcal{A}$ with $\mathcal{A}^{\text{op}}$ we get the functor $\underline{\text{hom}}_{\mathcal{A}^{\text{op}}}(A, \bullet) = \underline{\text{hom}}_{\mathcal{A}}(\bullet, A): \mathcal{A}^{\text{op}} \rightarrow \mathcal{V}$. These then form the partial functors of a functor

$$\underline{\text{hom}}_{\mathcal{A}}: \mathcal{A}^{\text{op}} \times \mathcal{A} \rightarrow \mathcal{V}.$$

Now consider the ordinary functor

$$\text{hom}_{\mathcal{A}}: \mathcal{A}_0^{\text{op}} \times \mathcal{A}_0 \rightarrow (\mathcal{A}^{\text{op}} \otimes \mathcal{A})_0 \xrightarrow{(\underline{\text{hom}}_{\mathcal{A}})_0} \mathcal{V}_0.$$

We know that the partial functors of $\text{hom}_{\mathcal{A}}$ are the underlying ordinary functors of the partial functors of $\underline{\text{hom}}_{\mathcal{A}}$. Thus for $g: B \rightarrow B'$ in $\mathcal{A}_0$ (corresponding to $\mathbf{1} \rightarrow \underline{\text{hom}}_{\mathcal{A}}(B, B')$), $\text{hom}_{\mathcal{A}}(A, g)$ is the composite

$$\text{hom}_{\mathcal{A}}(A, g) = \underline{\text{hom}}_{\mathcal{A}}(A, B) \xrightarrow{\lambda^{-1}} \mathbf{1} \otimes \underline{\text{hom}}_{\mathcal{A}}(A, B) \xrightarrow{g \otimes \text{id}} \underline{\text{hom}}_{\mathcal{A}}(B, B') \otimes \underline{\text{hom}}_{\mathcal{A}}(A, B) \xrightarrow{c} \underline{\text{hom}}_{\mathcal{A}}(A, B').$$

Similarly $\text{hom}_{\mathcal{A}}(f, B): \underline{\text{hom}}_{\mathcal{A}}(A, B) \rightarrow \underline{\text{hom}}_{\mathcal{A}}(A', B)$ is

$$\text{hom}_{\mathcal{A}}(f, B) = \underline{\text{hom}}_{\mathcal{A}}(A, B) \xrightarrow{\rho^{-1}} \underline{\text{hom}}_{\mathcal{A}}(A, B) \otimes \mathbf{1} \xrightarrow{\text{id} \otimes f} \underline{\text{hom}}_{\mathcal{A}}(A, B) \otimes \underline{\text{hom}}_{\mathcal{A}}(A', A) \xrightarrow{c} \underline{\text{hom}}_{\mathcal{A}}(A', B).$$

Thus the following commutes

$$\begin{array}{ccc} \mathcal{A}_0^{\text{op}} \times \mathcal{A}_0 & \xrightarrow{\text{hom}_{\mathcal{A}}} & \mathcal{V}_0 \\ & \searrow \text{hom}_{\mathcal{A}_0} & \downarrow \mathcal{A}_1 \\ & & \text{Set} \end{array}$$

When $\mathcal{A} = \mathcal{V}$ we see that $\text{hom}_{\mathcal{V}}$ is just the functor $[\bullet, \bullet]$.

Similarly we can make the monoidal product into an enriched functor $\text{ten}: \mathcal{V} \otimes \mathcal{V} \rightarrow \mathcal{V}$ mapping $\text{ten}(X, Y) = X \otimes Y$ and

$$\text{ten}_{(X, Y), (X', Y')}: [X, X'] \otimes [Y, Y'] \rightarrow [X \otimes Y, X' \otimes Y']$$

corresponds under the adjunction to

$$([X, X'] \otimes [Y, Y']) \otimes (X \otimes Y) \xrightarrow{\sim} ([X, X'] \otimes X) \otimes ([Y, Y'] \otimes Y) \xrightarrow{\text{ev} \otimes \text{ev}} X' \otimes Y'.$$

Its underlying ordinary functor is seen to be the monoidal product $\otimes$.

3 ∞ -Categories

3.1 2-categories

Note that **Cat**, the category of (small) categories, contains more structure than just an ordinary category. Indeed, between two morphisms (functors) there exist additional morphisms (natural transformations). This, as we will see, means that **Cat** is a **2-category**.